

Results — revised submission — original style

1 Datasets

#	Name	N	P	K	MeanIR	CVIR
1	GpositivePseAAC	519	440	4	3.86	1.15
2	CHD_49	555	49	6	5.77	1.60
3	Emotions	593	72	6	1.48	0.16
4	Scene	2407	294	6	1.25	0.11
5	PlantPseAAC	978	440	12	6.69	0.68
6	Water-quality	1060	16	14	1.77	0.29
7	Yeast	2417	103	14	7.20	1.82
8	HumanPseAAC	3106	440	14	15.29	1.05
9	enron	1702	1001	53	10.45	0.80

Table 1: (Table 3) Datasets used in the revised experiments. N , P , K are the number of instances, features, labels. MeanIR and CVIR report per-label imbalance (higher = more imbalanced).

Pending follow-ups (to fill before final submission)

- **LGBM run on enron.** Sequentially training (2–3h per task due to $K=53$ pairwise classifiers); will be merged into `full_enron_split_lgbm_summary/` when complete and the LGBM aggregate panels regenerated.
- **Extended PartialAbstention metrics.** The evaluator has been extended with 11 new PA metrics (`jaccard_pa`, `{example,macro,micro}-{precision,recall,f1}_pa`, `mfrd_pa`, `afrd_pa`). Existing runs were summarised with the previous metric set; the new columns will populate automatically on the next end-to-end training+evaluation pass.

2 Method encoding

For every panel that follows, the x-axis encodes the classifier as one of 12 methods, grouped into three blocks:

Block A — Order-based predictors (ours, indices 1–8). Each name has the form `<order>-<target>-<height>`; $order \in \{\text{PA} = \text{partial order}, \text{PR} = \text{pre-order}\}$; $target \in \{\text{H} = \text{Hamming}, \text{S} = \text{Subset } 0/1\}$; $height \in \{2, \infty \text{ (omitted in the label)}\}$.

- **1 = PA-H-2:** partial order, Hamming target, height ≤ 2 .

- 2 = **PA-H**: partial order, Hamming target, unrestricted height.
- 3 = **PA-S-2**: partial order, Subset 0/1 target, height ≤ 2 .
- 4 = **PA-S**: partial order, Subset 0/1, unrestricted.
- 5 = **PR-H-2**: pre-order, Hamming, height ≤ 2 .
- 6 = **PR-H**: pre-order, Hamming, unrestricted.
- 7 = **PR-S-2**: pre-order, Subset 0/1, height ≤ 2 .
- 8 = **PR-S**: pre-order, Subset 0/1, unrestricted.

Block B — Standard MLC baselines from the original paper (indices 9–11).

- 9 = **BR** (Binary Relevance): K independent binary classifiers, one per label.
- 10 = **CC** (Classifier Chain): sequential chain that uses previous labels as features.
- 11 = **CLR** (Calibrated Label Ranking): pairwise label ranking with a calibration label that separates relevant from irrelevant.

Block C — Extended baseline added for the revision (index 12).

- 12 = **ECC** (Ensemble of Classifier Chains): averages over multiple CCs with random chain orders.

Reading the panels. Each panel plots one dotted line with markers per noise level, colored as in the original paper’s Table 5: $\alpha = 0.0$, $\alpha = 0.1$, $\alpha = 0.2$, $\alpha = 0.3$. *PartialAbstention* and *ScoreVector* panels show only indices 1–8 (the remaining baselines do not export a partial-abstention prediction nor a probabilistic score vector). Vertical dashed lines at $x = 4.5$, $x = 8.5$, and $x = 11.5$ separate the four method groups: PA (1–4), PR (5–8), standard baselines BR/CC/CLR (9–11), and ECC (12). Missing methods show as gaps in the line (with a small grey “x” on the axis when the method has no data at all for that panel).

3 Paper results (RF base learner)

Reproduces the layout of the original paper’s main experiment tables (Tables 5–7): 3 datasets across, 4 metrics down, one table per prediction type (binary vector, partial abstention, score vector). Per the revision, $f_{\text{MLC}}^{\text{ham}}$ and $f_{\text{MLC}}^{\text{sub}}$ are dropped in favour of $f_{\text{MLC}}^{\text{Jacc}}$.

#	Name	N	P	K	MeanIR	CVIR
1	GpositivePseAAC	519	440	4	3.86	1.15
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Table 2: (Table 3) Datasets used in the revised experiments. N , P , K are the number of instances, features, labels. MeanIR and CVIR report per-label imbalance (higher = more imbalanced).

3.1 Standard MLC predictions (BinaryVector)

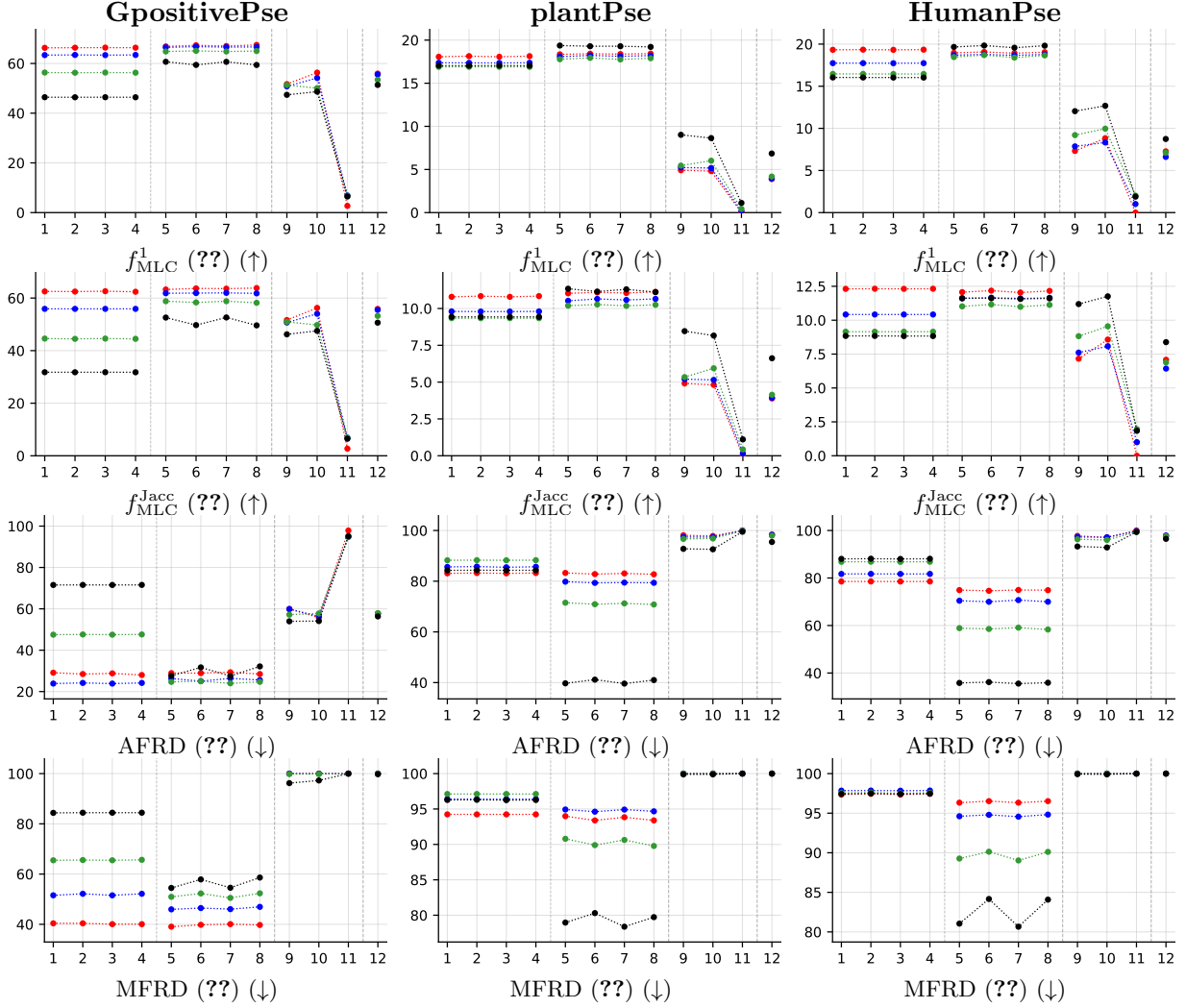


Table 3: (Table 4) Average scores (in %, y -axis) over 5×5 cross-validation folds, plotted against the encoded number of the classifiers (x -axis): (PA-H-2, PA-H, PA-S-2, PA-S), (PR-H-2, PR-H, PR-S-2, PR-S), and (BR, CC, CLR) are encoded as (1, 2, 3, 4), (5, 6, 7, 8), and (9, 10, 11), respectively. Results are color-coded with respect to the noisy level α as follows: $\alpha = 0.0$, $\alpha = 0.1$, $\alpha = 0.2$ and $\alpha = 0.3$.

3.2 Partial abstention (PartialAbstention)

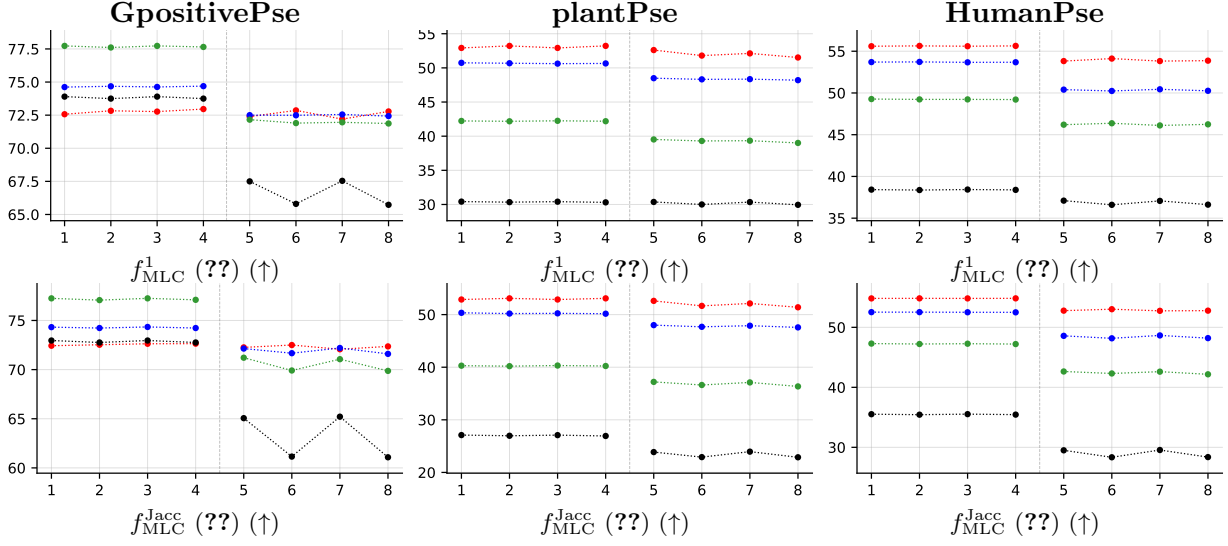


Table 4: (Table 5) Average scores (in %, y -axis) over 5×5 cross-validation folds, plotted against the encoded number of the classifiers (x -axis): (PA-H-2, PA-H, PA-S-2, PA-S) and (PR-H-2, PR-H, PR-S-2, PR-S) are encoded as (1, 2, 3, 4) and (5, 6, 7, 8), respectively. Results are color-coded with respect to the noisy level α as follows: $\alpha = 0.0$, $\alpha = 0.1$, $\alpha = 0.2$ and $\alpha = 0.3$.

3.3 Score-vector ranking metrics (AUROC, AUPRC, etc.) (ScoreVector)

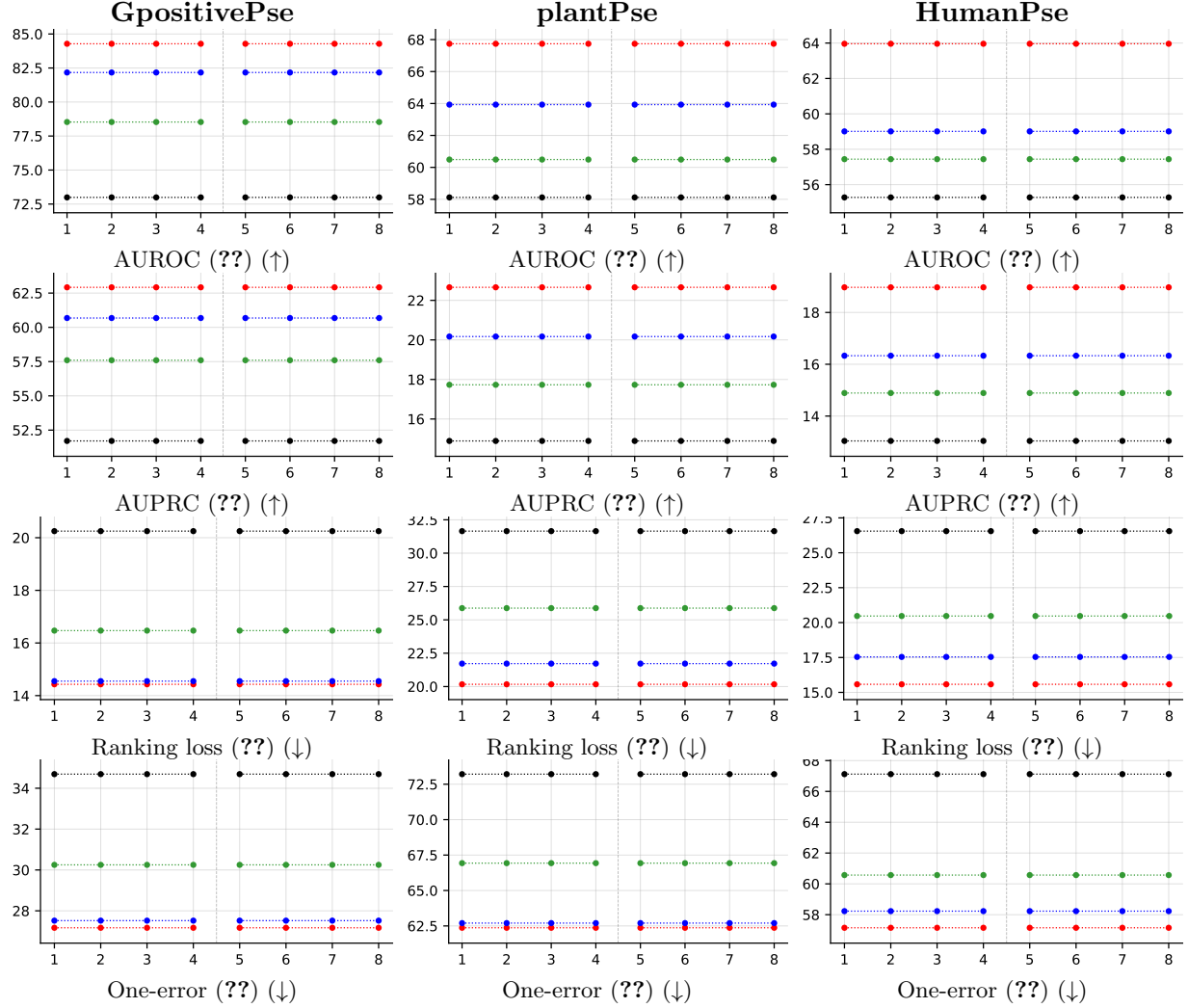


Table 5: (new) Average scores (in %, y -axis) over 5×5 cross-validation folds, plotted against the encoded number of the classifiers (x -axis): (PA-H-2, PA-H, PA-S-2, PA-S) and (PR-H-2, PR-H, PR-S-2, PR-S) are encoded as (1, 2, 3, 4) and (5, 6, 7, 8), respectively. Results are color-coded with respect to the noisy level α as follows: $\alpha = 0.0$, $\alpha = 0.1$, $\alpha = 0.2$ and $\alpha = 0.3$.

3.4 Experimental results — appendix tables (RF base learner)

Appendix counterpart of the main Paper results section. Tables G2–G5 mirror the original paper’s appendix exactly (same metric sets, with the 6 non-main datasets split into imbalanced vs. balanced buckets following the original layout). Tables G6–G8 are revision additions reporting the new metrics ($f_{\text{MLC}}^{\text{Jacc}}$, macro- f_1 , micro- f_1) and ScoreVector results.

3.4.1 Standard MLC predictions (BinaryVector)

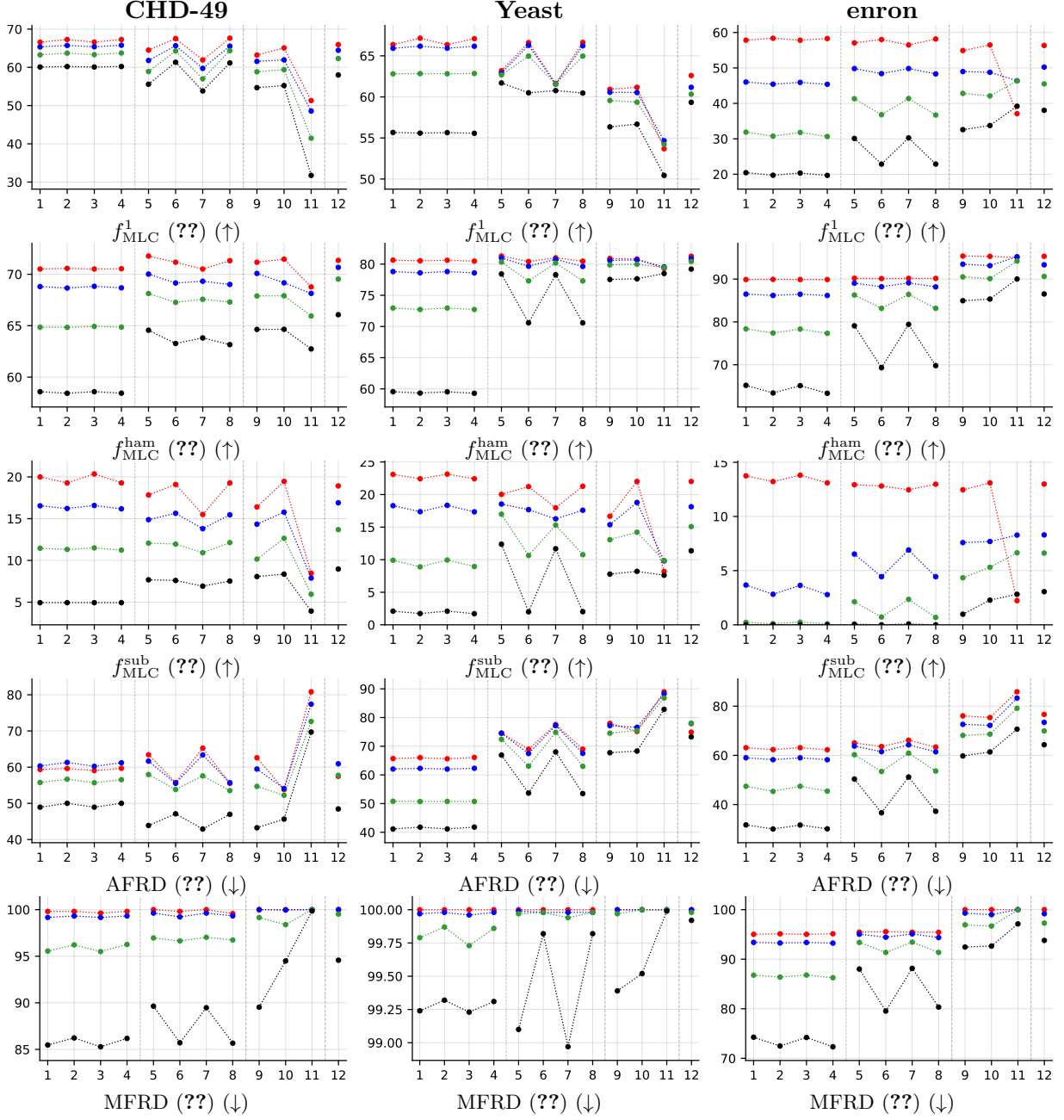


Table 6: (Table G2) Average scores (in %, y -axis) over 5×5 cross-validation folds, plotted against the encoded number of the classifiers (x -axis): (PA-H-2, PA-H, PA-S-2, PA-S), (PR-H-2, PR-H, PR-S-2, PR-S), and (BR, CC, CLR) are encoded as (1, 2, 3, 4), (5, 6, 7, 8), and (9, 10, 11), respectively. Results are color-coded with respect to the noisy level α as follows: $\alpha = 0.0$, $\alpha = 0.1$, $\alpha = 0.2$ and $\alpha = 0.3$.

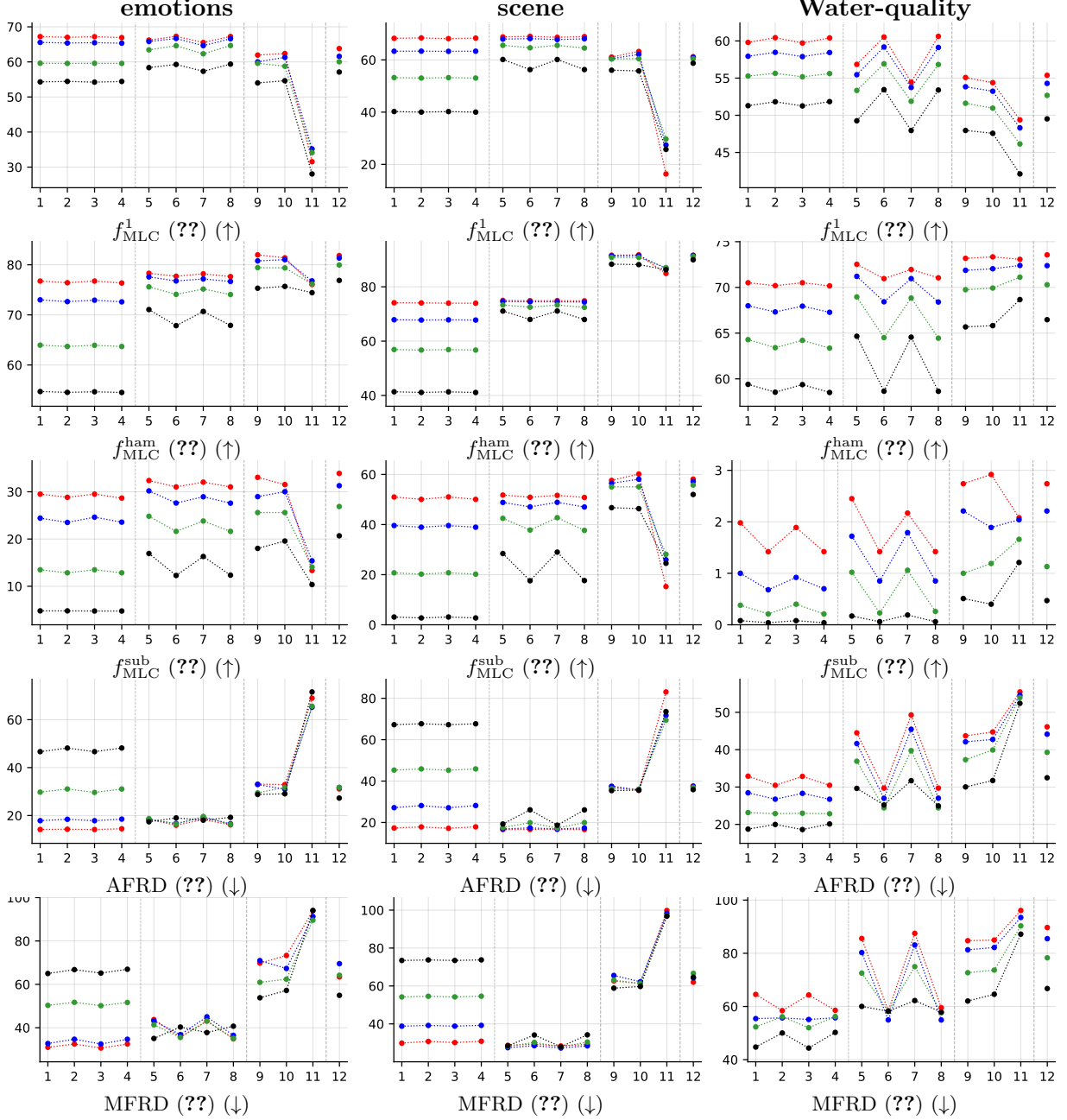


Table 7: (Table G3) Average scores (in %, y -axis) over 5×5 cross-validation folds, plotted against the encoded number of the classifiers (x -axis): (PA-H-2, PA-H, PA-S-2, PA-S), (PR-H-2, PR-H, PR-S-2, PR-S), and (BR, CC, CLR) are encoded as (1, 2, 3, 4), (5, 6, 7, 8), and (9, 10, 11), respectively. Results are color-coded with respect to the noisy level α as follows: $\alpha = 0.0$, $\alpha = 0.1$, $\alpha = 0.2$ and $\alpha = 0.3$.

3.4.2 Partial abstention (PartialAbstention)

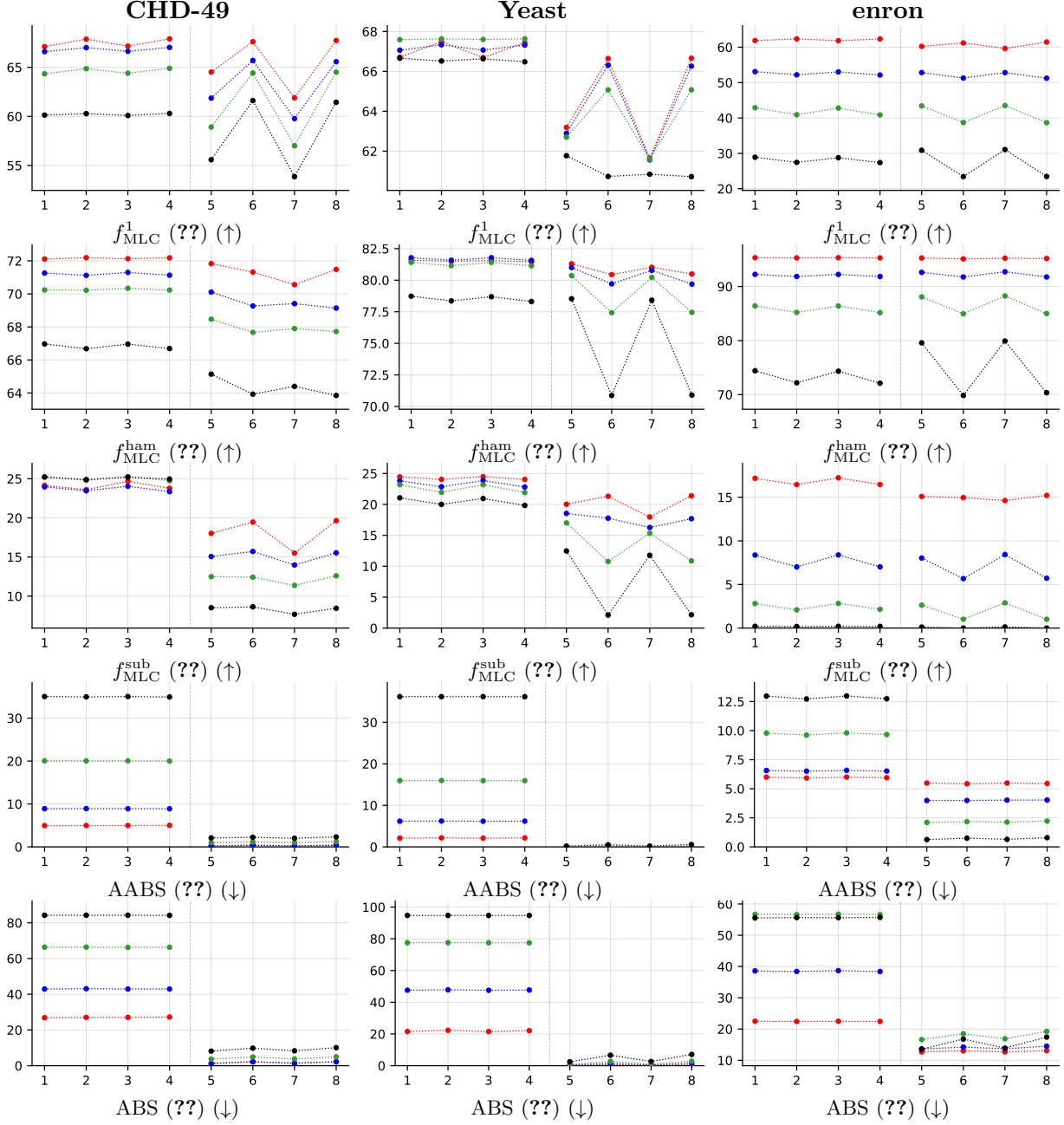


Table 8: (Table G4) Average scores (in %, y -axis) over 5×5 cross-validation folds, plotted against the encoded number of the classifiers (x -axis): (PA-H-2, PA-H, PA-S-2, PA-S) and (PR-H-2, PR-H, PR-S-2, PR-S) are encoded as (1, 2, 3, 4) and (5, 6, 7, 8), respectively. Results are color-coded with respect to the noisy level α as follows: $\alpha = 0.0$, $\alpha = 0.1$, $\alpha = 0.2$ and $\alpha = 0.3$.

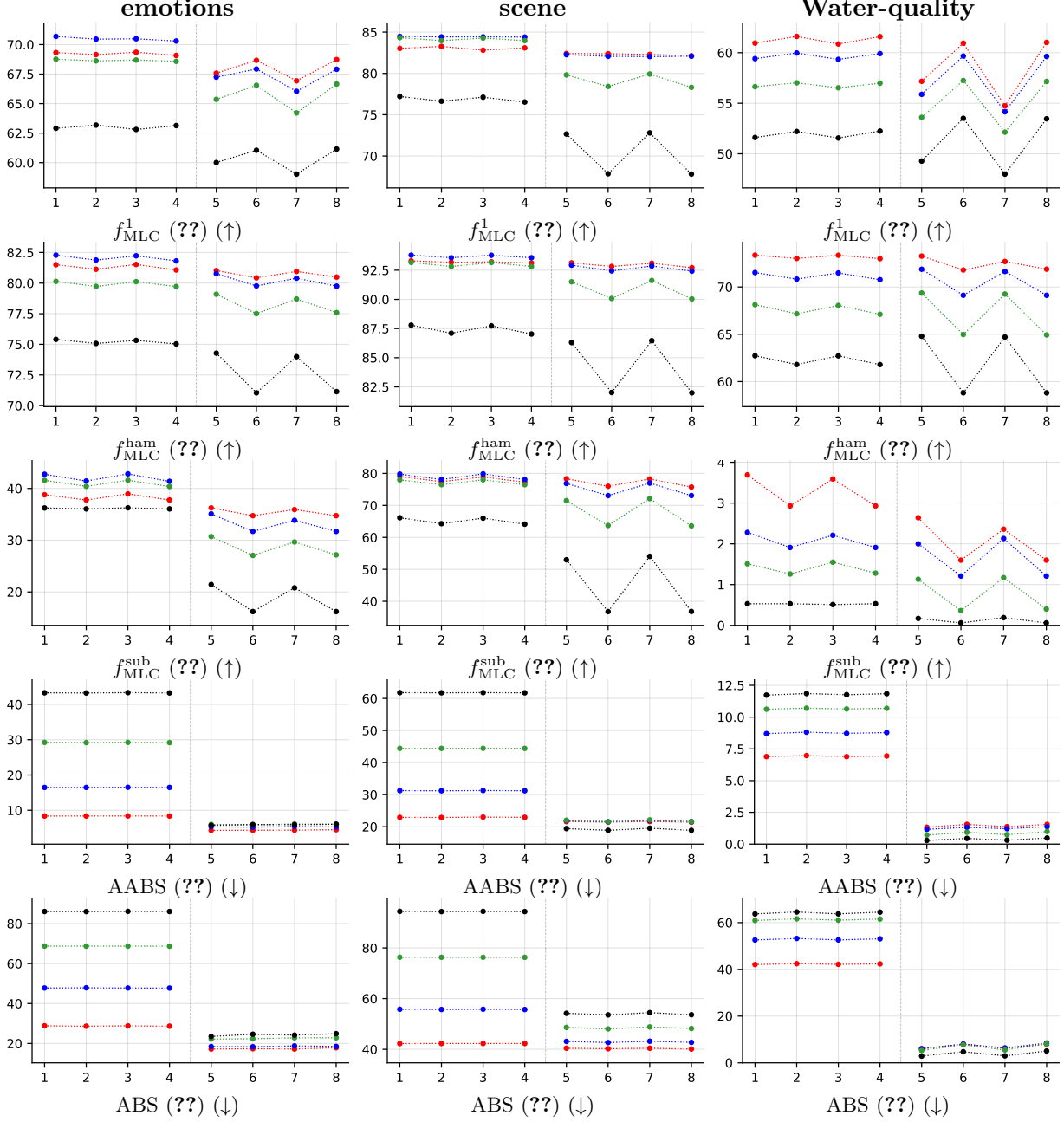


Table 9: (Table G5) Average scores (in %, y -axis) over 5×5 cross-validation folds, plotted against the encoded number of the classifiers (x -axis): (PA-H-2, PA-H, PA-S-2, PA-S) and (PR-H-2, PR-H, PR-S-2, PR-S) are encoded as (1, 2, 3, 4) and (5, 6, 7, 8), respectively. Results are color-coded with respect to the noisy level α as follows: $\alpha = 0.0$, $\alpha = 0.1$, $\alpha = 0.2$ and $\alpha = 0.3$.

3.4.3 Standard MLC predictions (BinaryVector)

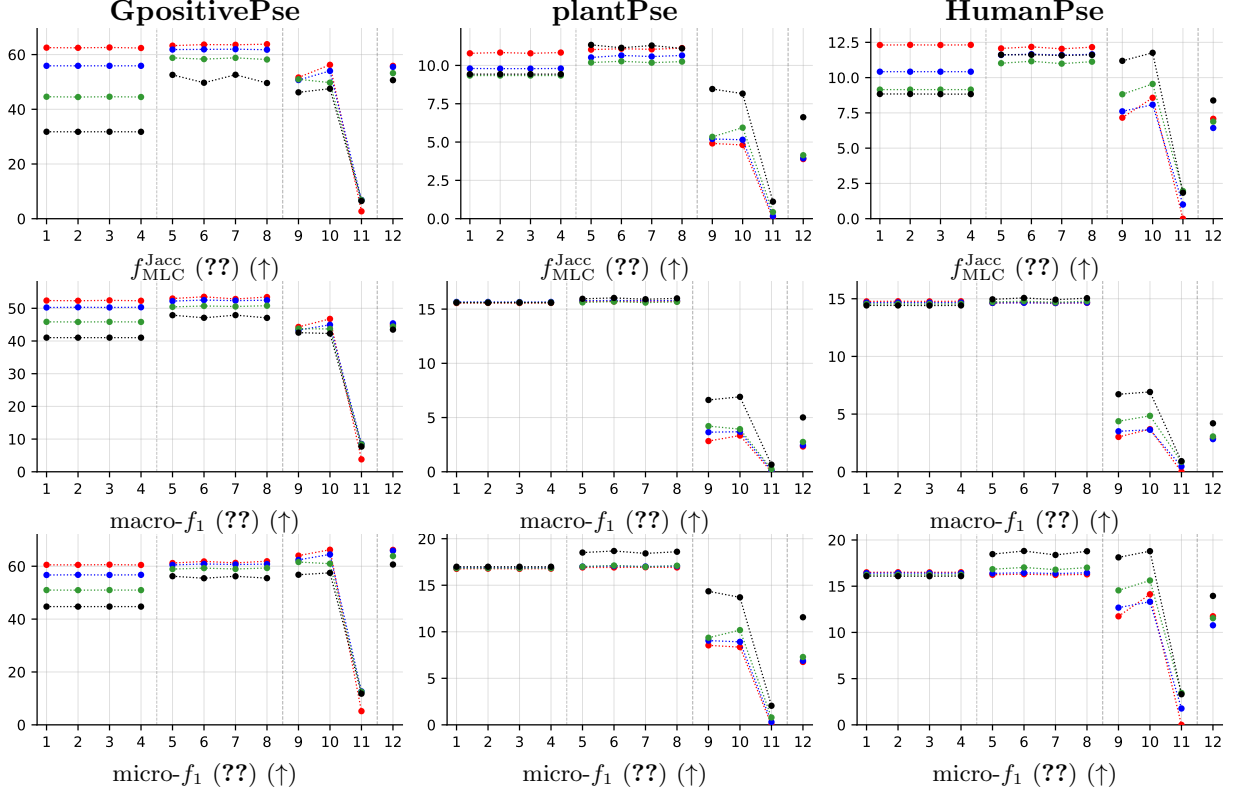


Table 10: (Table G6, new metrics — main datasets) Average scores (in %, y -axis) over 5×5 cross-validation folds, plotted against the encoded number of the classifiers (x -axis): (PA-H-2, PA-H, PA-S-2, PA-S), (PR-H-2, PR-H, PR-S-2, PR-S), and (BR, CC, CLR) are encoded as (1, 2, 3, 4), (5, 6, 7, 8), and (9, 10, 11), respectively. Results are color-coded with respect to the noisy level α as follows: $\alpha = 0.0$, $\alpha = 0.1$, $\alpha = 0.2$ and $\alpha = 0.3$.

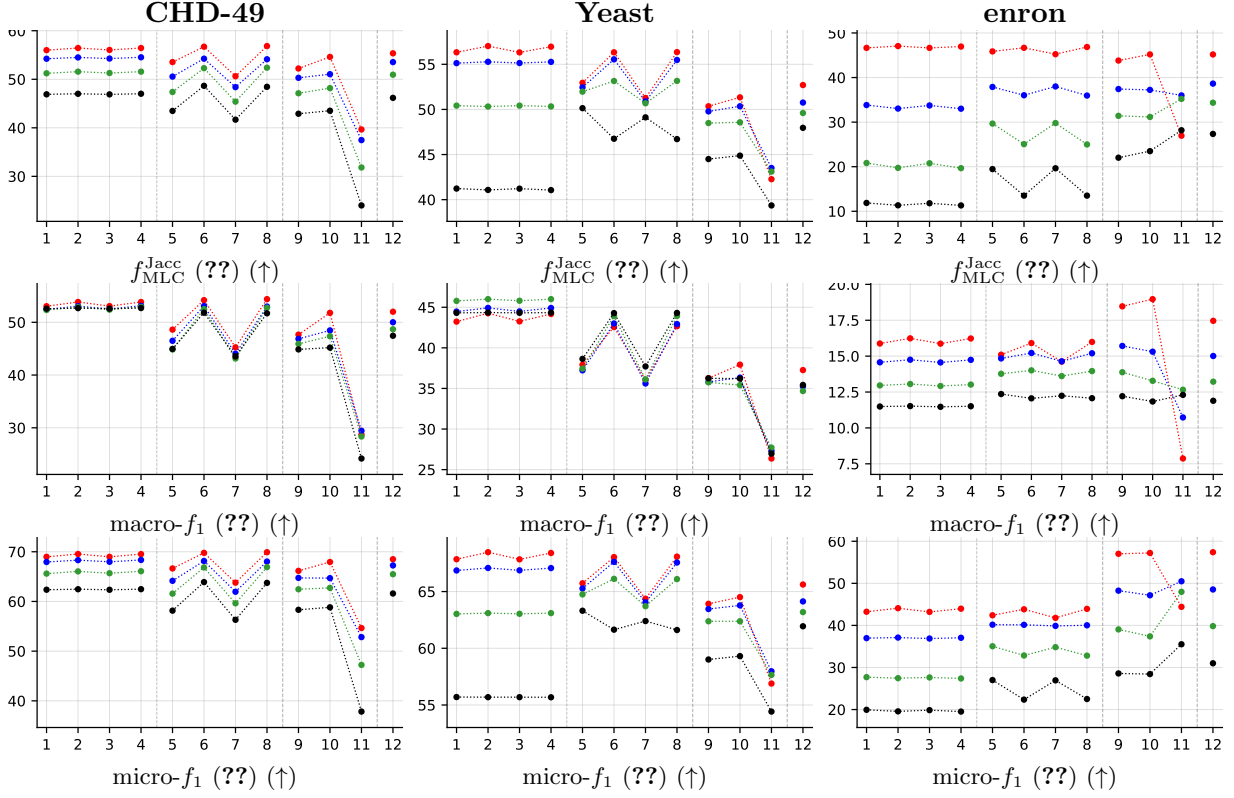


Table 11: (Table G6, new metrics — imbalanced) Average scores (in %, y -axis) over 5×5 cross-validation folds, plotted against the encoded number of the classifiers (x -axis): (PA-H-2, PA-H, PA-S-2, PA-S), (PR-H-2, PR-H, PR-S-2, PR-S), and (BR, CC, CLR) are encoded as (1, 2, 3, 4), (5, 6, 7, 8), and (9, 10, 11), respectively. Results are color-coded with respect to the noisy level α as follows: $\alpha = 0.0$, $\alpha = 0.1$, $\alpha = 0.2$ and $\alpha = 0.3$.

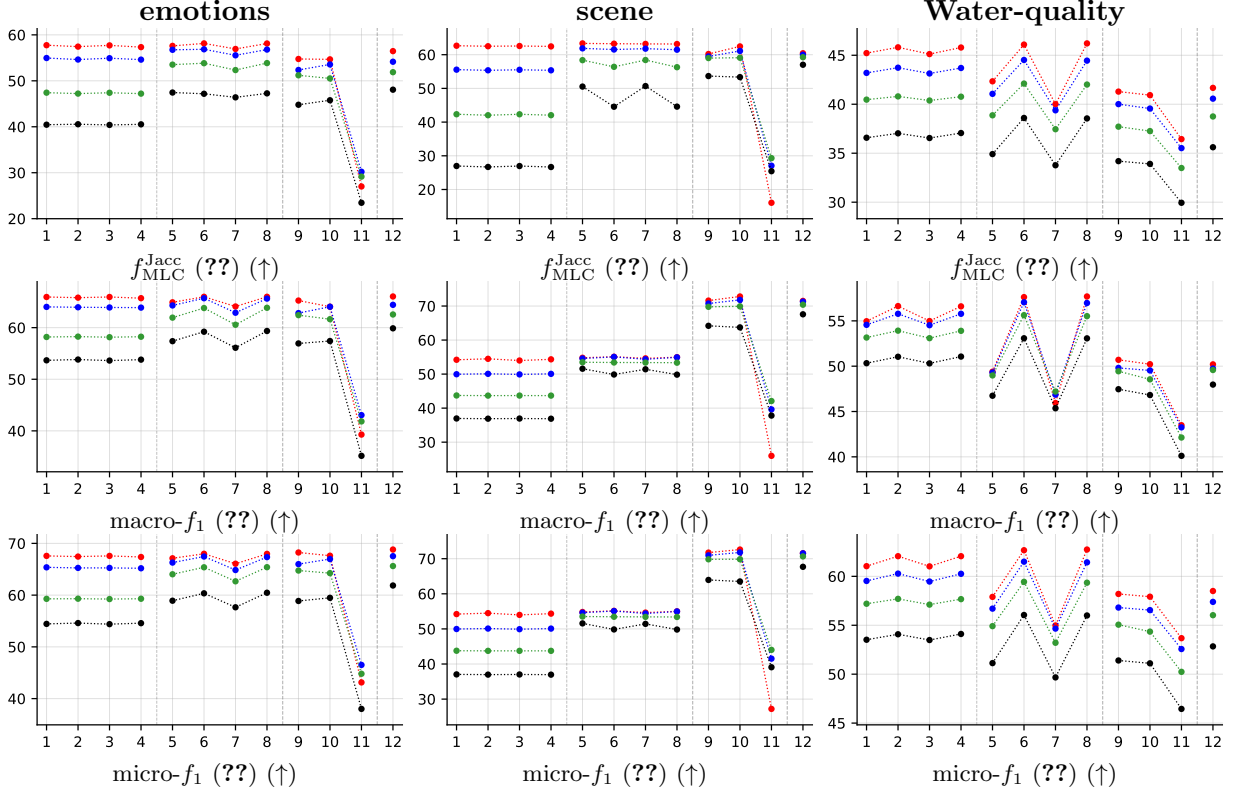


Table 12: (Table G6, new metrics — balanced) Average scores (in %, y -axis) over 5×5 cross-validation folds, plotted against the encoded number of the classifiers (x -axis): (PA-H-2, PA-H, PA-S-2, PA-S), (PR-H-2, PR-H, PR-S-2, PR-S), and (BR, CC, CLR) are encoded as (1, 2, 3, 4), (5, 6, 7, 8), and (9, 10, 11), respectively. Results are color-coded with respect to the noisy level α as follows: $\alpha = 0.0$, $\alpha = 0.1$, $\alpha = 0.2$ and $\alpha = 0.3$.

3.4.4 Partial abstention (PartialAbstention)

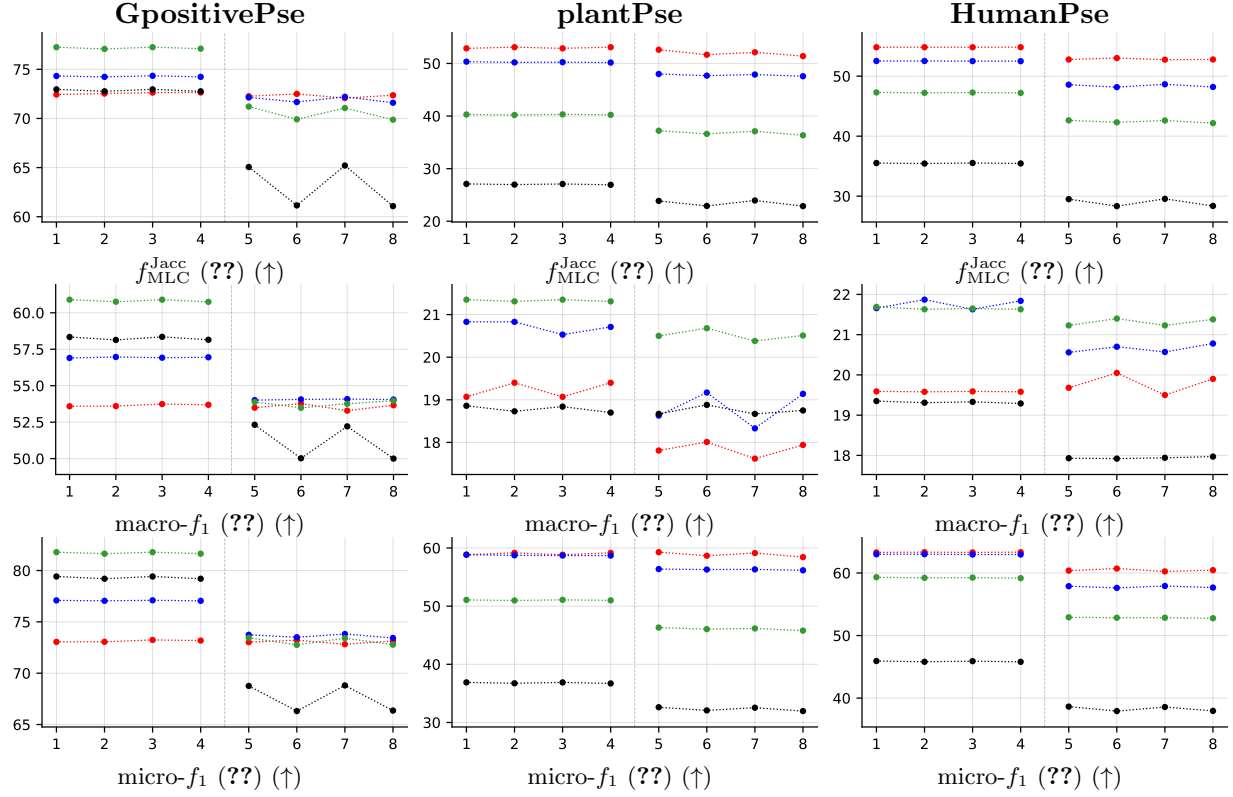


Table 13: (Table G7, new metrics — main datasets) Average scores (in %, y -axis) over 5×5 cross-validation folds, plotted against the encoded number of the classifiers (x -axis): (PA-H-2, PA-H, PA-S-2, PA-S) and (PR-H-2, PR-H, PR-S-2, PR-S) are encoded as (1, 2, 3, 4) and (5, 6, 7, 8), respectively. Results are color-coded with respect to the noisy level α as follows: $\alpha = 0.0$, $\alpha = 0.1$, $\alpha = 0.2$ and $\alpha = 0.3$.

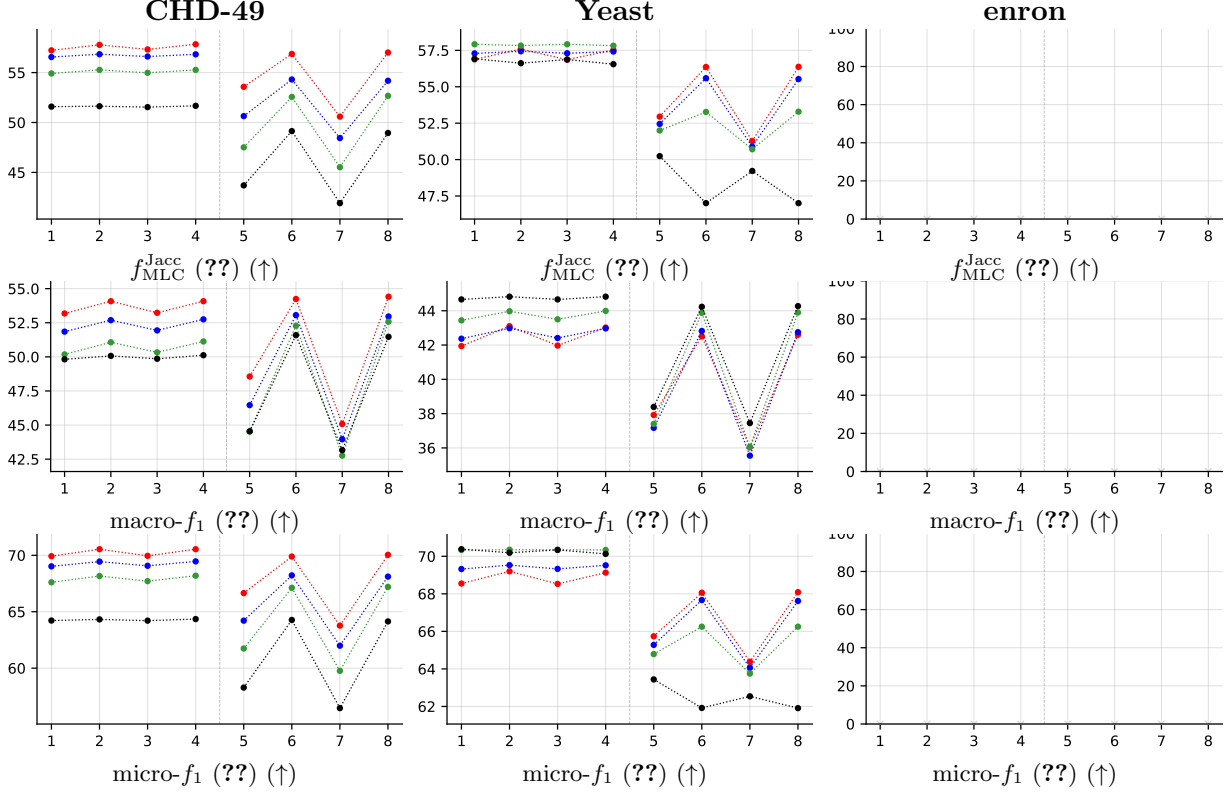


Table 14: (Table G7, new metrics — imbalanced) Average scores (in %, y -axis) over 5×5 cross-validation folds, plotted against the encoded number of the classifiers (x -axis): (PA-H-2, PA-H, PA-S-2, PA-S) and (PR-H-2, PR-H, PR-S-2, PR-S) are encoded as (1, 2, 3, 4) and (5, 6, 7, 8), respectively. Results are color-coded with respect to the noisy level α as follows: $\alpha = 0.0$, $\alpha = 0.1$, $\alpha = 0.2$ and $\alpha = 0.3$.

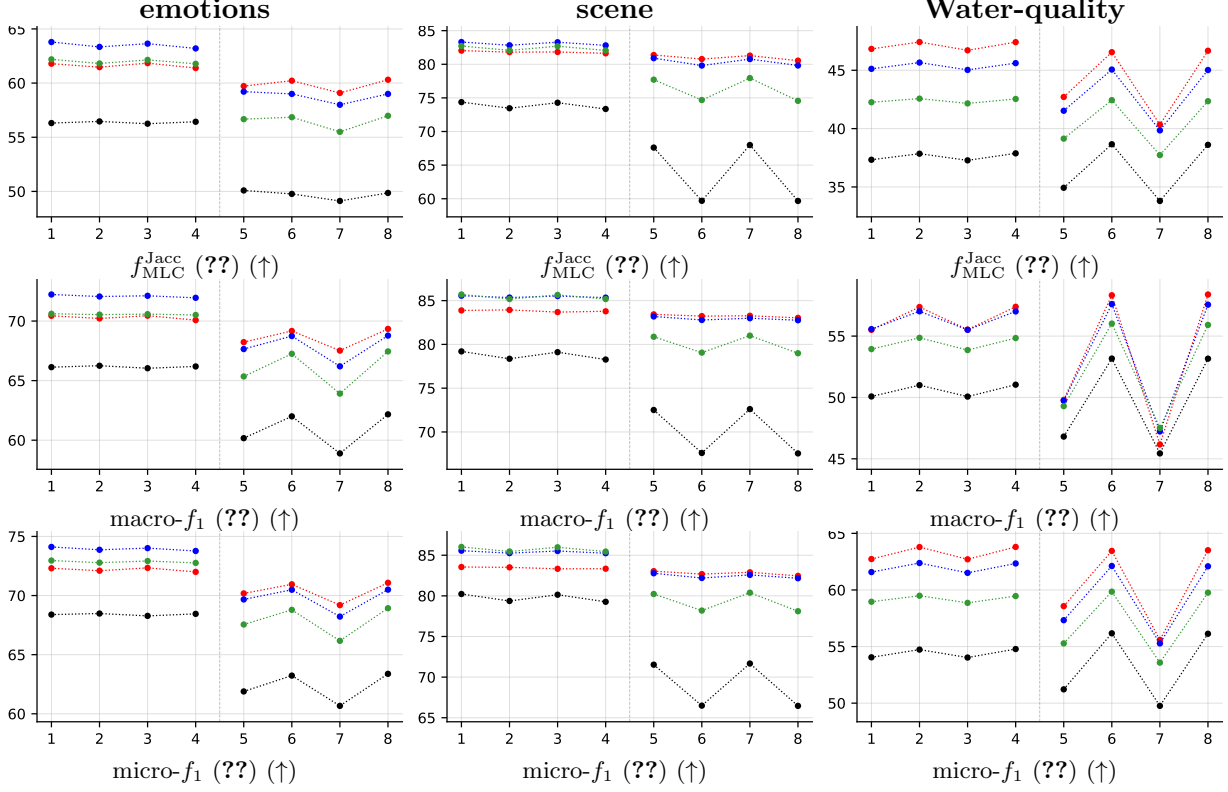


Table 15: (Table G7, new metrics — balanced) Average scores (in %, y -axis) over 5×5 cross-validation folds, plotted against the encoded number of the classifiers (x -axis): (PA-H-2, PA-H, PA-S-2, PA-S) and (PR-H-2, PR-H, PR-S-2, PR-S) are encoded as (1, 2, 3, 4) and (5, 6, 7, 8), respectively. Results are color-coded with respect to the noisy level α as follows: $\alpha = 0.0$, $\alpha = 0.1$, $\alpha = 0.2$ and $\alpha = 0.3$.

3.4.5 Score-vector ranking metrics (AUROC, AUPRC, etc.) (ScoreVector)

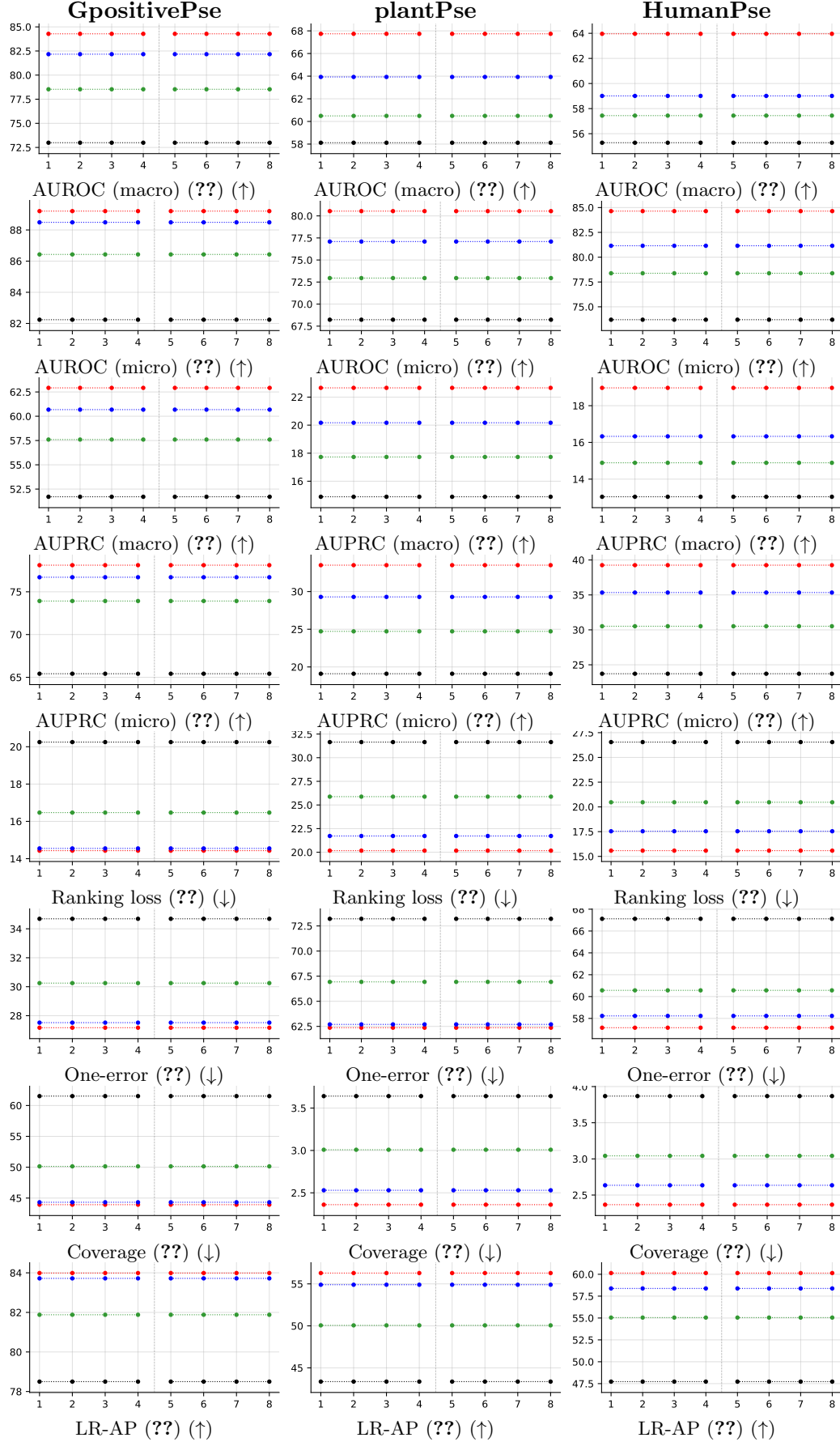


Table 16: (Table G8, new — main datasets) Average scores (in %, y -axis) over 5 × 5 cross-validation folds, plotted against the encoded number of the classifiers (x -axis): (PA-H-2, PA-H, PA-S-2, PA-S) and (PR-H-2, PR-H, PR-S-2, PR-S) are encoded as (1, 2, 3, 4) and (5, 6, 7, 8), respectively. Results

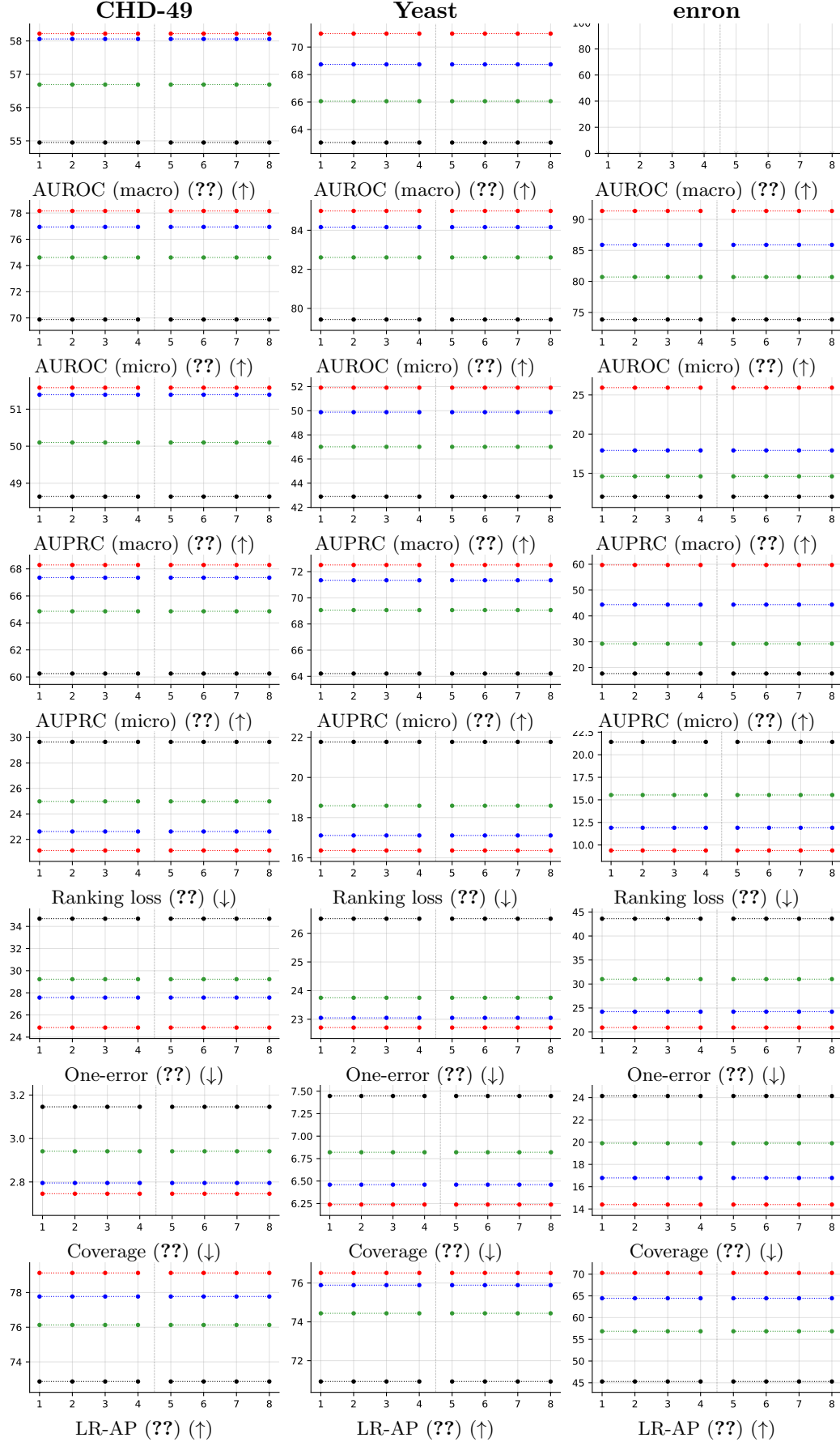


Table 17: (Table G8, new — imbalanced) Average scores (in %, y-axis) over 5 × 5 cross-validation folds, plotted against the encoded number of the classifiers (x-axis): (PA-H-2, PA-H, PA-S-2, PA-S) and (PR-H-2, PR-H, PR-S-2, PR-S) are encoded as (1, 2, 3, 4) and (5, 6, 7, 8), respectively. Results

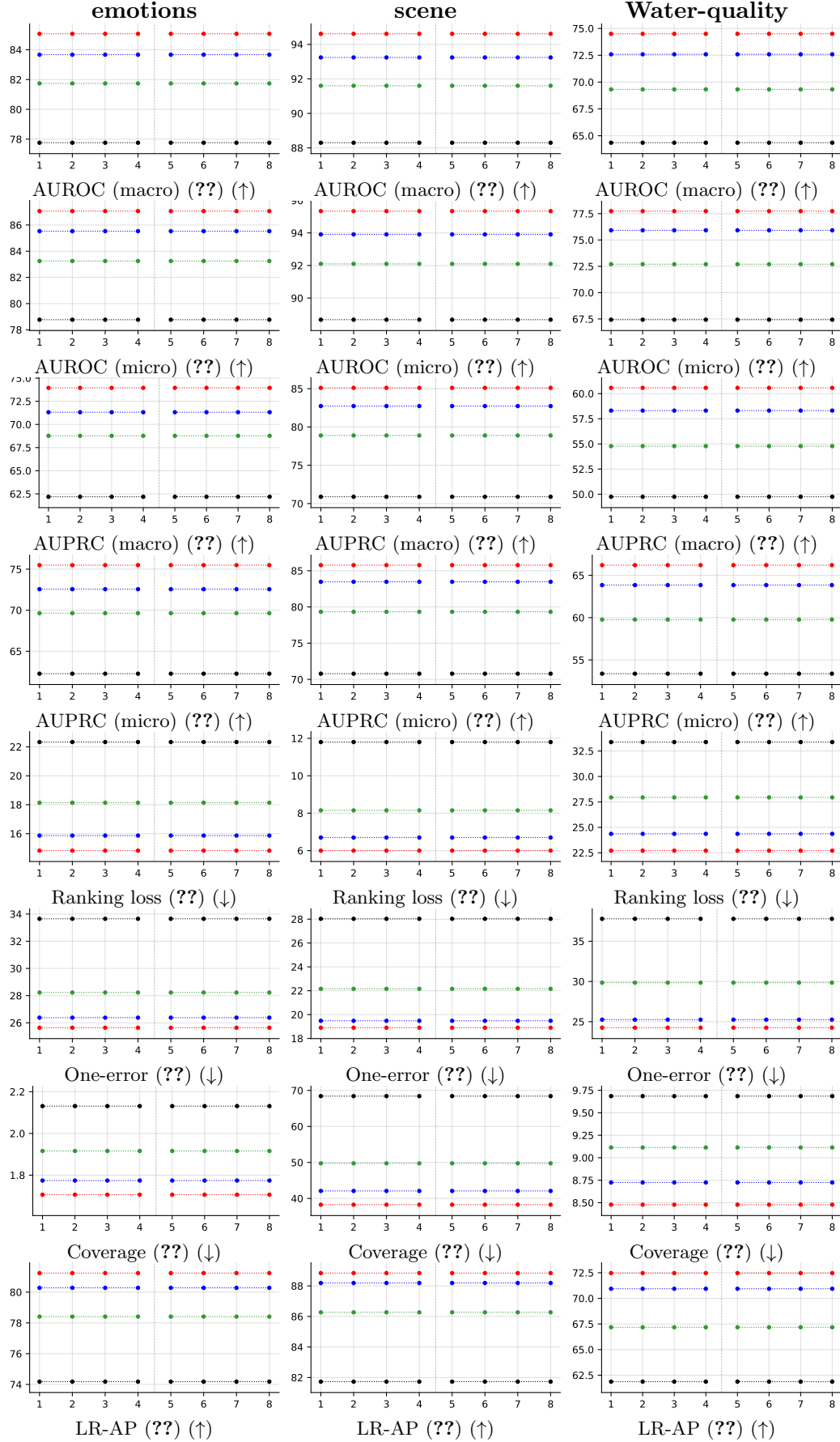


Table 18: (Table G8, new — balanced) Average scores (in %, y -axis) over 5×5 cross-validation folds, plotted against the encoded number of the classifiers (x -axis): (PA-H-2, PA-H, PA-S-2, PA-S) and (PR-H-2, PR-H, PR-S-2, PR-S) are encoded as (1, 2, 3, 4) and (5, 6, 7, 8), respectively. Results

3.5 Abstention overview: aggregate trends

Purpose. This compact section shows the headline abstention-vs-standard-MLC trend across noise levels, averaged over all datasets, for both base learners (RF, LGBM) and both retained quality metrics (f_1 , Jaccard). Use it as a quick visual summary before diving into the detailed per-method / per-dataset figures in Section 5.

How to read. In each panel: blue bar = standard MLC (no abstention) on its native quality metric; coloured bar = with-abstention quality on retained labels. Green “+ $x.y$ ” label = score-point gain of abstention over the standard-MLC bar of the same method. The four sub-panels per figure correspond to $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$.

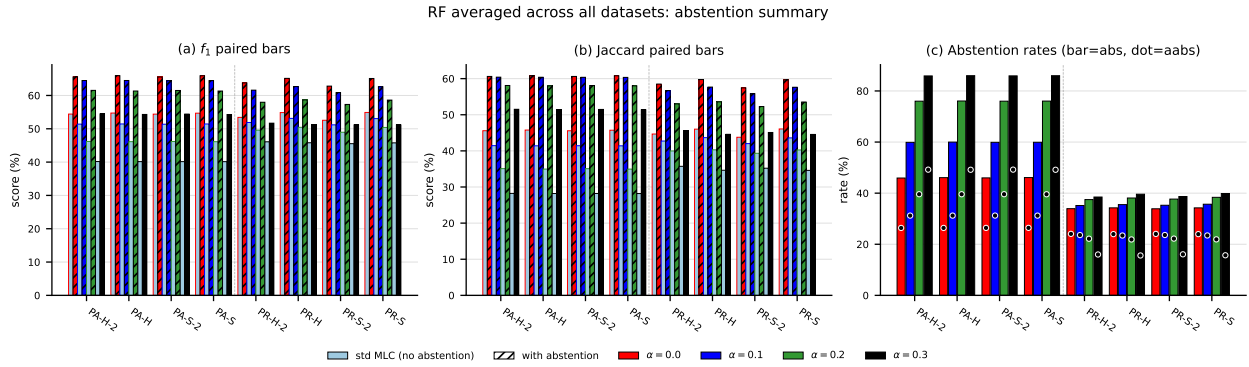


Figure 1: Figure 1.1: RF averaged across all datasets — paired bars (light blue = std MLC, hatched = with-abstention) for f_1 and Jaccard, plus abstention rates. Color = $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$.

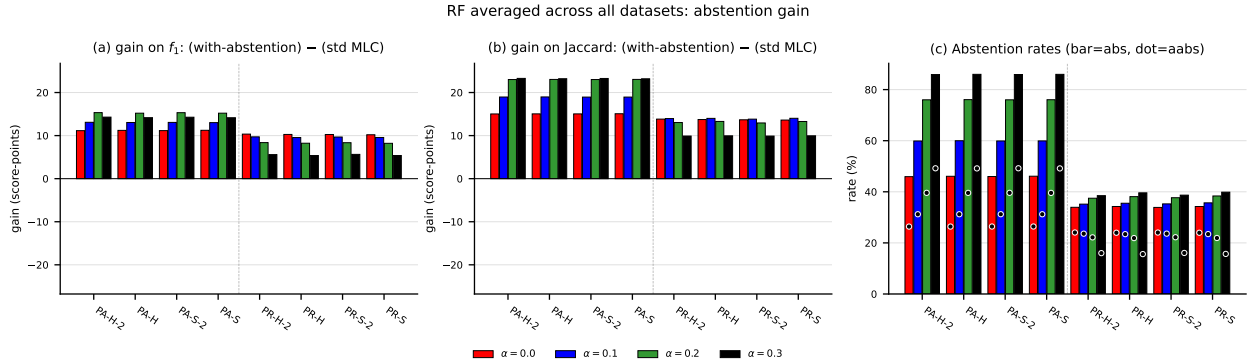


Figure 2: Figure 1.2: RF averaged across all datasets — gain-only view: bar height = (with-abstention) – (std MLC) in score points, so positive bars show the empirical benefit of abstaining. Right panel: abstention rates. Color = $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$.

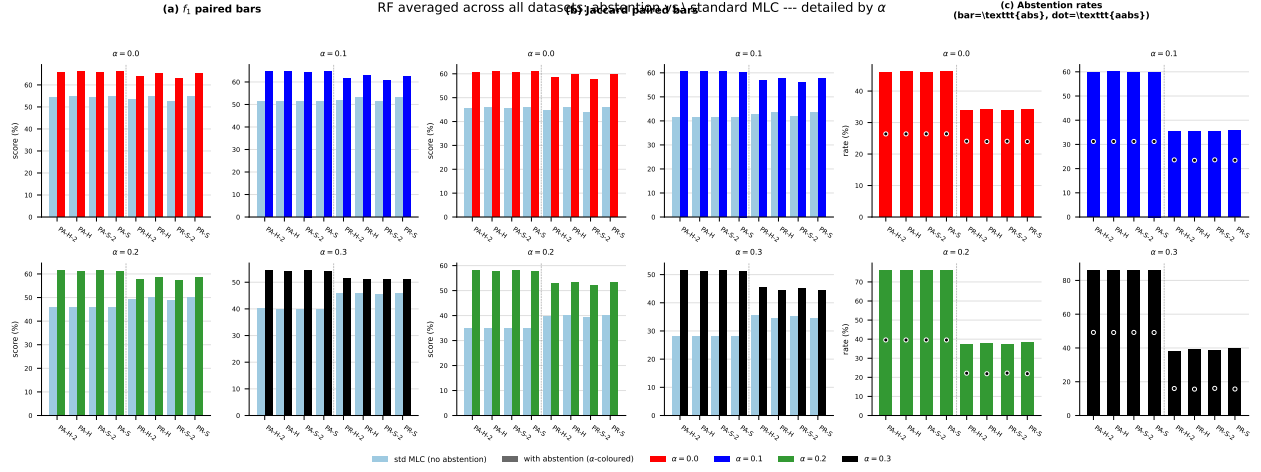


Figure 3: Figure 1.3: RF averaged across all datasets — detailed 2-row layout (low noise top, high noise bottom): (a) f_1 paired bars, (b) Jaccard paired bars, (c) abstention rates — bar = abs = % of instances with at least one abstained label (instance-level), dot = aabs = % of labels skipped (per-cell, label-level). The rates in panel (c) are independent of which quality metric is reported in (a)/(b) — they characterise the abstention decision rule, not its downstream effect on f_1 or Jaccard. Color = $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$.

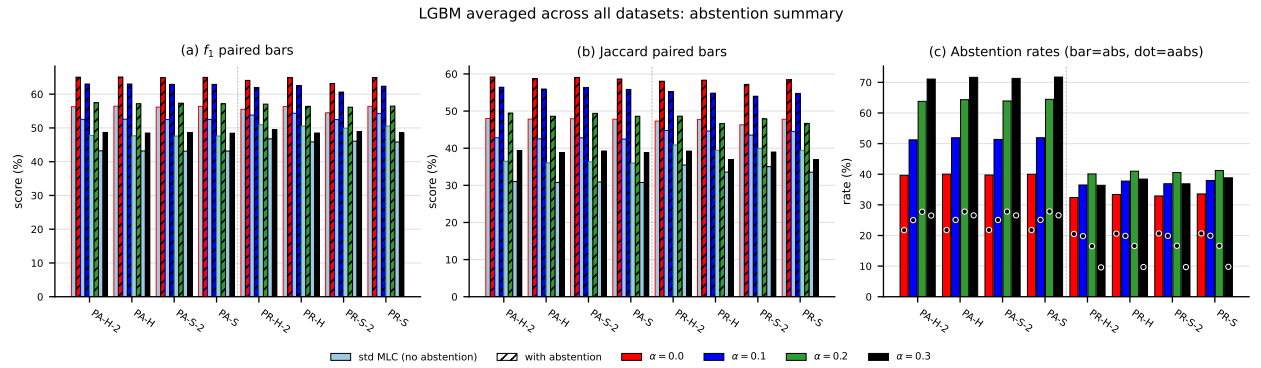


Figure 4: Figure 1.1: LGBM averaged across all datasets — paired bars (light blue = std MLC, hatched = with-abstention) for f_1 and Jaccard, plus abstention rates. Color = $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$.

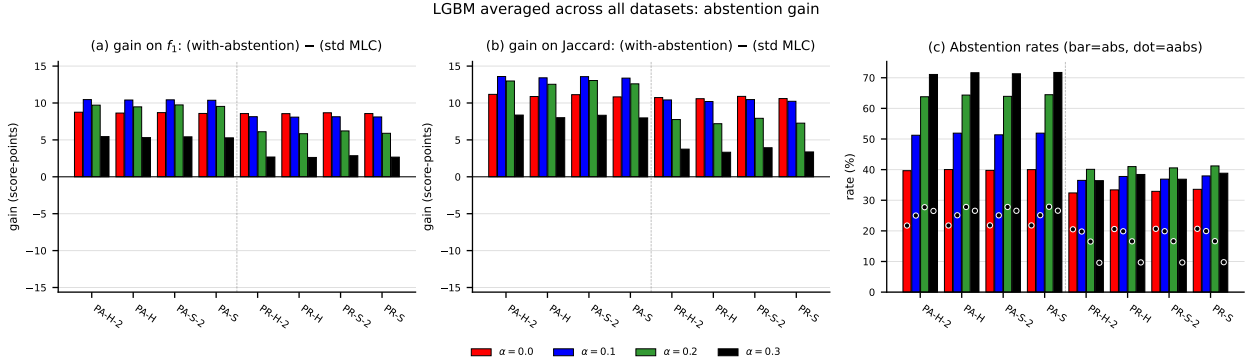


Figure 5: Figure 1.2: LGBM averaged across all datasets — gain-only view: bar height = (with-abstention) - (std MLC) in score points, so positive bars show the empirical benefit of abstaining. Right panel: abstention rates. Color = $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$.

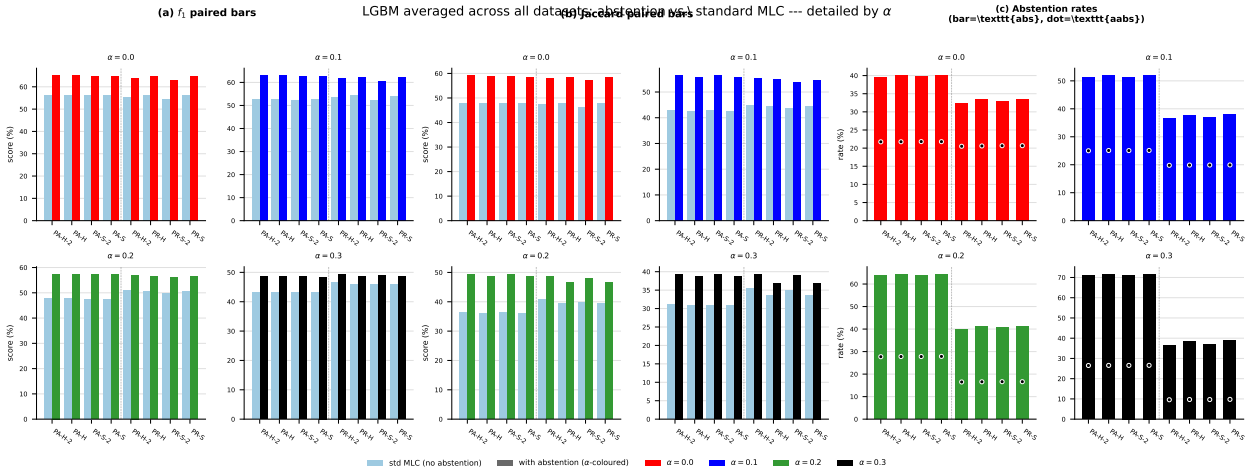


Figure 6: Figure 1.3: LGBM averaged across all datasets — detailed 2-row layout (low noise top, high noise bottom): (a) f_1 paired bars, (b) Jaccard paired bars, (c) abstention rates — bar = abs = % of instances with at least one abstained label (instance-level), dot = aabs = % of labels skipped (per-cell, label-level). The rates in panel (c) are independent of which quality metric is reported in (a)/(b) — they characterise the abstention decision rule, not its downstream effect on f_1 or Jaccard. Color = $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$.

4 Main results: RF base learner, aggregated

RF • Aggregate • BinaryVector — Standard MLC predictions

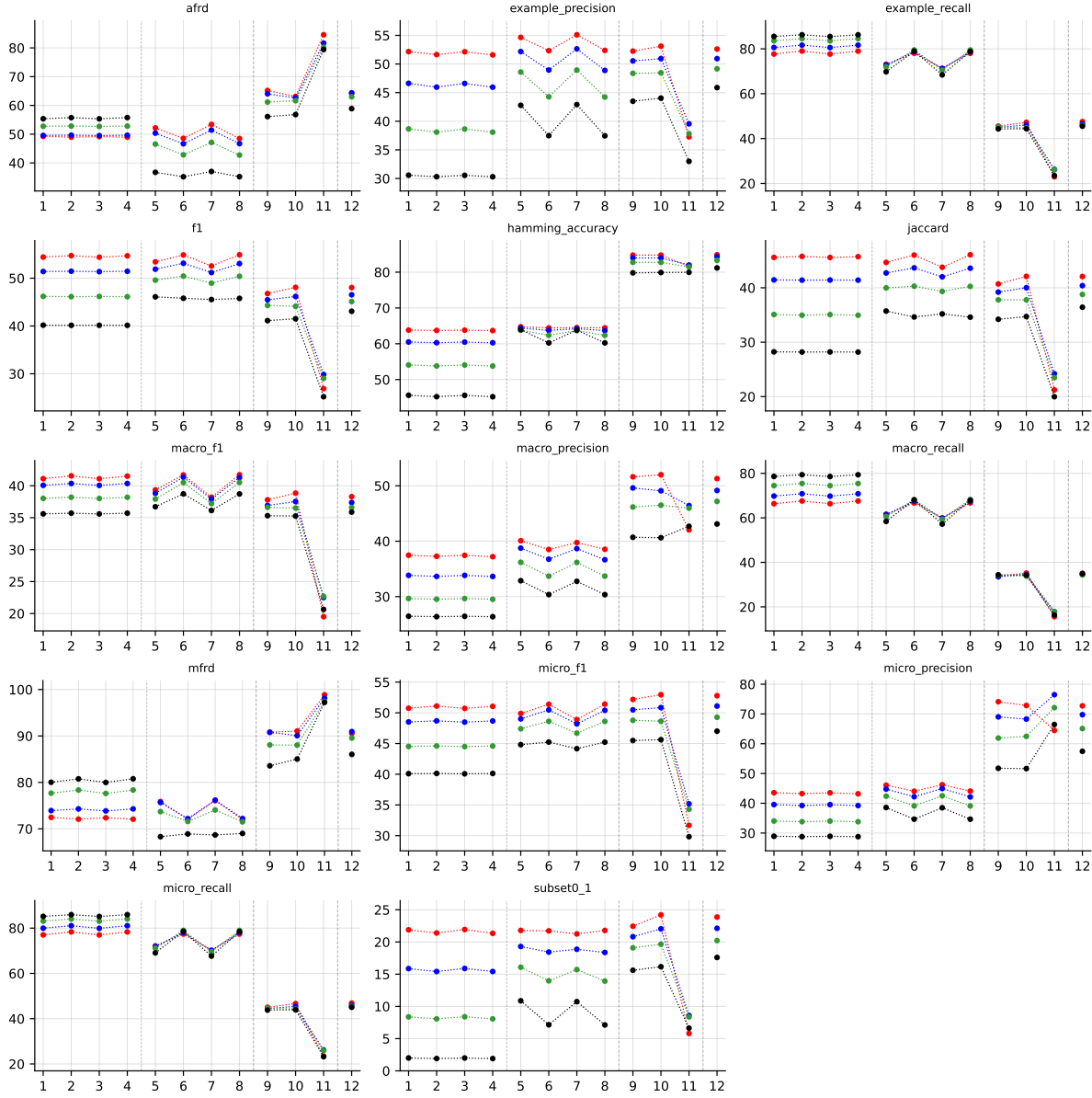


Table 19: Standard MLC predictions (BinaryVector), RF base learner, averaged across datasets. Method indices and line colors follow the encoding paragraph above.

RF • Aggregate • PartialAbstention — Partial abstention

RF • Aggregate • ScoreVector — Score-vector ranking metrics (AUROC, AUPRC, etc.)

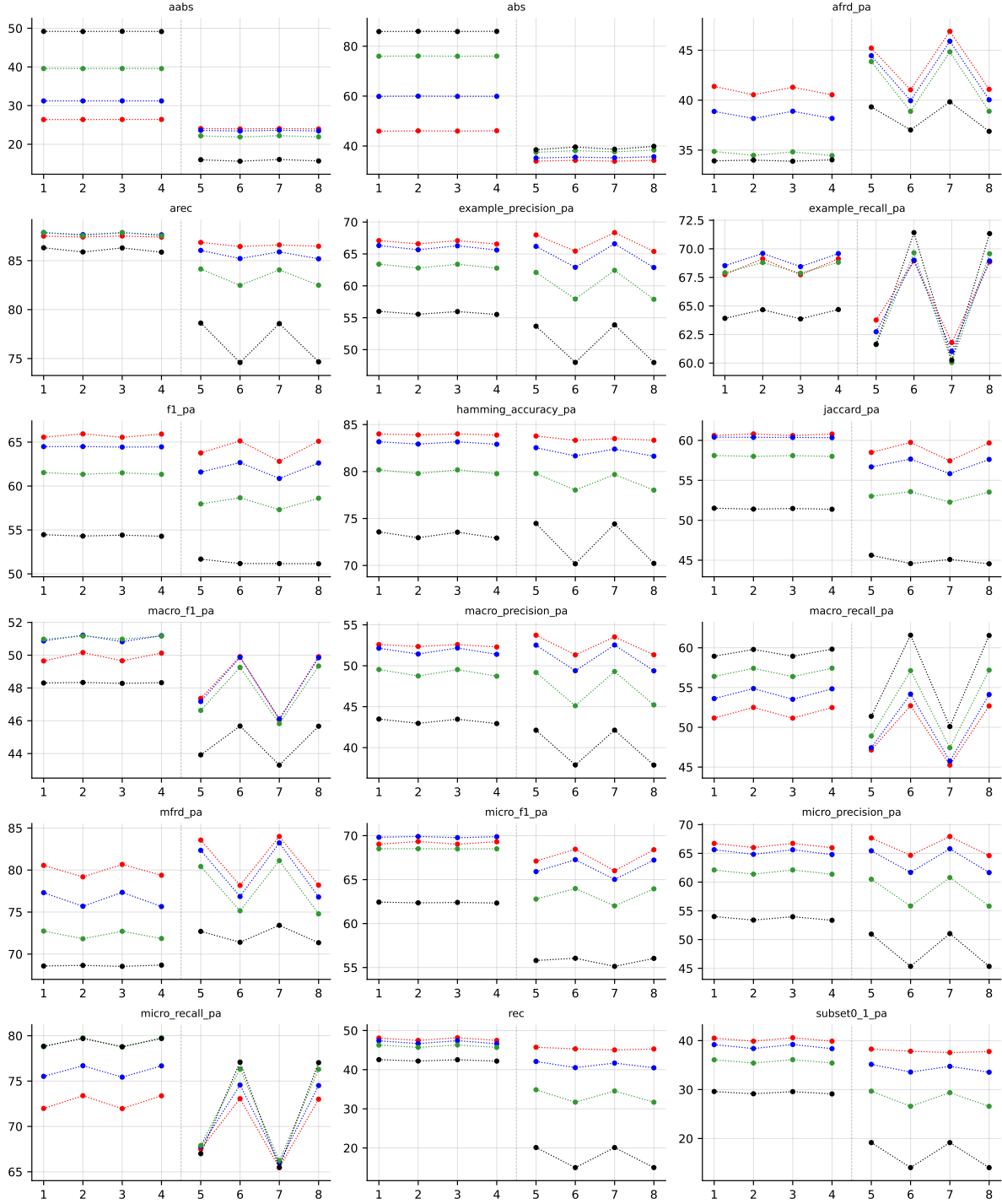


Table 20: Partial abstention (PartialAbstention), RF base learner, averaged across datasets. Method indices and line colors follow the encoding paragraph above.

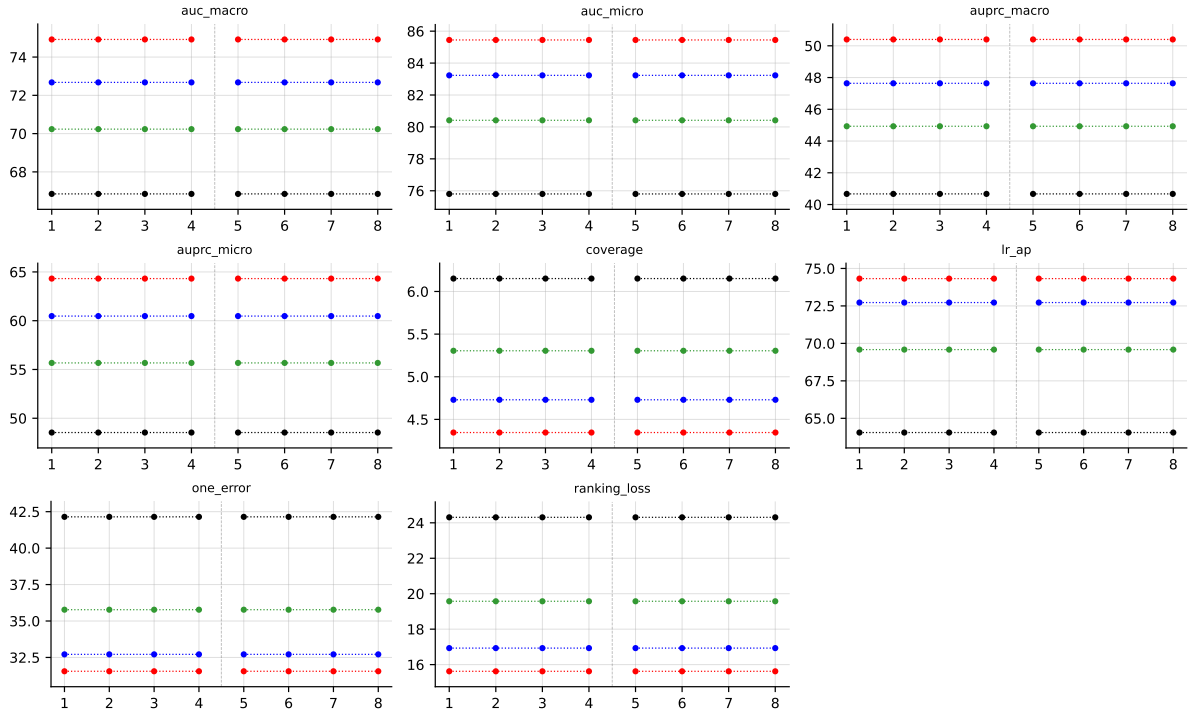


Table 21: Score-vector ranking metrics (AUROC, AUPRC, etc.) (ScoreVector), RF base learner, averaged across datasets. Method indices and line colors follow the encoding paragraph above.

5 Abstention vs standard MLC

Why this section exists. The paper’s central claim is that an order-based predictor can *abstain* on labels it is unsure about and, on the labels it does predict, achieve higher quality than a standard MLC method that is forced to predict every label. The two chart families below make that claim visible.

Terminology. ‘*Standard MLC*’ (a.k.a. **BinaryVector** / BV) means each method outputs a 0/1 prediction for every one of the K labels. ‘*With abstention*’ (a.k.a. **PartialAbstention** / PA) means an order-based predictor may return $\perp$ (abstain) on some labels; the PA quality metrics (**f1_pa**, **jaccard_pa**, ...) are computed only over the non-abstained labels. Only the 8 PA/PR methods (indices 1–8) can abstain; BR/CC/CLR/ECC cannot.

Reading the coverage-risk chart. Each point is one (PA/PR method, noise level α) pair. The x-axis is the abstention rate (fraction of labels the method skipped); the y-axis is the quality metric on the labels that *were* predicted (higher is better). Marker shape encodes method, colour encodes noise level. The dashed grey line is the best standard-MLC baseline (BR / CC / CLR / ECC) on the BV quality metric — it is the bar that “no abstention” has to clear. **Any point above the dashed line is direct evidence that abstaining improved quality beyond what any standard MLC method achieves.** Points further right traded coverage for that gain.

Reading the paired-bars chart. Same data, different view. The four sub-panels correspond to the four noise levels. Inside each sub-panel: 12 method positions, two bars per position — blue is the standard-MLC quality (**f1**, no abstention), warm-colour is the with-abstention quality on retained labels. Only the 8 PA/PR methods (1–8) have a warm bar; the standard MLC baselines (9–12) only have a blue bar by construction. The green “+ $x.y$ ” label above each warm bar is the gain in score-points over that method’s own BV bar.

5.1 RF, metric f1_pa — aggregate



Figure 7: RF (**f1_pa**) averaged across all datasets: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = **abs** (% of instances with at least one abstained label, instance-level); dot = **aabs** (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

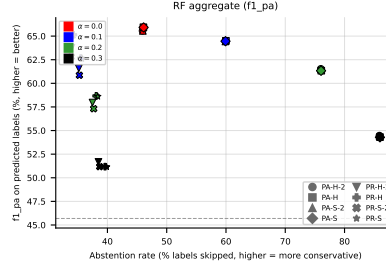


Figure 8: RF ($f1_{pa}$) averaged across all datasets: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.



Figure 9: RF ($f1_{pa}$) averaged across all datasets: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($f1_{pa} - f1$) heatmap (right).



Figure 10: RF ($f1_{pa}$) averaged across all datasets — robustness view. Left: efficiency-quality scatter (x = coverage = $1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective $f1_{pa} = (1 - \text{abs}) \cdot f1_{pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

5.2 RF, metric f_{1_pa} — per dataset



Figure 11: RF (f_{1_pa}) on **GpositivePseAAC**: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = **abs** (% of instances with at least one abstained label, instance-level); dot = **aabs** (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

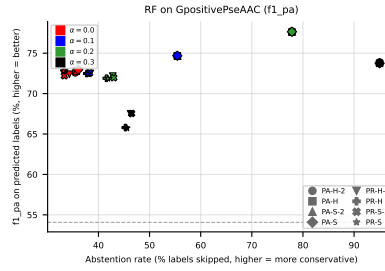


Figure 12: RF (f_{1_pa}) on **GpositivePseAAC**: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.



Figure 13: RF (f_{1_pa}) on **GpositivePseAAC**: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($f_{1_pa} - f_1$) heatmap (right).

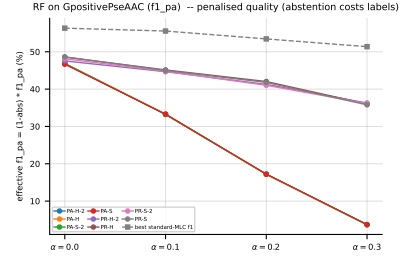
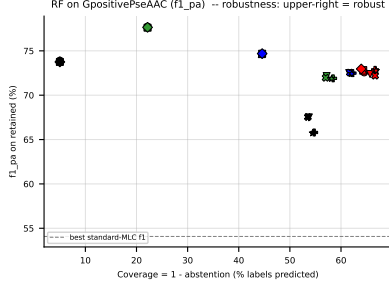


Figure 14: RF (f1_pa) on GpositivePseAAC — robustness view. Left: efficiency-quality scatter (x = coverage = $1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective f1_pa = $(1 - \text{abs}) \cdot f1_pa$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

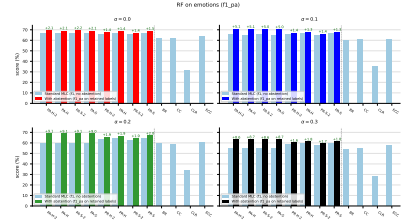
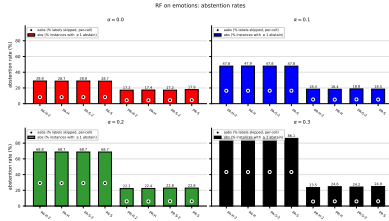


Figure 15: RF (f1_pa) on emotions: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = **abs** (% of instances with at least one abstained label, instance-level); dot = **aabs** (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

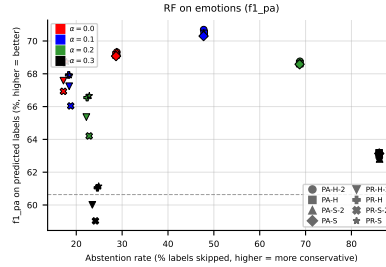


Figure 16: RF (f1_pa) on emotions: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

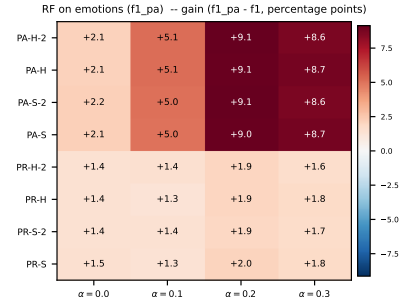
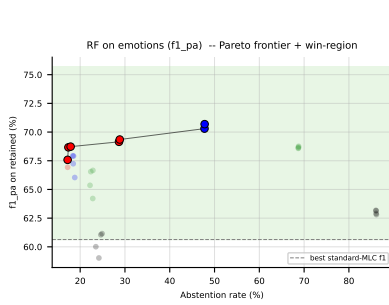


Figure 17: RF ($f1_{pa}$) on **emotions**: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($f1_{pa} - f_1$) heatmap (right).

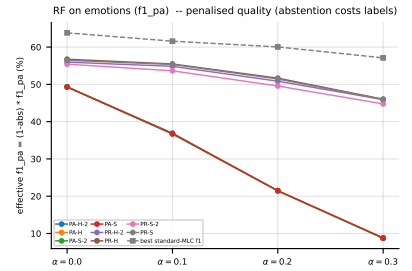
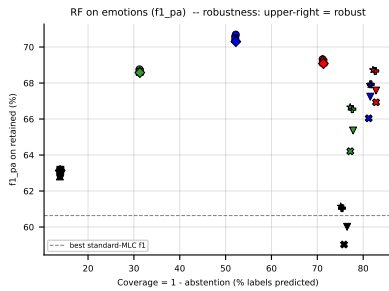


Figure 18: RF ($f1_{pa}$) on **emotions** — robustness view. Left: efficiency-quality scatter ($x = \text{coverage} = 1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective $f1_{pa} = (1 - \text{abs}) \cdot f1_{pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

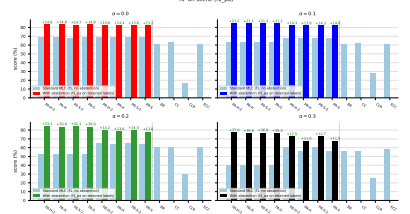
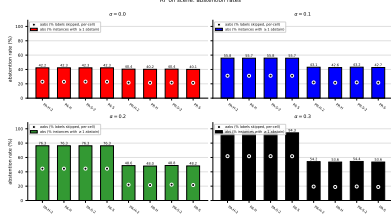


Figure 19: RF ($f1_{pa}$) on **scene**: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = **abs** (% of instances with at least one abstained label, instance-level); dot = **aabs** (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

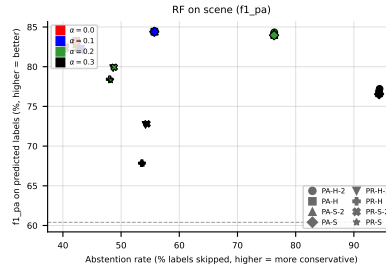


Figure 20: RF ($f1_{pa}$) on **scene**: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

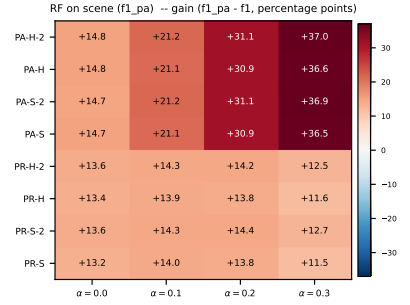
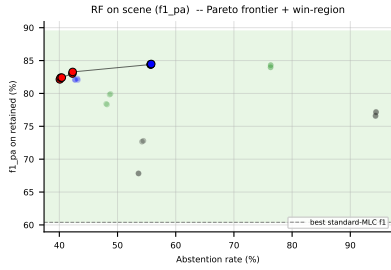


Figure 21: RF ($f1_{pa}$) on **scene**: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($f1_{pa} - f1$) heatmap (right).

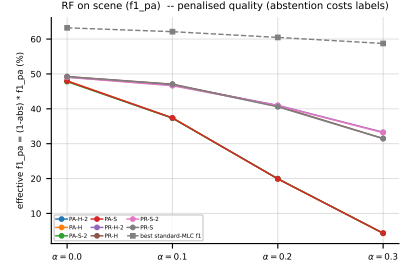
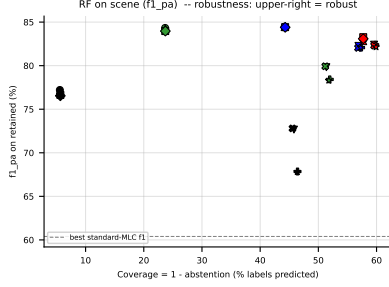


Figure 22: RF (f1_{pa}) on **scene** — robustness view. Left: efficiency-quality scatter ($x = \text{coverage} = 1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective f1_{pa} = $(1 - \text{abs}) \cdot f1_{pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

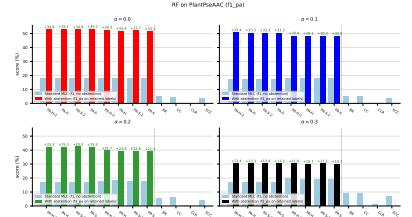
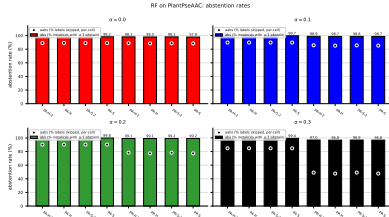


Figure 23: RF (f1_{pa}) on **PlantPseAAC**: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = **abs** (% of instances with at least one abstained label, instance-level); dot = **aabs** (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

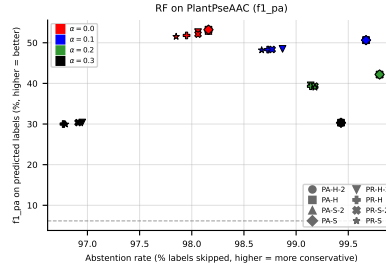


Figure 24: RF (f1_{pa}) on **PlantPseAAC**: coverage-risk scatter. Each point is one (method, noise) pair; $x = \text{abstention rate}$, $y = \text{quality on retained labels}$. Points above the dashed line beat the best standard-MLC baseline.



Figure 25: RF (f1_pa) on PlantPseAAC: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($f1_{pa} - f1$) heatmap (right).



Figure 26: RF (f1_pa) on PlantPseAAC — robustness view. Left: efficiency-quality scatter ($x = \text{coverage} = 1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective $f1_{pa} = (1 - \text{abs}) \cdot f1_{pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

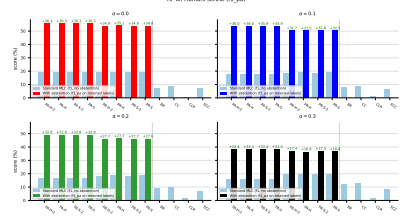
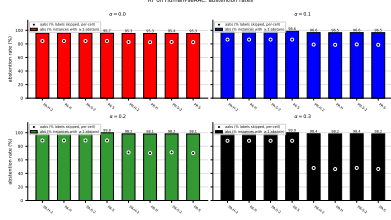


Figure 27: RF ($f1_{pa}$) on HumanPseAAC: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = **abs** (% of instances with at least one abstained label, instance-level); dot = **aabs** (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

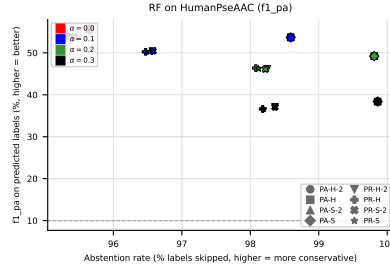


Figure 28: RF ($f1_{pa}$) on HumanPseAAC: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

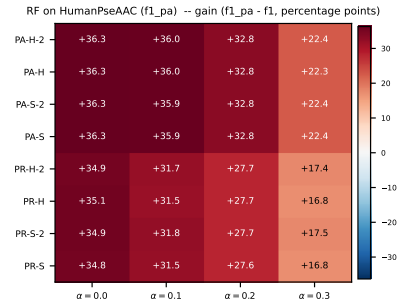
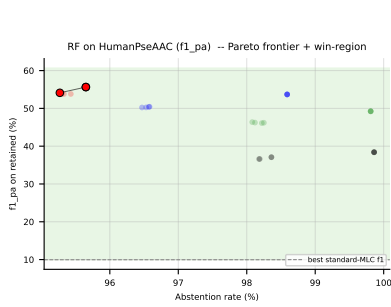


Figure 29: RF ($f1_{pa}$) on HumanPseAAC: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($f1_{pa} - f_1$) heatmap (right).

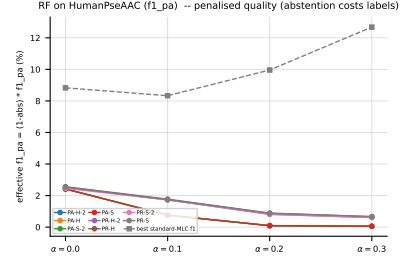
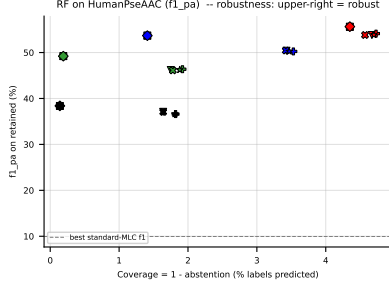


Figure 30: RF ($f1_{pa}$) on **HumanPseAAC** — robustness view. Left: efficiency-quality scatter ($x = \text{coverage} = 1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective $f1_{pa} = (1 - \text{abs}) \cdot f1_{pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

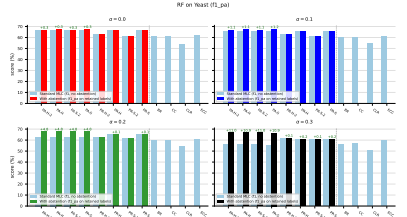
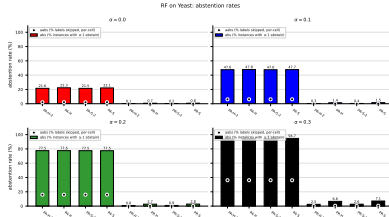


Figure 31: RF ($f1_{pa}$) on **Yeast**: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = **abs** (% of instances with at least one abstained label, instance-level); dot = **aabs** (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

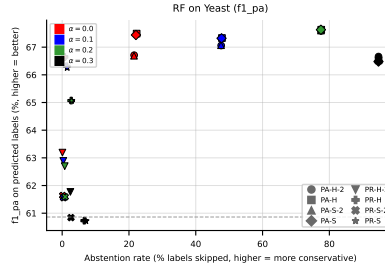


Figure 32: RF ($f1_{pa}$) on **Yeast**: coverage-risk scatter. Each point is one (method, noise) pair; $x = \text{abstention rate}$, $y = \text{quality on retained labels}$. Points above the dashed line beat the best standard-MLC baseline.



Figure 33: RF (f1_pa) on Yeast: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($f1_pa - f1$) heatmap (right).



Figure 34: RF (f1_pa) on Yeast — robustness view. Left: efficiency-quality scatter ($x = \text{coverage} = 1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective $f1_pa = (1 - \text{abs}) \cdot f1_pa$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

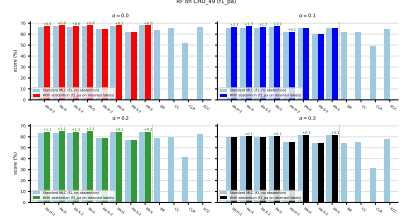
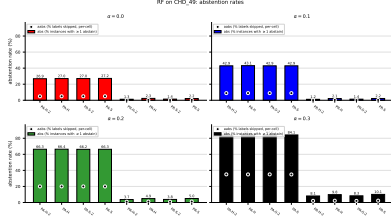


Figure 35: RF ($f1_pa$) on CHD_49: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = **abs** (% of instances with at least one abstained label, instance-level); dot = **aabs** (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

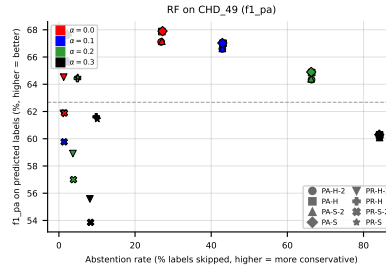


Figure 36: RF ($f1_pa$) on CHD_49: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

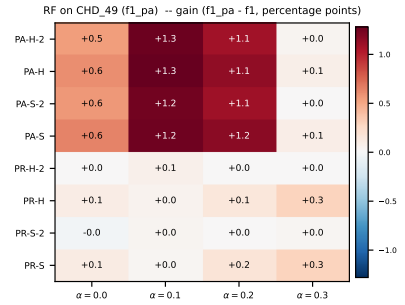
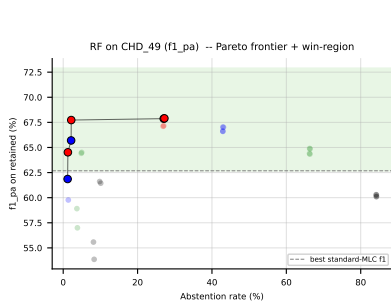


Figure 37: RF ($f1_pa$) on CHD_49: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($f1_pa - f1$, percentage points) heatmap (right).

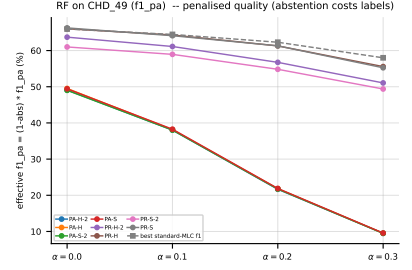
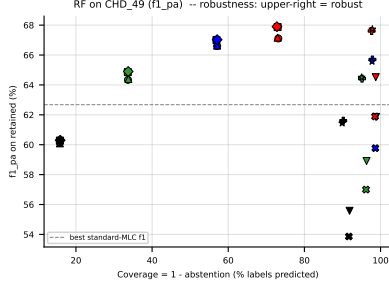


Figure 38: RF ($f1_{pa}$) on CHD_49 — robustness view. Left: efficiency-quality scatter ($x = \text{coverage} = 1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective $f1_{pa} = (1 - \text{abs}) \cdot f1_{pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

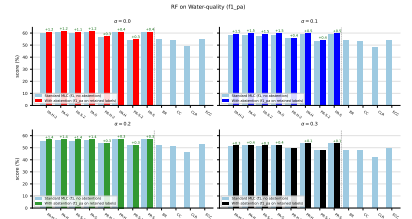
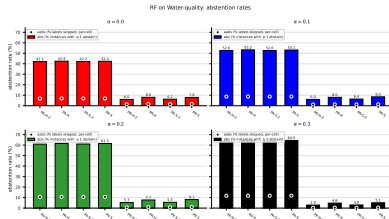


Figure 39: RF ($f1_{pa}$) on Water-quality: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = abs (% of instances with at least one abstained label, instance-level); dot = aabs (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

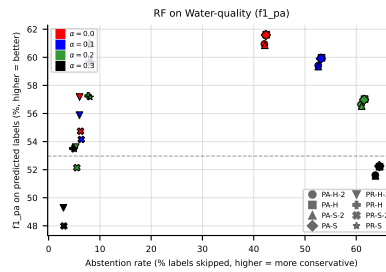


Figure 40: RF ($f1_{pa}$) on Water-quality: coverage-risk scatter. Each point is one (method, noise) pair; $x = \text{abstention rate}$, $y = \text{quality on retained labels}$. Points above the dashed line beat the best standard-MLC baseline.



Figure 41: RF (f1_pa) on Water-quality: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($f1_{pa} - f1$) heatmap (right).



Figure 42: RF (f1_pa) on Water-quality — robustness view. Left: efficiency-quality scatter ($x = \text{coverage} = 1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective $f1_{pa} = (1 - \text{abs}) \cdot f1_{pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

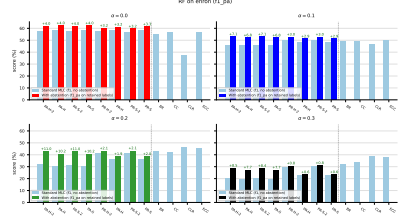
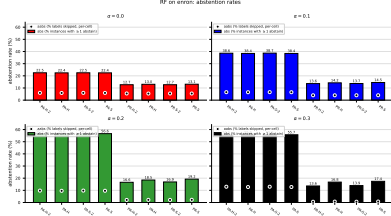


Figure 43: RF ($f1_pa$) on **enron**: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = **abs** (% of instances with at least one abstained label, instance-level); dot = **aabs** (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

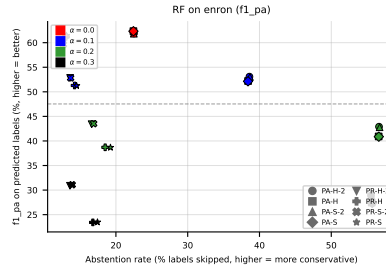


Figure 44: RF ($f1_pa$) on **enron**: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

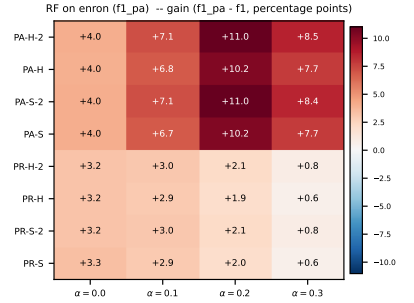
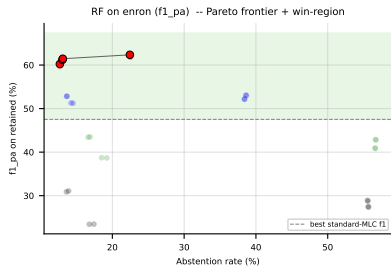


Figure 45: RF ($f1_pa$) on **enron**: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($f1_pa - f1$) heatmap (right).

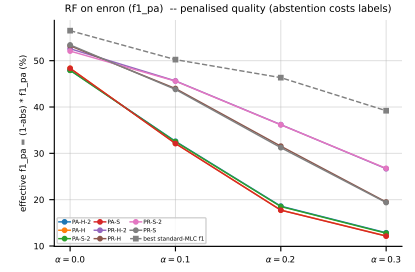
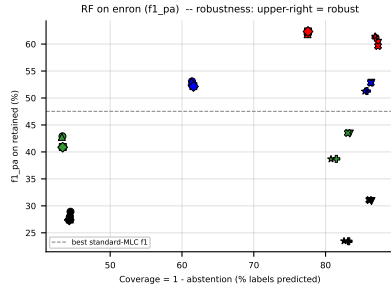


Figure 46: RF ($f1_{pa}$) on **enron** — robustness view. Left: efficiency-quality scatter ($x = \text{coverage} = 1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective $f1_{pa} = (1 - \text{abs}) \cdot f1_{pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

5.3 RF, metric jaccard_pa — aggregate



Figure 47: RF (jaccard_pa) averaged across all datasets: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = **abs** (% of instances with at least one abstained label, instance-level); dot = **aabs** (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

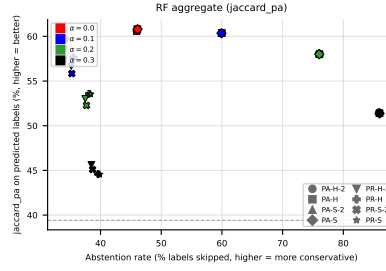


Figure 48: RF (jaccard_pa) averaged across all datasets: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.



Figure 49: RF (jaccard_pa) averaged across all datasets: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($jaccard_pa - f_1$) heatmap (right).

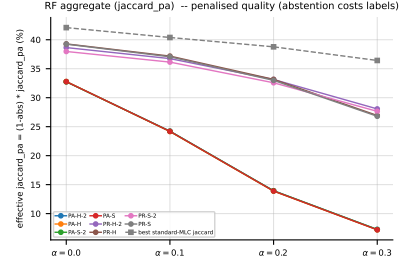
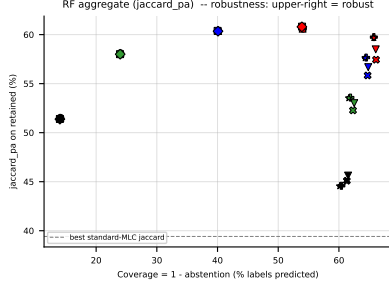


Figure 50: RF (jaccard_{pa}) averaged across all datasets — robustness view. Left: efficiency-quality scatter ($x = \text{coverage} = 1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective jaccard_{pa} = $(1 - \text{abs}) \cdot \text{jaccard}_{pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

5.4 RF, metric jaccard_{pa} — per dataset

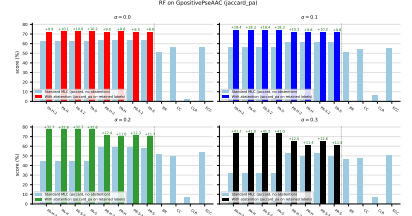
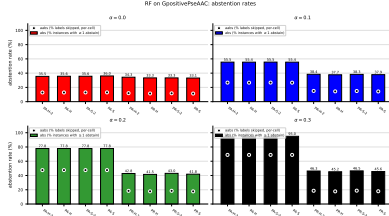


Figure 51: RF (jaccard_{pa}) on GpositivePseAAC: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = **abs** (% of instances with at least one abstained label, instance-level); dot = **aabs** (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

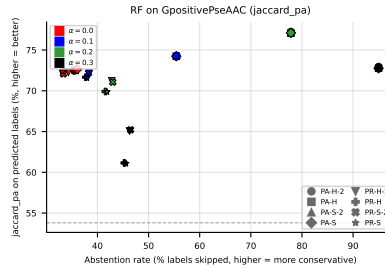


Figure 52: RF (jaccard_{pa}) on GpositivePseAAC: coverage-risk scatter. Each point is one (method, noise) pair; $x = \text{abstention rate}$, $y = \text{quality on retained labels}$. Points above the dashed line beat the best standard-MLC baseline.

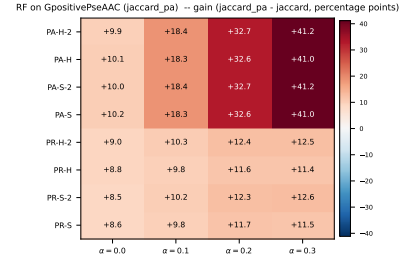
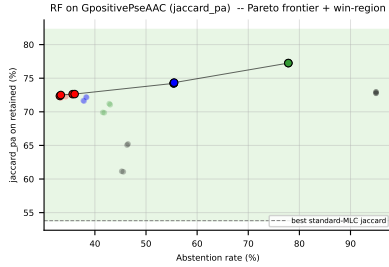


Figure 53: RF (jaccard_{pa}) on GpositivePseAAC: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (jaccard_{pa} − f_1) heatmap (right).

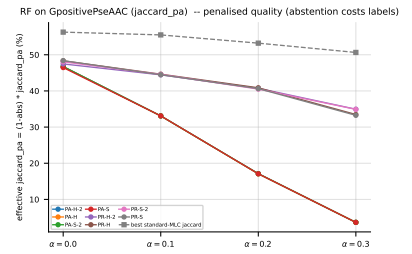
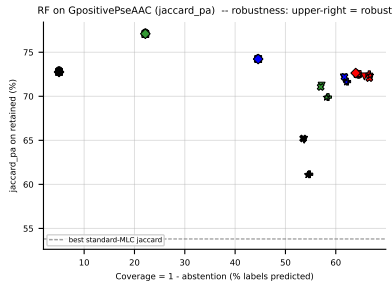


Figure 54: RF (jaccard_{pa}) on GpositivePseAAC — robustness view. Left: efficiency-quality scatter (x = coverage = 1 − abs); upper-right is robust (predicts many labels and keeps quality high). Right: effective jaccard_{pa} = (1 − abs) · jaccard_{pa}, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

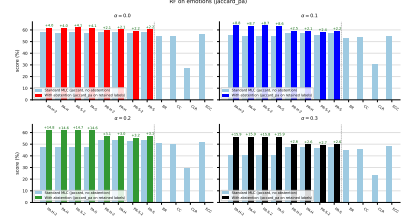
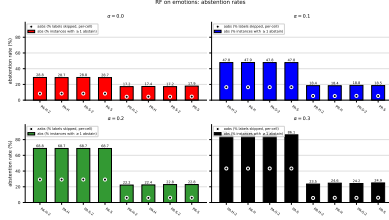


Figure 55: RF (jaccard_pa) on **emotions**: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = **abs** (% of instances with at least one abstained label, instance-level); dot = **aabs** (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

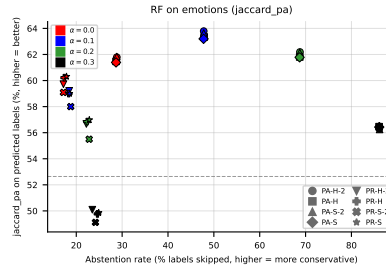


Figure 56: RF (jaccard_pa) on **emotions**: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

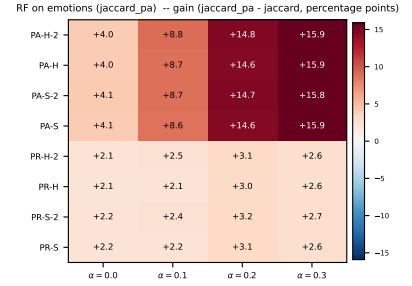
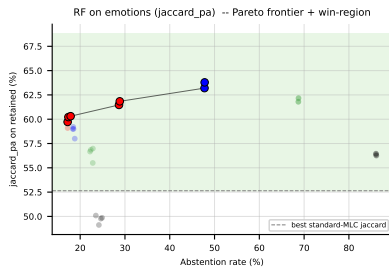


Figure 57: RF (jaccard_pa) on **emotions**: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($\text{jaccard_pa} - f_1$) heatmap (right).

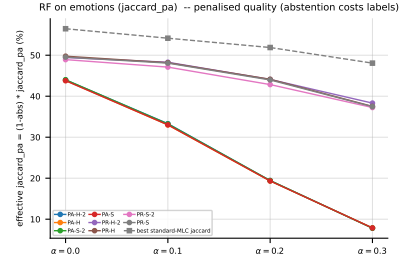
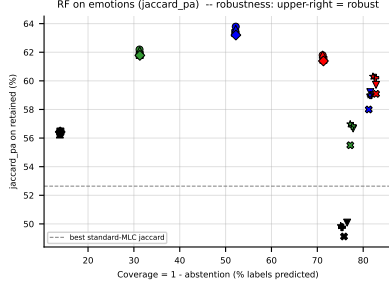


Figure 58: RF (jaccard_pa) on **emotions** — robustness view. Left: efficiency-quality scatter ($x = \text{coverage} = 1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective jaccard_pa = $(1 - \text{abs}) \cdot \text{jaccard_pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

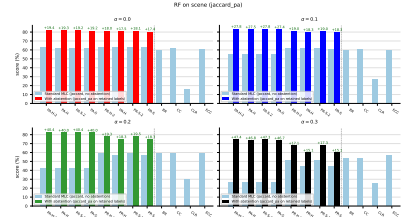
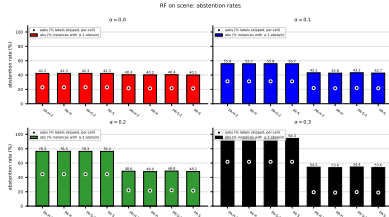


Figure 59: RF (jaccard_pa) on **scene**: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = **abs** (% of instances with at least one abstained label, instance-level); dot = **aabs** (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

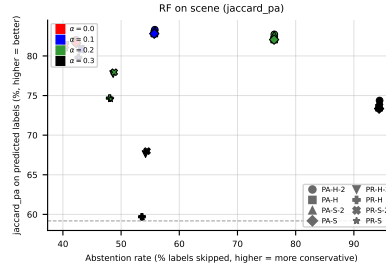


Figure 60: RF (jaccard_pa) on **scene**: coverage-risk scatter. Each point is one (method, noise) pair; $x = \text{abstention rate}$, $y = \text{quality on retained labels}$. Points above the dashed line beat the best standard-MLC baseline.

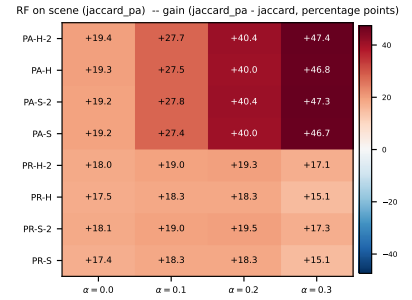
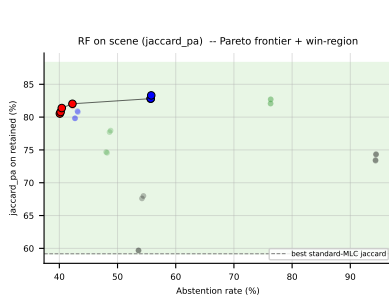


Figure 61: RF (jaccard_{pa}) on **scene**: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (jaccard_{pa} − f_1) heatmap (right).

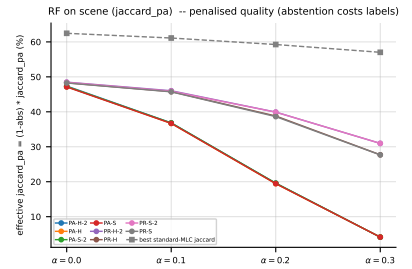
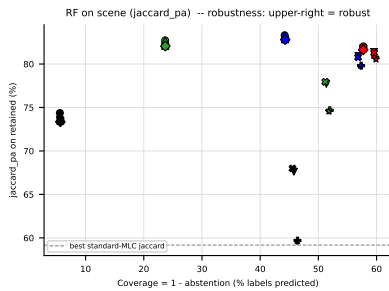


Figure 62: RF (jaccard_{pa}) on **scene** — robustness view. Left: efficiency-quality scatter ($x = \text{coverage} = 1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective jaccard_{pa} = $(1 - \text{abs}) \cdot \text{jaccard}_{pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

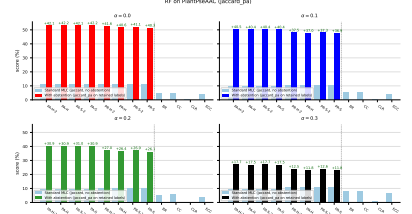
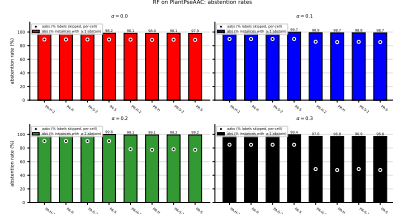


Figure 63: RF (jaccard_pa) on PlantPseAAC: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = **abs** (% of instances with at least one abstained label, instance-level); dot = **aabs** (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

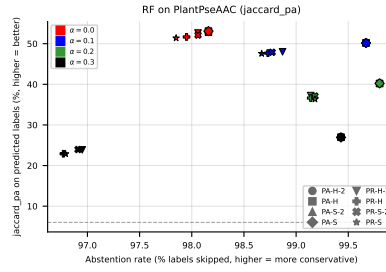


Figure 64: RF (jaccard_pa) on PlantPseAAC: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

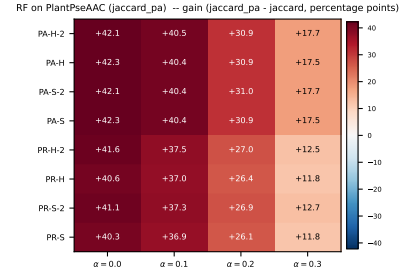
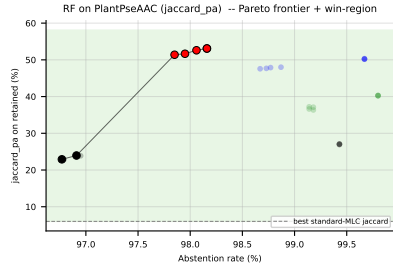


Figure 65: RF (jaccard_pa) on PlantPseAAC: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($jaccard_pa - f_1$) heatmap (right).

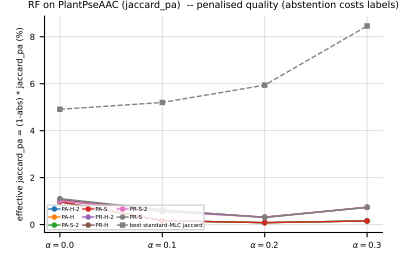
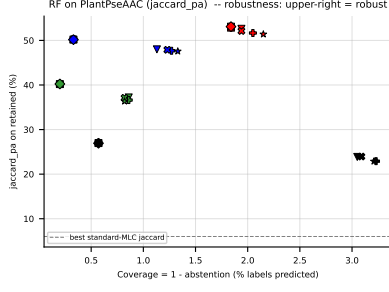


Figure 66: RF (jaccard_{pa}) on PlantPseAAC — robustness view. Left: efficiency-quality scatter (x = coverage = $1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective jaccard_{pa} = $(1 - \text{abs}) \cdot \text{jaccard}_{pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

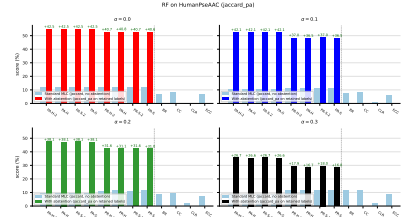
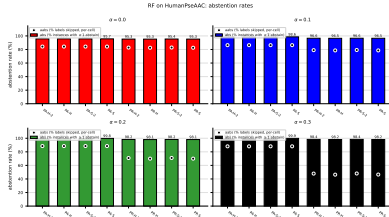


Figure 67: RF (jaccard_{pa}) on HumanPseAAC: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = **abs** (% of instances with at least one abstained label, instance-level); dot = **aabs** (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

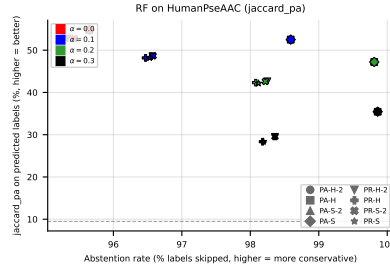


Figure 68: RF (jaccard_{pa}) on HumanPseAAC: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

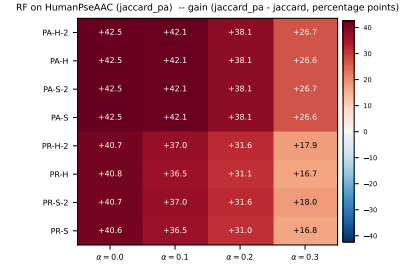
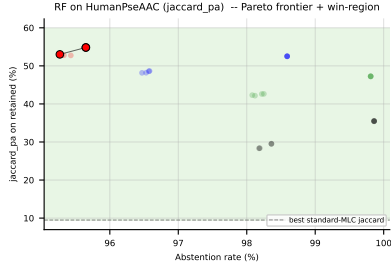


Figure 69: RF (jaccard_pa) on HumanPseAAC: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($jaccard_pa - f_1$) heatmap (right).

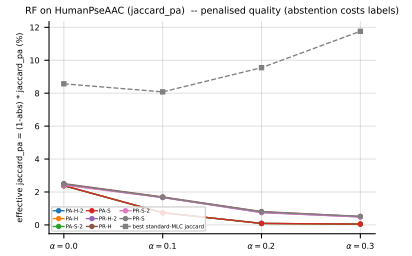
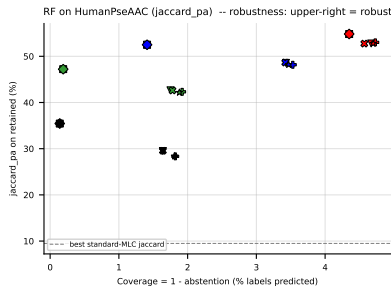


Figure 70: RF (jaccard_pa) on HumanPseAAC — robustness view. Left: efficiency-quality scatter ($x = coverage = 1 - abs$); upper-right is robust (predicts many labels and keeps quality high). Right: effective jaccard_pa = $(1 - abs) \cdot jaccard_pa$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

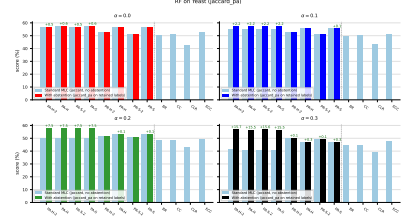
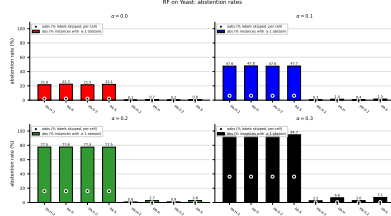


Figure 71: RF (jaccard_pa) on **Yeast**: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = **abs** (% of instances with at least one abstained label, instance-level); dot = **aabs** (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

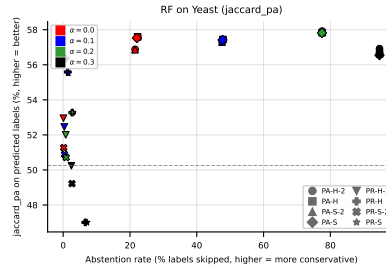


Figure 72: RF (jaccard_pa) on **Yeast**: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

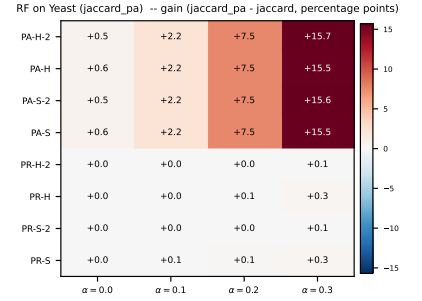
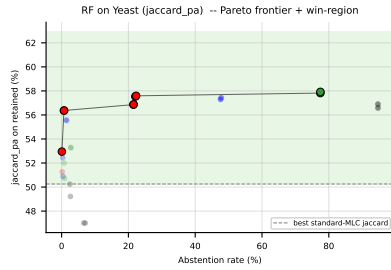


Figure 73: RF (jaccard_pa) on **Yeast**: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($\text{jaccard_pa} - f_1$) heatmap (right).

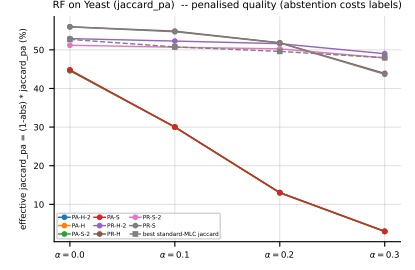
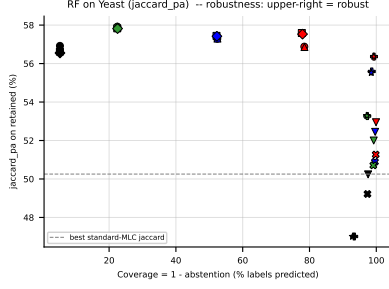


Figure 74: RF (jaccard_pa) on Yeast — robustness view. Left: efficiency-quality scatter ($x = \text{coverage} = 1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective jaccard_pa = $(1 - \text{abs}) \cdot \text{jaccard_pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

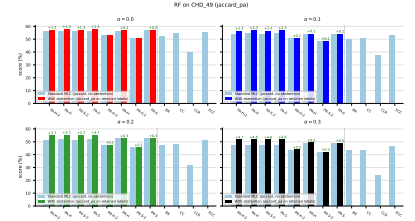
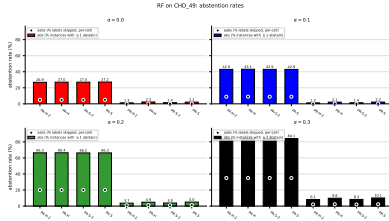


Figure 75: RF (jaccard_pa) on CHD_49: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = **abs** (% of instances with at least one abstained label, instance-level); dot = **aabs** (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

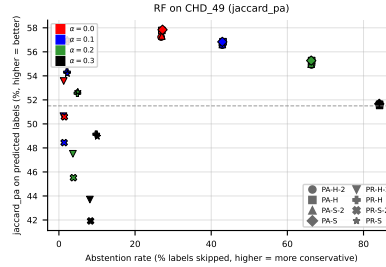


Figure 76: RF (jaccard_pa) on CHD_49: coverage-risk scatter. Each point is one (method, noise) pair; $x = \text{abstention rate}$, $y = \text{quality on retained labels}$. Points above the dashed line beat the best standard-MLC baseline.

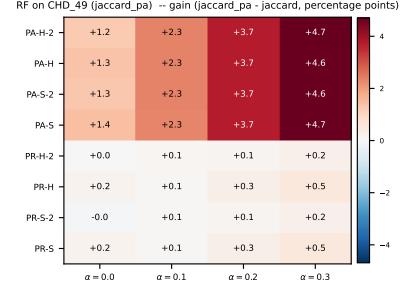
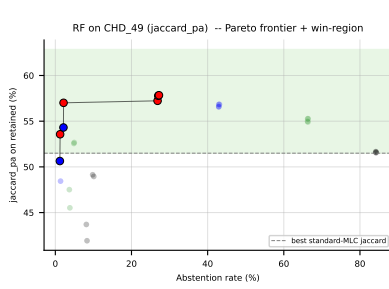


Figure 77: RF (jaccard.pa) on CHD_49: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($jaccard.pa - f_1$) heatmap (right).

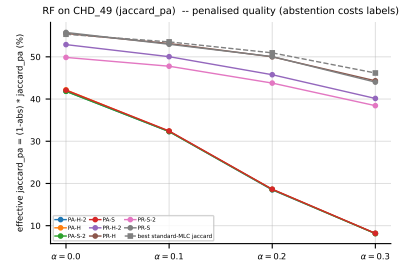
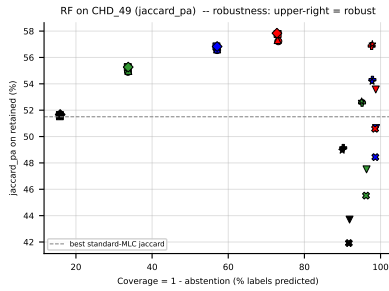


Figure 78: RF (jaccard.pa) on CHD_49 — robustness view. Left: efficiency-quality scatter ($x = coverage = 1 - abs$); upper-right is robust (predicts many labels and keeps quality high). Right: effective jaccard.pa = $(1 - abs) \cdot jaccard.pa$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

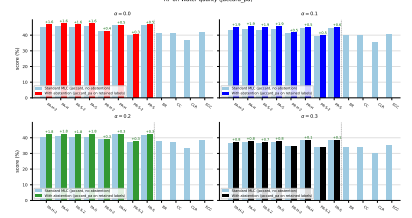
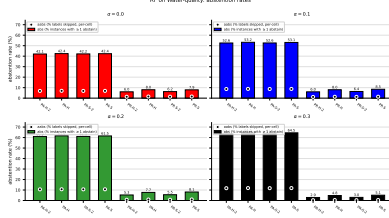


Figure 79: RF (jaccard_pa) on **Water-quality**: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = **abs** (% of instances with at least one abstained label, instance-level); dot = **aabs** (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

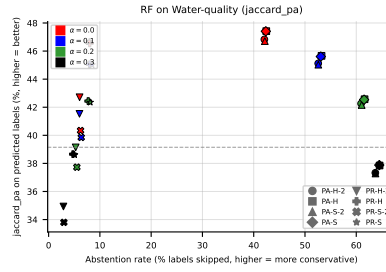


Figure 80: RF (jaccard_pa) on **Water-quality**: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

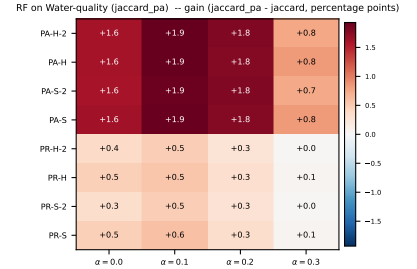
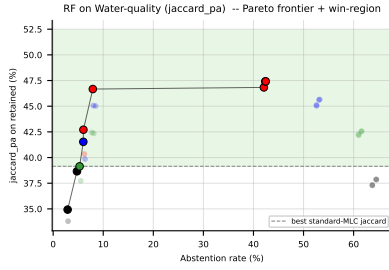


Figure 81: RF (jaccard_pa) on **Water-quality**: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($\text{jaccard_pa} - f_1$) heatmap (right).

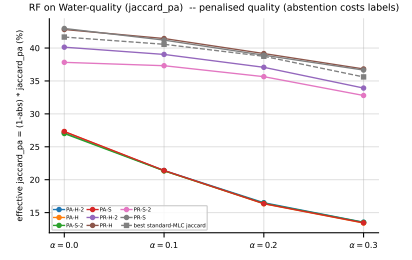
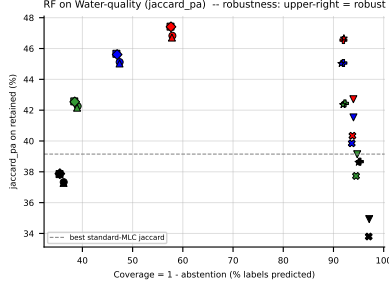


Figure 82: RF (jaccard_pa) on **Water-quality** — robustness view. Left: efficiency-quality scatter ($x = \text{coverage} = 1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective jaccard_pa = $(1 - \text{abs}) \cdot \text{jaccard_pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

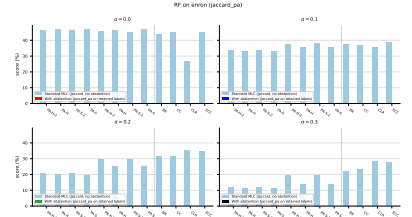
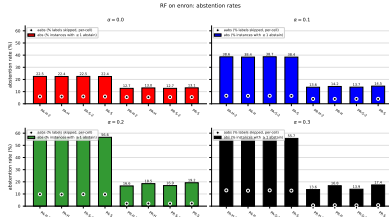


Figure 83: RF (jaccard_pa) on **enron**: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = **abs** (% of instances with at least one abstained label, instance-level); dot = **aabs** (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

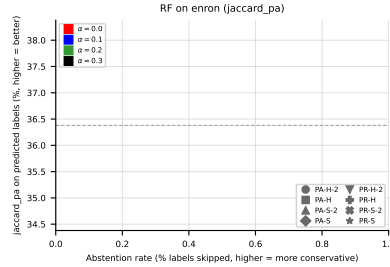


Figure 84: RF (jaccard_pa) on **enron**: coverage-risk scatter. Each point is one (method, noise) pair; $x = \text{abstention rate}$, $y = \text{quality on retained labels}$. Points above the dashed line beat the best standard-MLC baseline.



Figure 85: RF (jaccard_pa) on **enron**: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (jaccard_pa - f_1) heatmap (right).



Figure 86: RF (jaccard_pa) on **enron** — robustness view. Left: efficiency-quality scatter (x = coverage = 1 - abs); upper-right is robust (predicts many labels and keeps quality high). Right: effective jaccard_pa = (1 - abs) · jaccard_pa, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

5.5 LGBM, metric f1_pa — aggregate



Figure 87: LGBM (f1_pa) averaged across all datasets: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = **abs** (% of instances with at least one abstained label, instance-level); dot = **aabs** (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

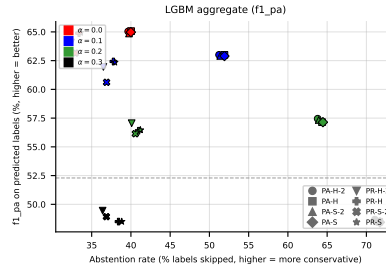


Figure 88: LGBM (f1_pa) averaged across all datasets: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.



Figure 89: LGBM (f1_pa) averaged across all datasets: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($f1_{pa} - f1$) heatmap (right).

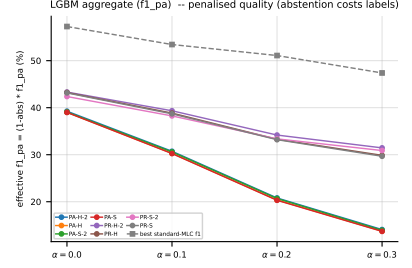
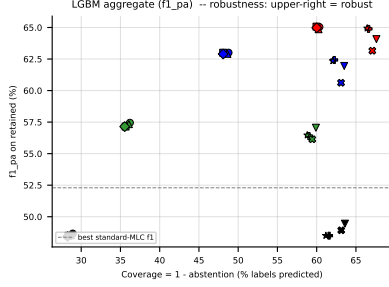


Figure 90: LGBM ($f1_{pa}$) averaged across all datasets — robustness view. Left: efficiency-quality scatter ($x = \text{coverage} = 1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective $f1_{pa} = (1 - \text{abs}) \cdot f1_{pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

5.6 LGBM, metric $f1_{pa}$ — per dataset

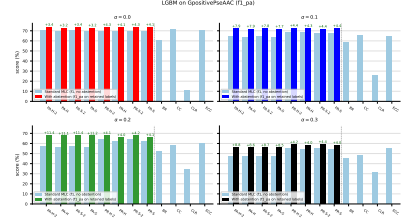
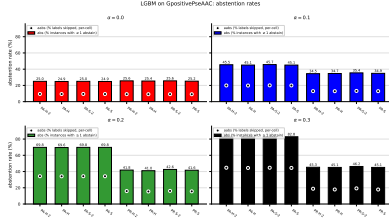


Figure 91: LGBM ($f1_{pa}$) on **GpositivePseAAC**: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = **abs** (% of instances with at least one abstained label, instance-level); dot = **aabs** (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

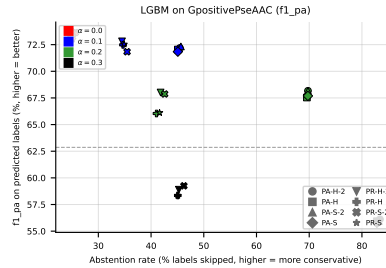


Figure 92: LGBM ($f1_{pa}$) on **GpositivePseAAC**: coverage-risk scatter. Each point is one (method, noise) pair; $x = \text{abstention rate}$, $y = \text{quality on retained labels}$. Points above the dashed line beat the best standard-MLC baseline.



Figure 93: LGBM ($f1_{pa}$) on **GpositivePseAAC**: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($f1_{pa} - f1$) heatmap (right).



Figure 94: LGBM ($f1_{pa}$) on **GpositivePseAAC** — robustness view. Left: efficiency-quality scatter ($x = \text{coverage} = 1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective $f1_{pa} = (1 - \text{abs}) \cdot f1_{pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

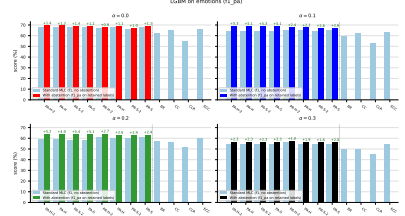
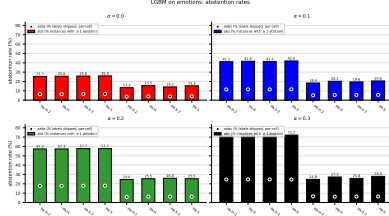


Figure 95: LGBM ($f1_{pa}$) on **emotions**: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = **abs** (% of instances with at least one abstained label, instance-level); dot = **aabs** (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

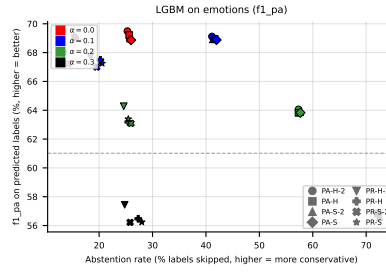


Figure 96: LGBM ($f1_{pa}$) on **emotions**: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

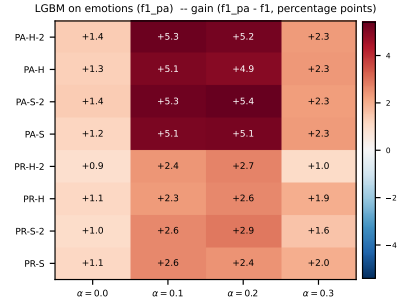
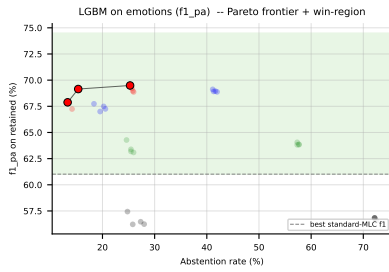


Figure 97: LGBM ($f1_{pa}$) on **emotions**: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($f1_{pa} - f_1$) heatmap (right).

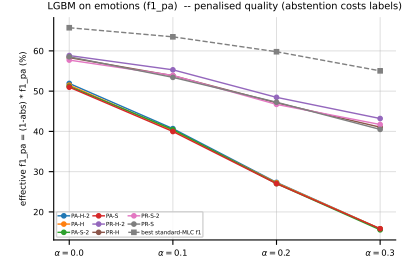
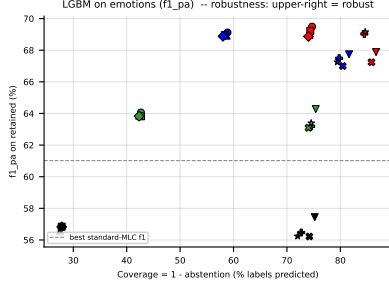


Figure 98: LGBM ($f1_{pa}$) on **emotions** — robustness view. Left: efficiency-quality scatter ($x = \text{coverage} = 1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective $f1_{pa} = (1 - \text{abs}) \cdot f1_{pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

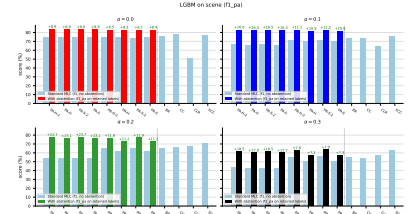
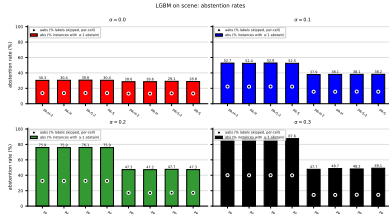


Figure 99: LGBM ($f1_{pa}$) on **scene**: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = **abs** (% of instances with at least one abstained label, instance-level); dot = **aabs** (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

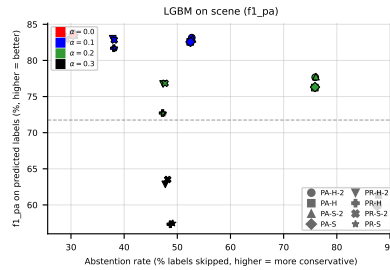


Figure 100: LGBM ($f1_{pa}$) on **scene**: coverage-risk scatter. Each point is one (method, noise) pair; $x = \text{abstention rate}$, $y = \text{quality on retained labels}$. Points above the dashed line beat the best standard-MLC baseline.

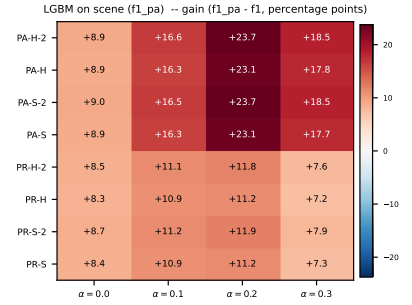
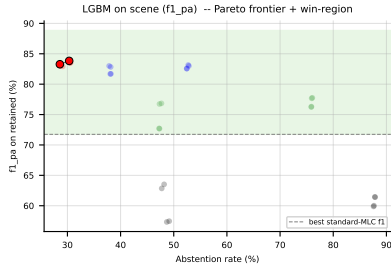


Figure 101: LGBM ($f1_{pa}$) on **scene**: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($f1_{pa} - f1$) heatmap (right).

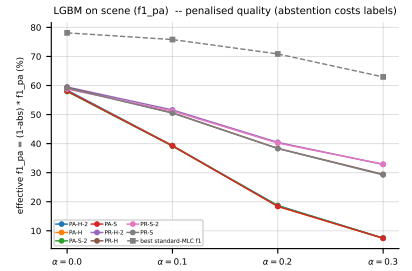
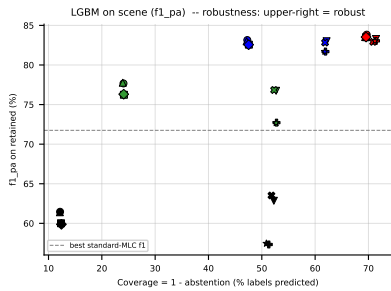


Figure 102: LGBM ($f1_{pa}$) on **scene** — robustness view. Left: efficiency-quality scatter ($x = \text{coverage} = 1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective $f1_{pa} = (1 - \text{abs}) \cdot f1_{pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

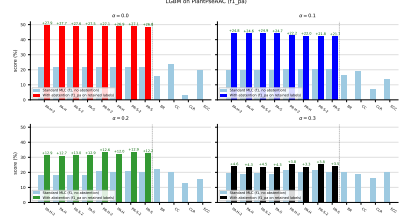
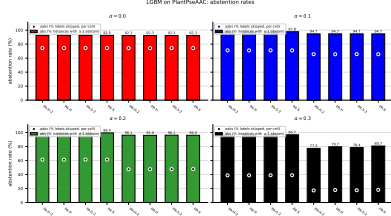


Figure 103: LGBM ($f1_pa$) on PlantPseAAC: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = **abs** (% of instances with at least one abstained label, instance-level); dot = **aabs** (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

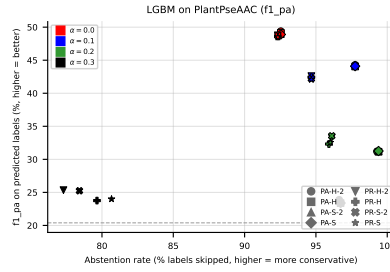


Figure 104: LGBM ($f1_pa$) on PlantPseAAC: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

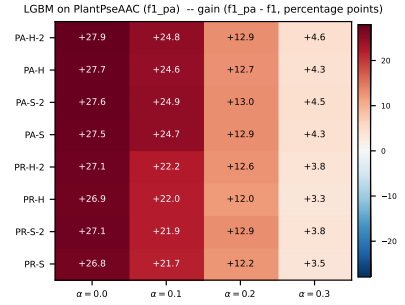
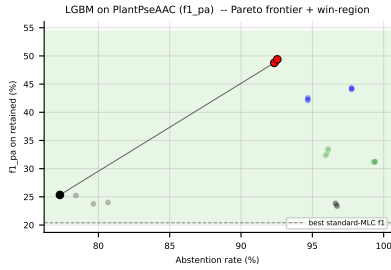


Figure 105: LGBM ($f1_pa$) on PlantPseAAC: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($f1_pa - f1$) heatmap (right).

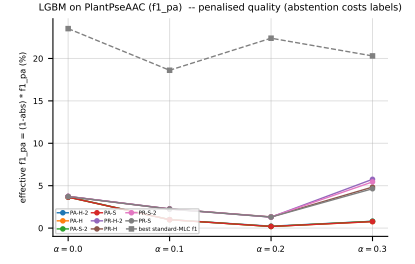
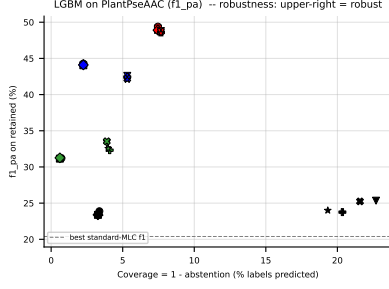


Figure 106: LGBM ($f1_{pa}$) on PlantPseAAC — robustness view. Left: efficiency-quality scatter (x = coverage = $1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective $f1_{pa} = (1 - \text{abs}) \cdot f1_{pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

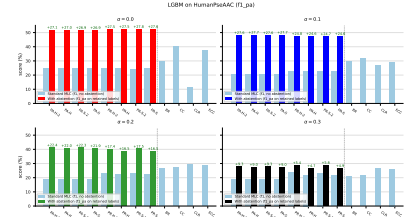
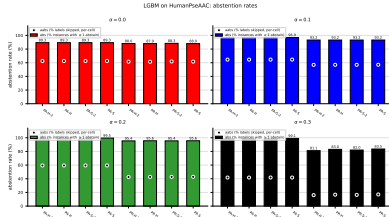


Figure 107: LGBM ($f1_{pa}$) on HumanPseAAC: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = **abs** (% of instances with at least one abstained label, instance-level); dot = **aabs** (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

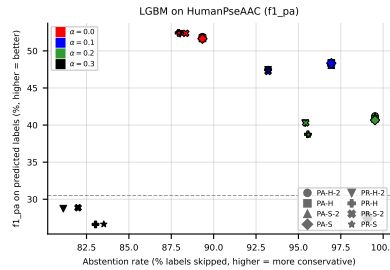


Figure 108: LGBM ($f1_{pa}$) on HumanPseAAC: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.



Figure 109: LGBM ($f1_{pa}$) on HumanPseAAC: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($f1_{pa} - f1$) heatmap (right).



Figure 110: LGBM ($f1_{pa}$) on HumanPseAAC — robustness view. Left: efficiency-quality scatter ($x = \text{coverage} = 1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective $f1_{pa} = (1 - \text{abs}) \cdot f1_{pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

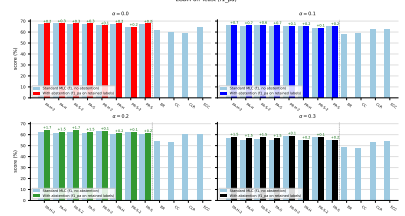
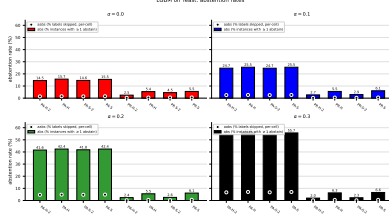


Figure 111: LGBM ($f1_{pa}$) on **Yeast**: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = **abs** (% of instances with at least one abstained label, instance-level); dot = **aabs** (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

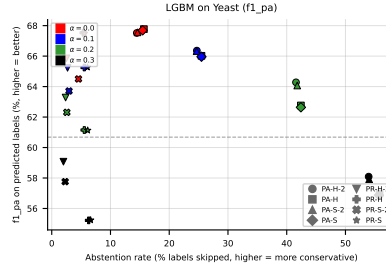


Figure 112: LGBM ($f1_{pa}$) on **Yeast**: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

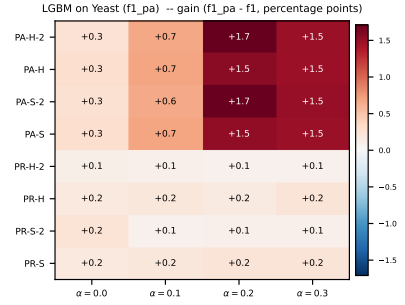
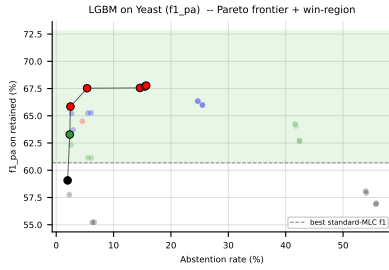


Figure 113: LGBM ($f1_{pa}$) on **Yeast**: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($f1_{pa} - f_1$) heatmap (right).

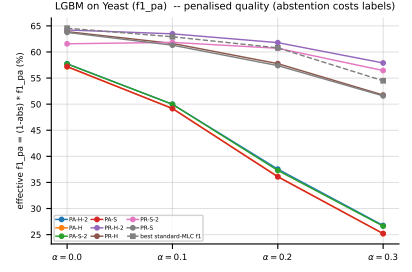
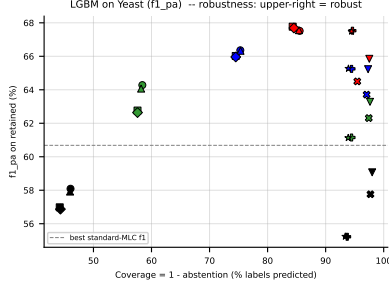


Figure 114: LGBM ($f1_{pa}$) on Yeast — robustness view. Left: efficiency-quality scatter ($x = \text{coverage} = 1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective $f1_{pa} = (1 - \text{abs}) \cdot f1_{pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

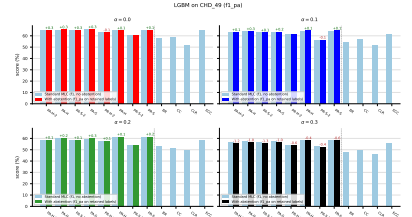
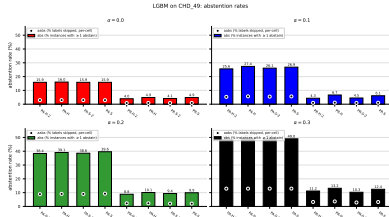


Figure 115: LGBM ($f1_{pa}$) on CHD_49: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = **abs** (% of instances with at least one abstained label, instance-level); dot = **aabs** (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

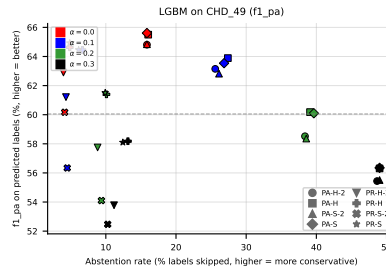


Figure 116: LGBM ($f1_{pa}$) on CHD_49: coverage-risk scatter. Each point is one (method, noise) pair; $x = \text{abstention rate}$, $y = \text{quality on retained labels}$. Points above the dashed line beat the best standard-MLC baseline.

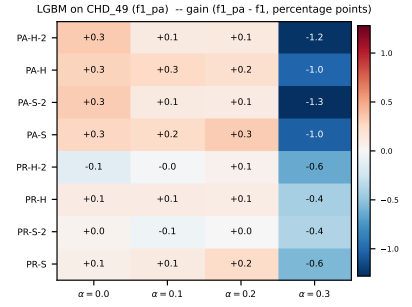
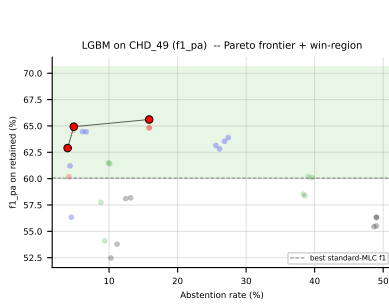


Figure 117: LGBM ($f1_{pa}$) on CHD_49: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($f1_{pa} - f1$) heatmap (right).

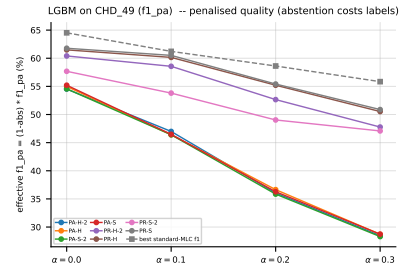
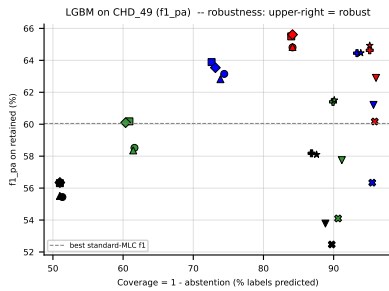


Figure 118: LGBM ($f1_{pa}$) on CHD_49 — robustness view. Left: efficiency-quality scatter ($x = \text{coverage} = 1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective $f1_{pa} = (1 - \text{abs}) \cdot f1_{pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

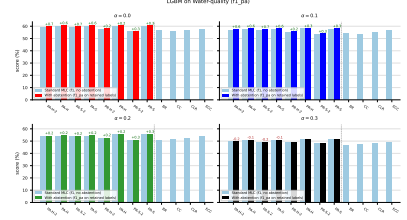
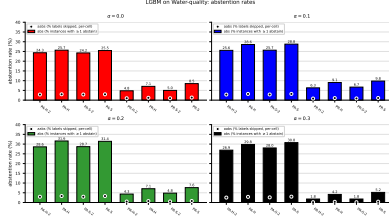


Figure 119: LGBM (f1_pa) on Water-quality: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = **abs** (% of instances with at least one abstained label, instance-level); dot = **aabs** (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

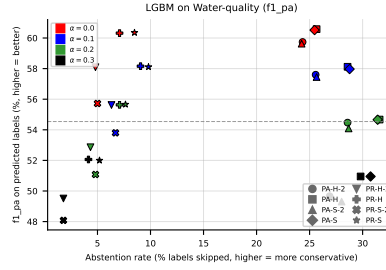


Figure 120: LGBM (f1_pa) on Water-quality: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

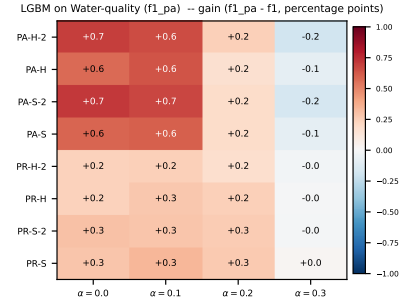
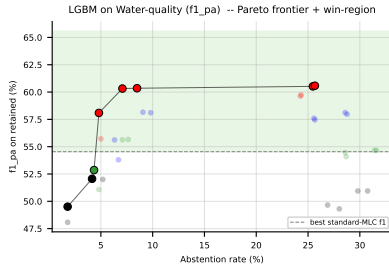


Figure 121: LGBM (f1_pa) on Water-quality: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($f1_{pa} - f1$) heatmap (right).

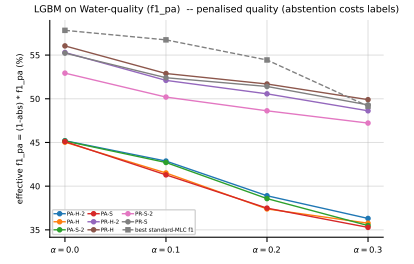
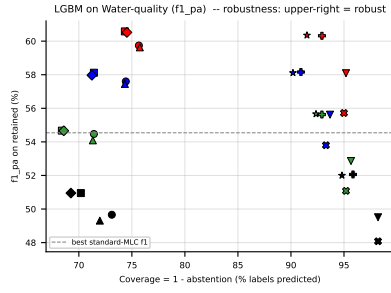


Figure 122: LGBM ($f1_{pa}$) on **Water-quality** — robustness view. Left: efficiency-quality scatter ($x = \text{coverage} = 1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective $f1_{pa} = (1 - \text{abs}) \cdot f1_{pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

5.7 LGBM, metric jaccard_pa — aggregate



Figure 123: LGBM (jaccard_pa) averaged across all datasets: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = **abs** (% of instances with at least one abstained label, instance-level); dot = **aabs** (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

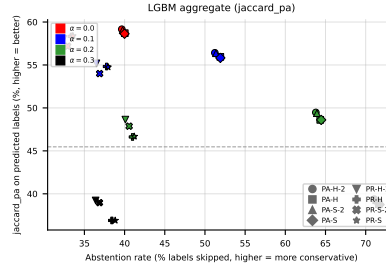


Figure 124: LGBM (jaccard_pa) averaged across all datasets: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.



Figure 125: LGBM (jaccard_pa) averaged across all datasets: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($\text{jaccard_pa} - f_1$) heatmap (right).

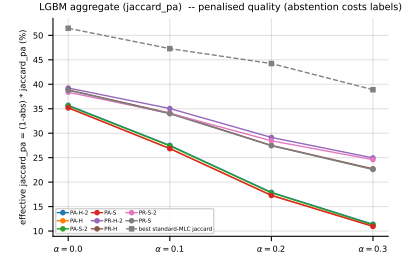
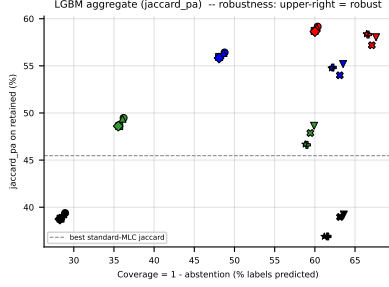


Figure 126: LGBM (*jaccard_pa*) averaged across all datasets — robustness view. Left: efficiency-quality scatter ($x = \text{coverage} = 1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective *jaccard_pa* = $(1 - \text{abs}) \cdot \text{jaccard_pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

5.8 LGBM, metric *jaccard_pa* — per dataset

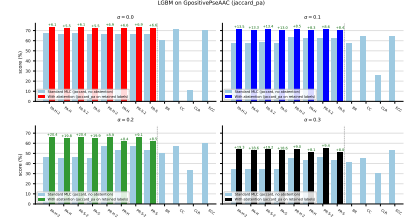
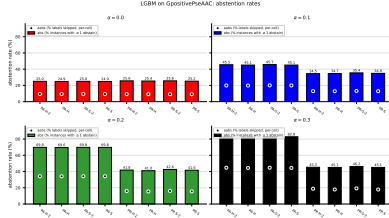


Figure 127: LGBM (*jaccard_pa*) on **GpositivePseAAC**: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = **abs** (% of instances with at least one abstained label, instance-level); dot = **aabs** (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

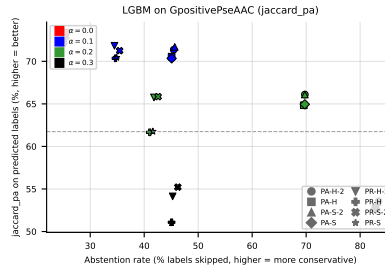


Figure 128: LGBM (*jaccard_pa*) on **GpositivePseAAC**: coverage-risk scatter. Each point is one (method, noise) pair; $x = \text{abstention rate}$, $y = \text{quality on retained labels}$. Points above the dashed line beat the best standard-MLC baseline.

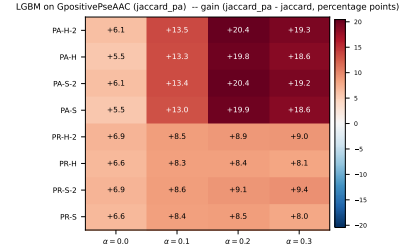
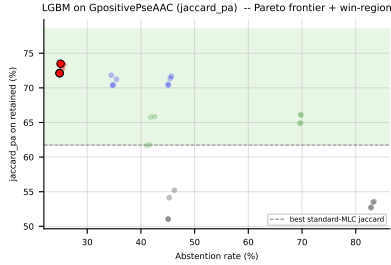


Figure 129: LGBM (jaccard_pa) on GpositivePseAAC: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (jaccard_pa - f_1) heatmap (right).

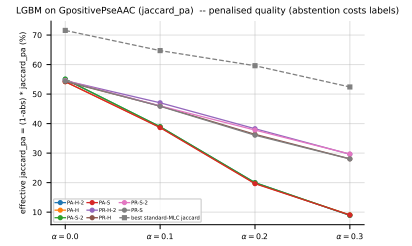
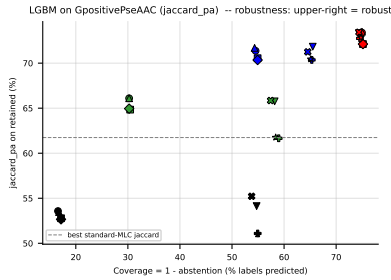


Figure 130: LGBM (jaccard_pa) on GpositivePseAAC — robustness view. Left: efficiency-quality scatter (x = coverage = 1 - abs); upper-right is robust (predicts many labels and keeps quality high). Right: effective jaccard_pa = (1 - abs) · jaccard_pa, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

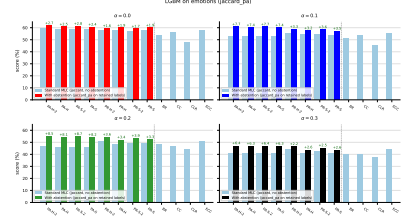
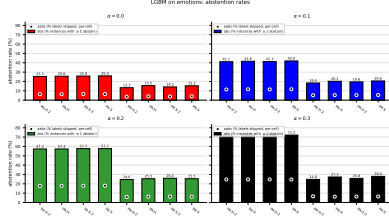


Figure 131: LGBM (jaccard_pa) on **emotions**: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = **abs** (% of instances with at least one abstained label, instance-level); dot = **aabs** (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

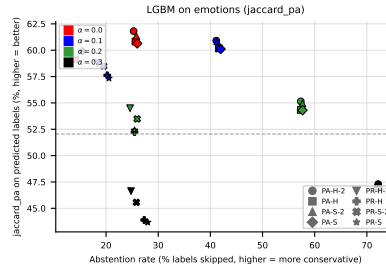


Figure 132: LGBM (jaccard_pa) on **emotions**: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

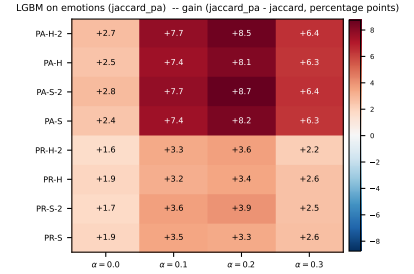
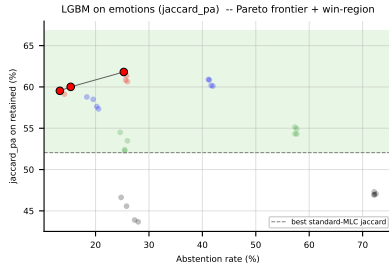


Figure 133: LGBM (jaccard_pa) on **emotions**: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($\text{jaccard_pa} - f_1$) heatmap (right).

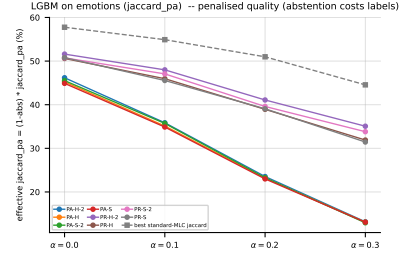
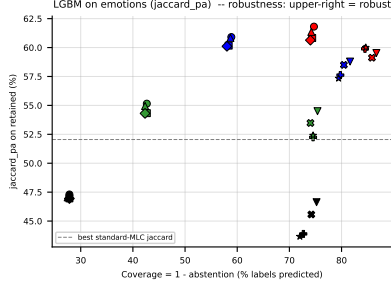


Figure 134: LGBM (jaccard_pa) on **emotions** — robustness view. Left: efficiency-quality scatter (x = coverage = $1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective $\text{jaccard_pa} = (1 - \text{abs}) \cdot \text{jaccard_pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

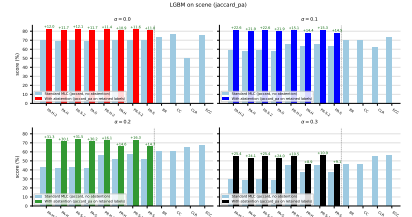
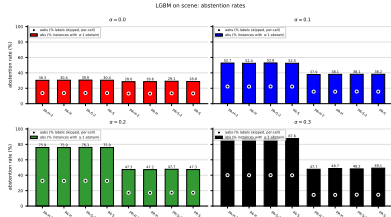


Figure 135: LGBM (jaccard_pa) on **scene**: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = **abs** (% of instances with at least one abstained label, instance-level); dot = **aabs** (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

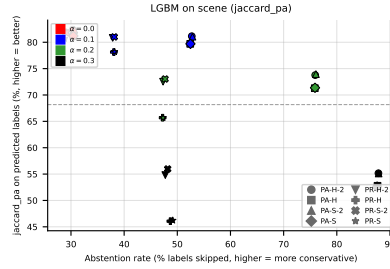


Figure 136: LGBM (jaccard_pa) on **scene**: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

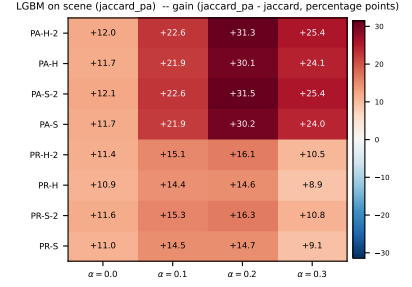
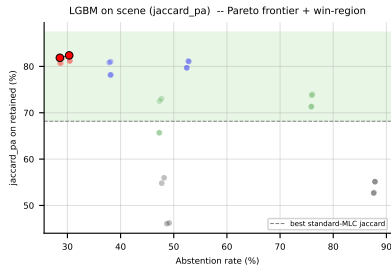


Figure 137: LGBM (jaccard_pa) on **scene**: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($jaccard_pa - f_1$) heatmap (right).

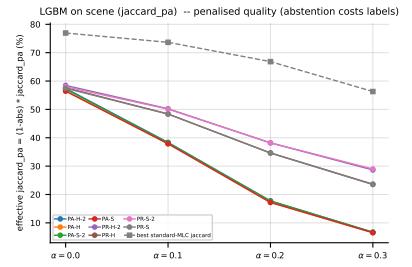
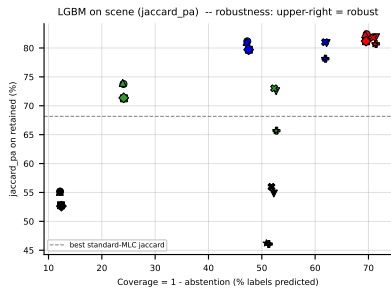


Figure 138: LGBM (jaccard_pa) on **scene** — robustness view. Left: efficiency-quality scatter ($x = \text{coverage} = 1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective jaccard_pa = $(1 - \text{abs}) \cdot jaccard_pa$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

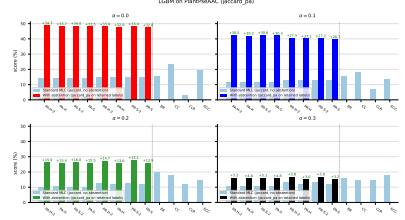
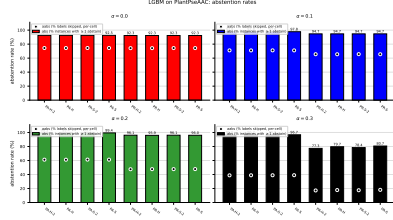


Figure 139: LGBM (jaccard_pa) on PlantPseAAC: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = **abs** (% of instances with at least one abstained label, instance-level); dot = **aabs** (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

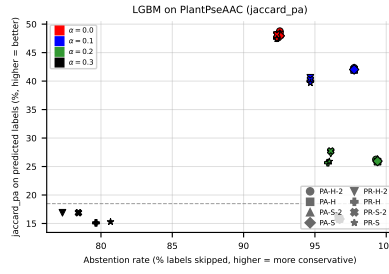


Figure 140: LGBM (jaccard_pa) on PlantPseAAC: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

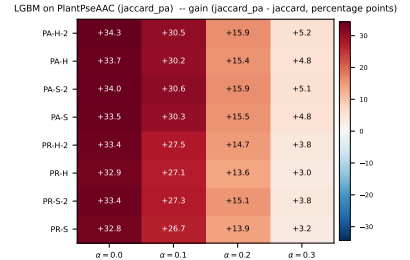
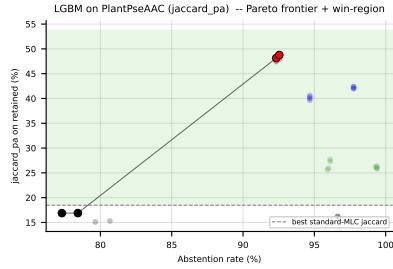


Figure 141: LGBM (jaccard_pa) on PlantPseAAC: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($\text{jaccard_pa} - f_1$) heatmap (right).

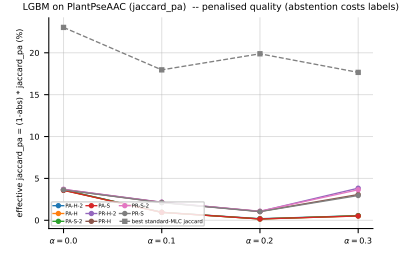
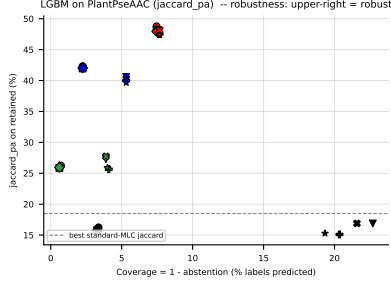


Figure 142: LGBM (jaccard_pa) on PlantPseAAC — robustness view. Left: efficiency-quality scatter ($x = \text{coverage} = 1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective jaccard_pa = $(1 - \text{abs}) \cdot \text{jaccard_pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

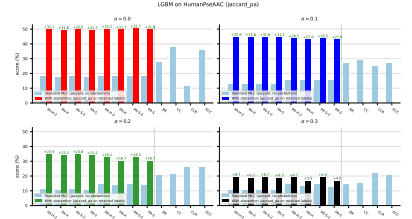
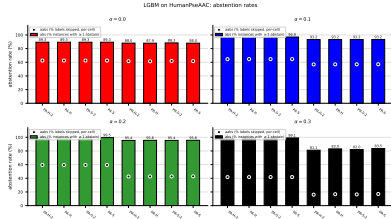


Figure 143: LGBM (jaccard_pa) on HumanPseAAC: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = **abs** (% of instances with at least one abstained label, instance-level); dot = **aabs** (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

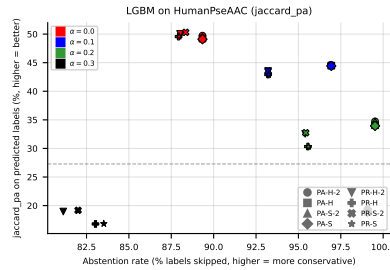


Figure 144: LGBM (jaccard_pa) on HumanPseAAC: coverage-risk scatter. Each point is one (method, noise) pair; $x = \text{abstention rate}$, $y = \text{quality on retained labels}$. Points above the dashed line beat the best standard-MLC baseline.

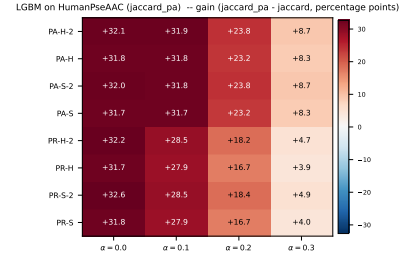
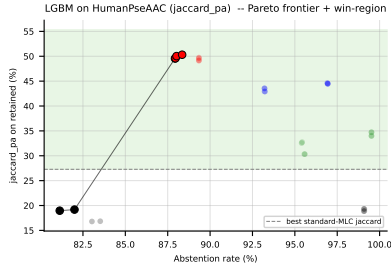


Figure 145: LGBM (jaccard_pa) on HumanPseAAC: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($jaccard_pa - f_1$) heatmap (right).

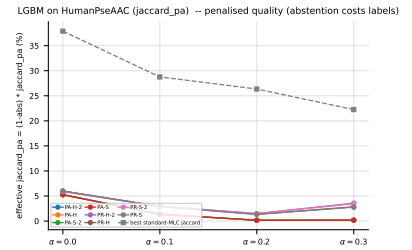
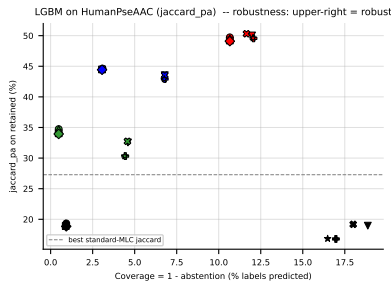


Figure 146: LGBM (jaccard_pa) on HumanPseAAC — robustness view. Left: efficiency-quality scatter ($x = coverage = 1 - abs$); upper-right is robust (predicts many labels and keeps quality high). Right: effective jaccard_pa = $(1 - abs) \cdot jaccard_pa$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

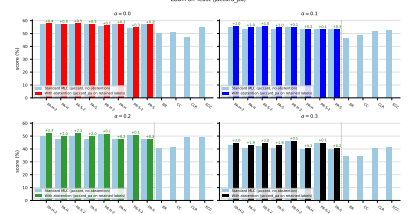
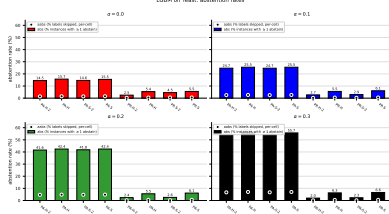


Figure 147: LGBM (jaccard_pa) on **Yeast**: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = **abs** (% of instances with at least one abstained label, instance-level); dot = **aabs** (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

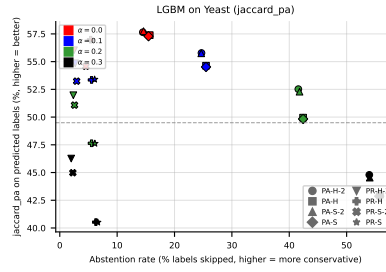


Figure 148: LGBM (jaccard_pa) on **Yeast**: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

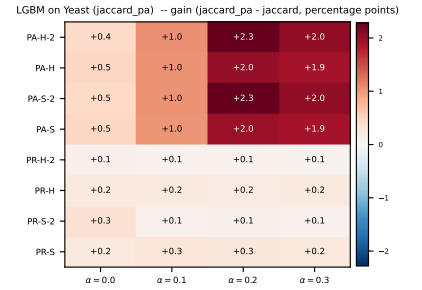
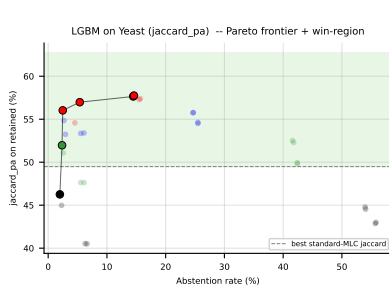


Figure 149: LGBM (jaccard_pa) on **Yeast**: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain (jaccard_pa - f_1) heatmap (right).

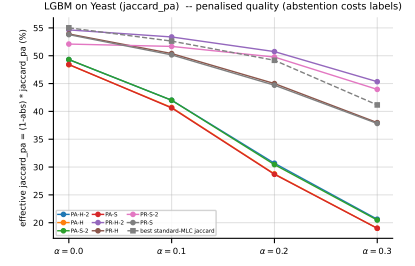
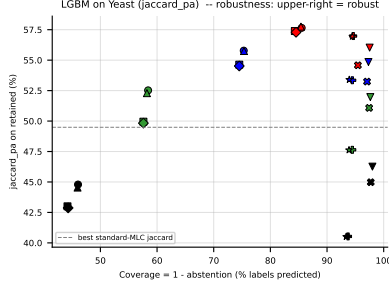


Figure 150: LGBM (jaccard_pa) on **Yeast** — robustness view. Left: efficiency-quality scatter ($x = \text{coverage} = 1 - \text{abs}$); upper-right is robust (predicts many labels and keeps quality high). Right: effective jaccard_pa = $(1 - \text{abs}) \cdot \text{jaccard_pa}$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

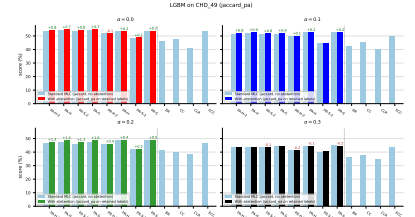
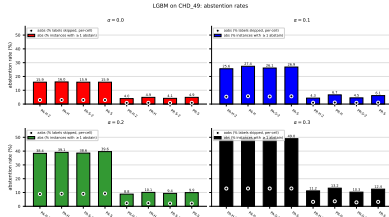


Figure 151: LGBM (jaccard_pa) on **CHD_49**: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = **abs** (% of instances with at least one abstained label, instance-level); dot = **aabs** (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

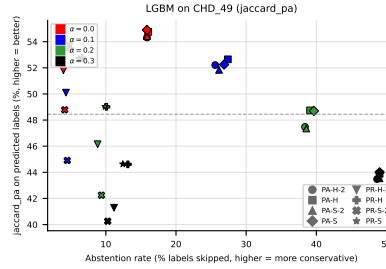


Figure 152: LGBM (jaccard_pa) on **CHD_49**: coverage-risk scatter. Each point is one (method, noise) pair; $x = \text{abstention rate}$, $y = \text{quality on retained labels}$. Points above the dashed line beat the best standard-MLC baseline.

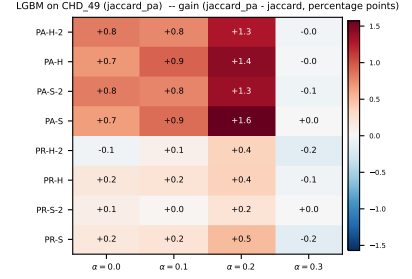
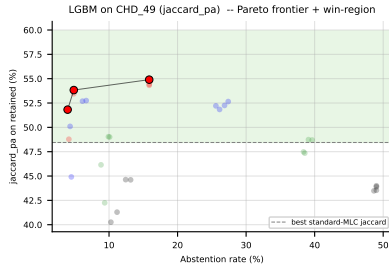


Figure 153: LGBM (jaccard_pa) on CHD_49: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($jaccard_pa - f_1$) heatmap (right).

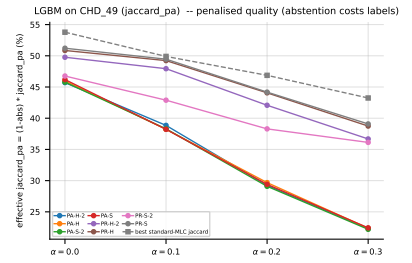
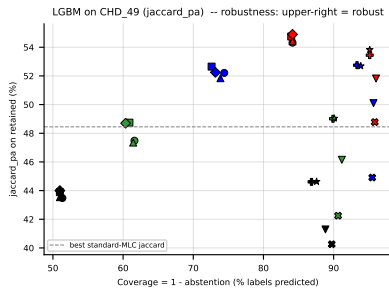


Figure 154: LGBM (jaccard_pa) on CHD_49 — robustness view. Left: efficiency-quality scatter ($x = coverage = 1 - abs$); upper-right is robust (predicts many labels and keeps quality high). Right: effective $jaccard_pa = (1 - abs) \cdot jaccard_pa$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

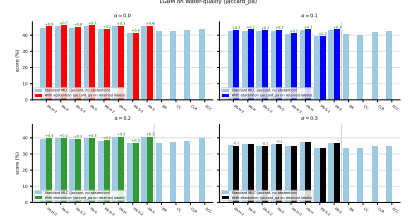
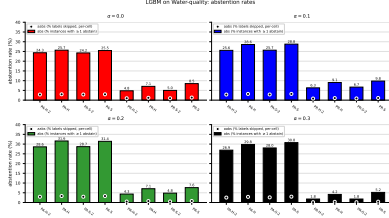


Figure 155: LGBM (jaccard_pa) on **Water-quality**: abstention rates (left) and standard-MLC-vs-abstention paired bars (right), across $\alpha \in \{0.0, 0.1, 0.2, 0.3\}$. Left: bar = **abs** (% of instances with at least one abstained label, instance-level); dot = **aabs** (% of labels skipped, per-cell label-level). These rates are independent of the quality metric. Right: blue = standard MLC quality (no abstention) on the matched BV metric; coloured = with-abstention quality on retained labels.

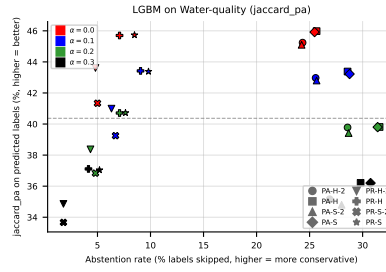


Figure 156: LGBM (jaccard_pa) on **Water-quality**: coverage-risk scatter. Each point is one (method, noise) pair; x = abstention rate, y = quality on retained labels. Points above the dashed line beat the best standard-MLC baseline.

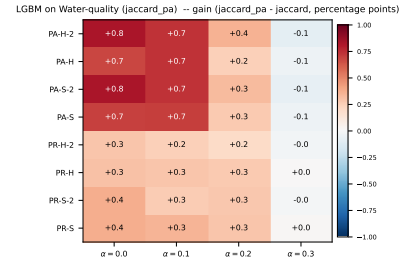
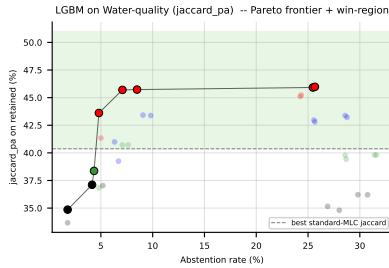


Figure 157: LGBM (jaccard_pa) on **Water-quality**: Pareto frontier of (abstention, retained quality) with the win-region shaded green (left); per-method per-noise gain ($\text{jaccard_pa} - f_1$) heatmap (right).

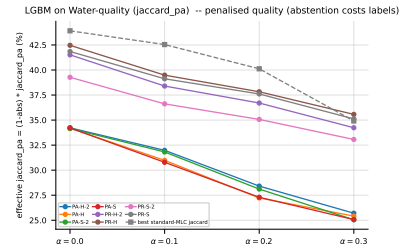
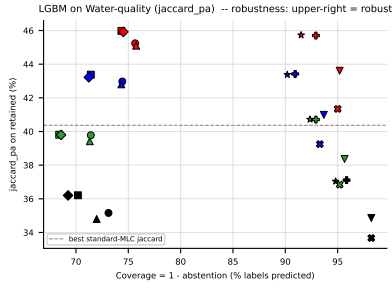


Figure 158: LGBM ($jaccard_pa$) on **Water-quality** — robustness view. Left: efficiency-quality scatter ($x = coverage = 1 - abs$); upper-right is robust (predicts many labels and keeps quality high). Right: effective $jaccard_pa = (1 - abs) \cdot jaccard_pa$, i.e. retained quality penalised by skipped coverage. A method whose effective curve stays above the dashed standard-MLC baseline at every α is robust to noise, not just selectively abstaining on hard labels.

A Experimental results — appendix tables (RF base learner)

Appendix counterpart of the main Paper results section. Tables G2–G5 mirror the original paper’s appendix exactly (same metric sets, with the 6 non-main datasets split into imbalanced vs. balanced buckets following the original layout). Tables G6–G8 are revision additions reporting the new metrics ($f_{\text{MLC}}^{\text{Jacc}}$, macro- f_1 , micro- f_1) and ScoreVector results.

A.1 Standard MLC predictions (BinaryVector)

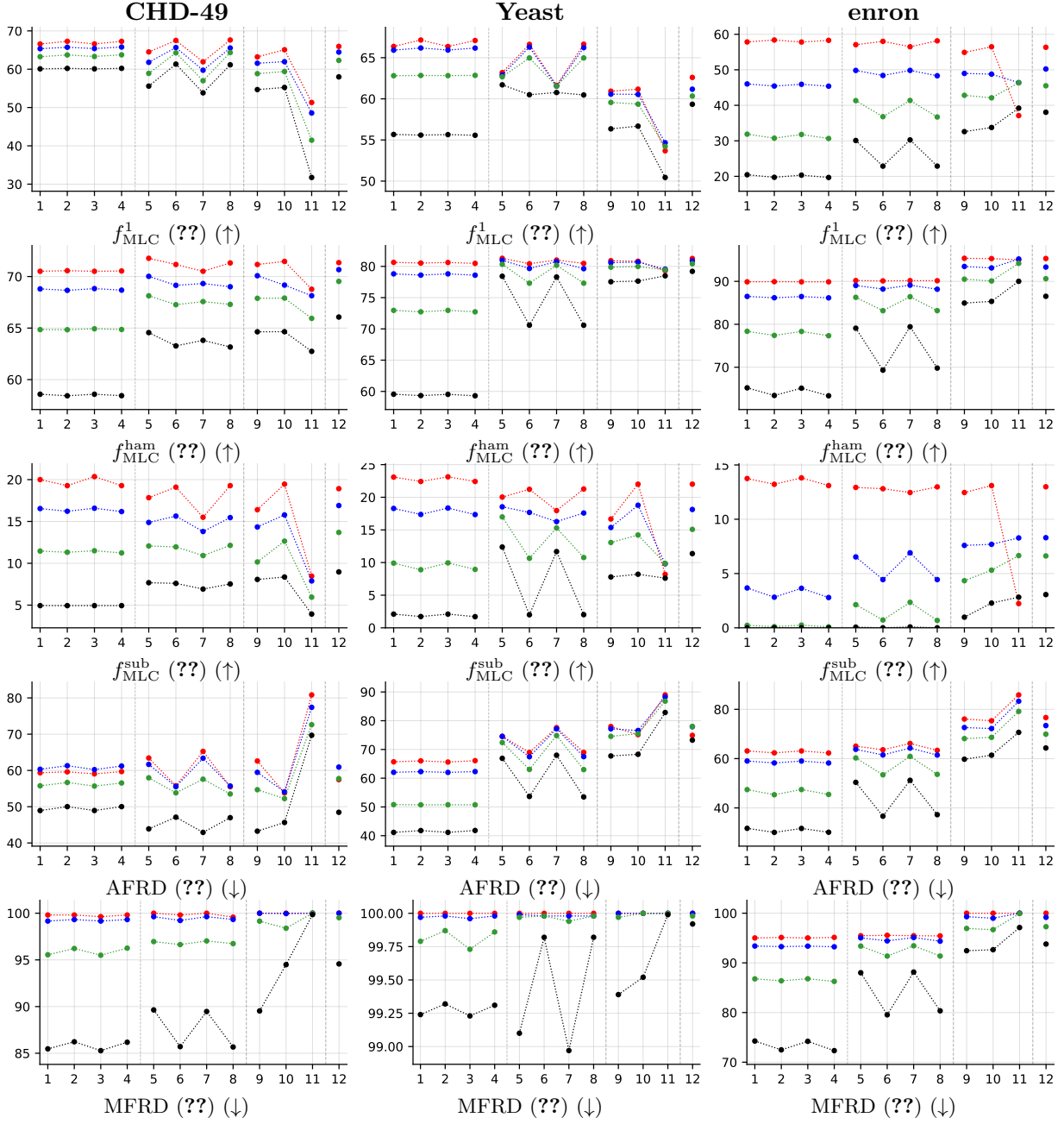


Table 22: (Table G2) Average scores (in %, y -axis) over 5×5 cross-validation folds, plotted against the encoded number of the classifiers (x -axis): (PA-H-2, PA-H, PA-S-2, PA-S), (PR-H-2, PR-H, PR-S-2, PR-S), and (BR, CC, CLR) are encoded as (1, 2, 3, 4), (5, 6, 7, 8), and (9, 10, 11), respectively. Results are color-coded with respect to the noisy level α as follows: $\alpha = 0.0$, $\alpha = 0.1$, $\alpha = 0.2$ and $\alpha = 0.3$.

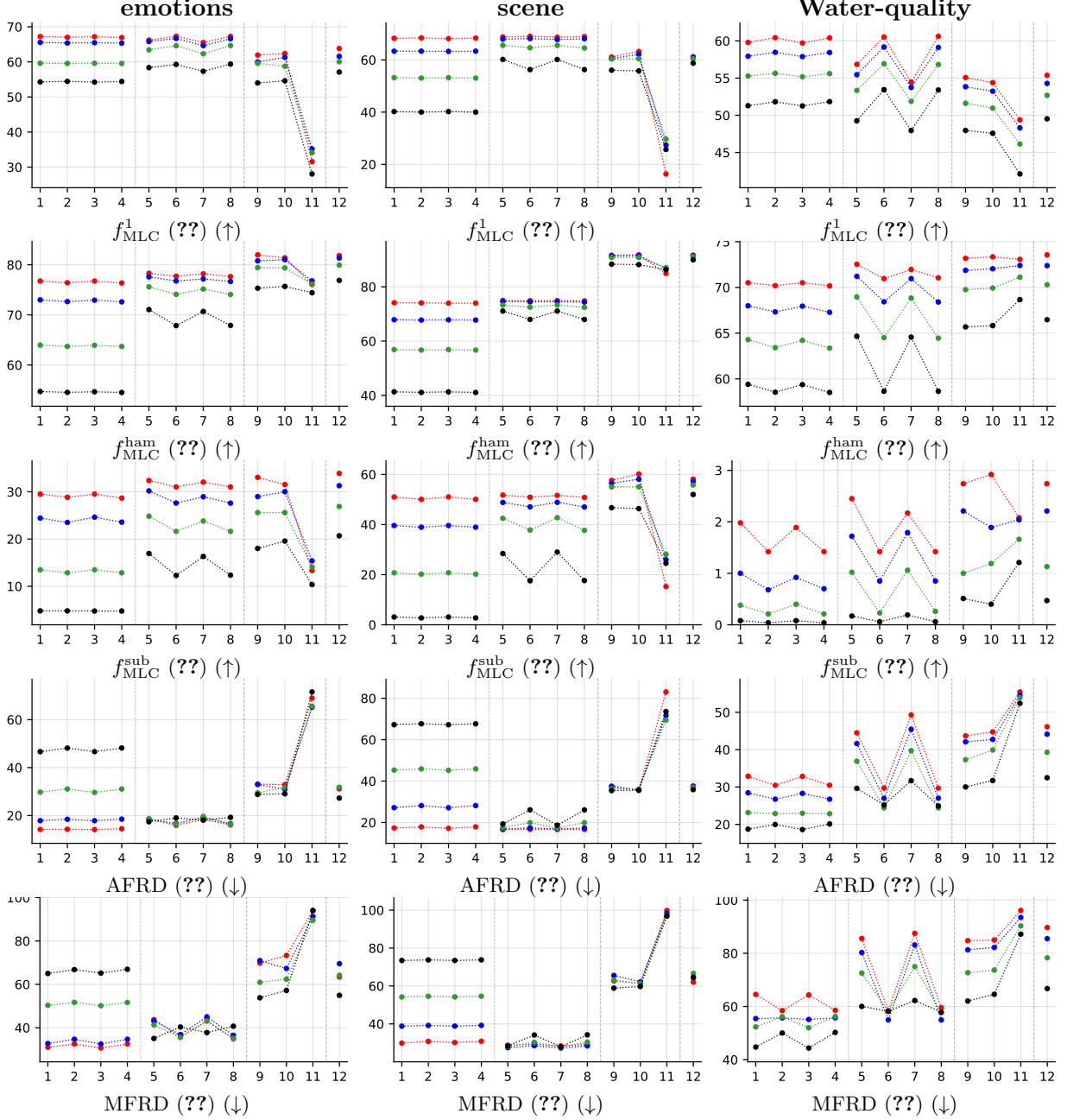


Table 23: (Table G3) Average scores (in %, y -axis) over 5×5 cross-validation folds, plotted against the encoded number of the classifiers (x -axis): (PA-H-2, PA-H, PA-S-2, PA-S), (PR-H-2, PR-H, PR-S-2, PR-S), and (BR, CC, CLR) are encoded as (1, 2, 3, 4), (5, 6, 7, 8), and (9, 10, 11), respectively. Results are color-coded with respect to the noisy level α as follows: $\alpha = 0.0$, $\alpha = 0.1$, $\alpha = 0.2$ and $\alpha = 0.3$.

A.2 Partial abstention (PartialAbstention)

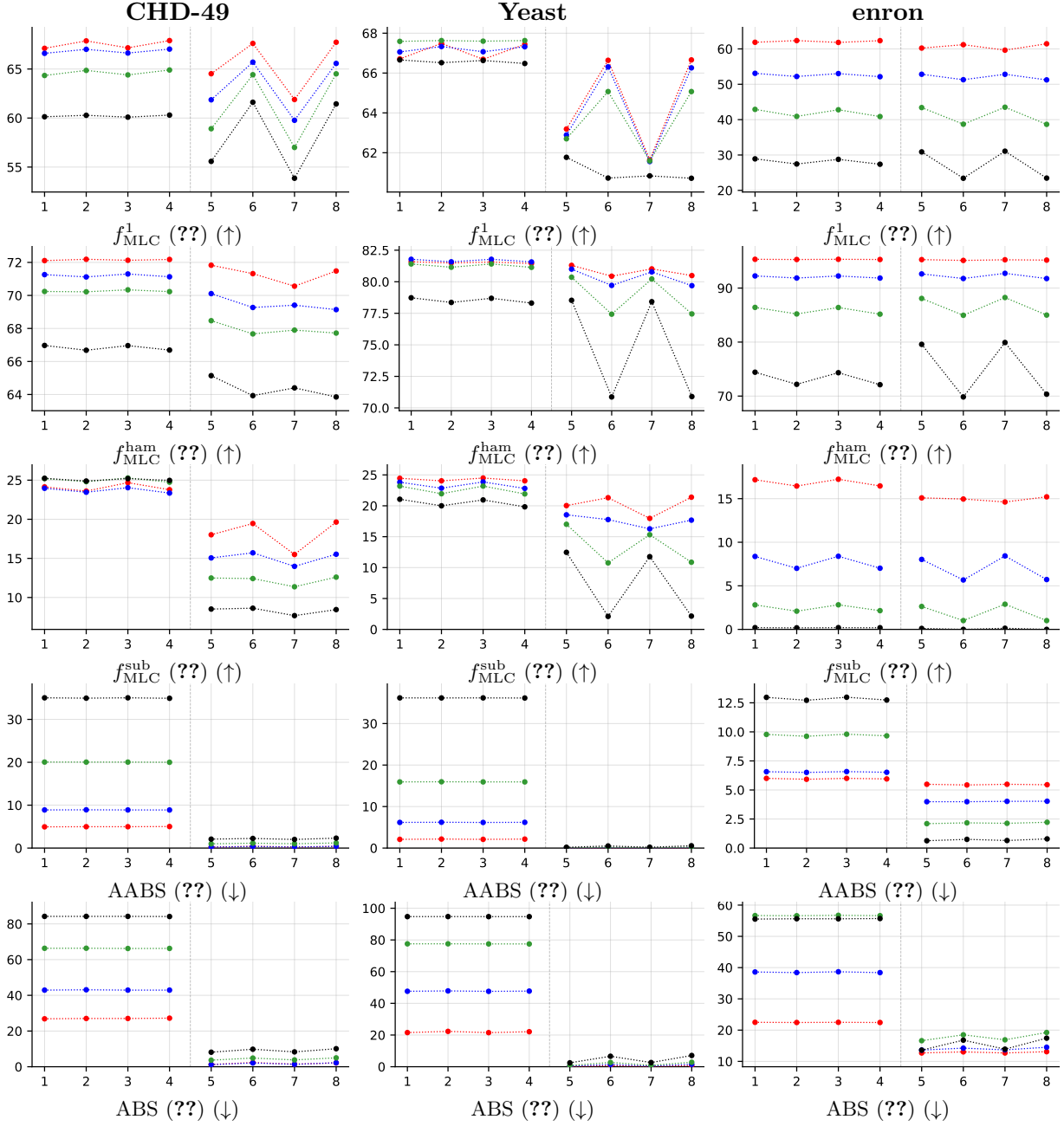


Table 24: (Table G4) Average scores (in %, y -axis) over 5×5 cross-validation folds, plotted against the encoded number of the classifiers (x -axis): (PA-H-2, PA-H, PA-S-2, PA-S) and (PR-H-2, PR-H, PR-S-2, PR-S) are encoded as (1, 2, 3, 4) and (5, 6, 7, 8), respectively. Results are color-coded with respect to the noisy level α as follows: $\alpha = 0.0$, $\alpha = 0.1$, $\alpha = 0.2$ and $\alpha = 0.3$.

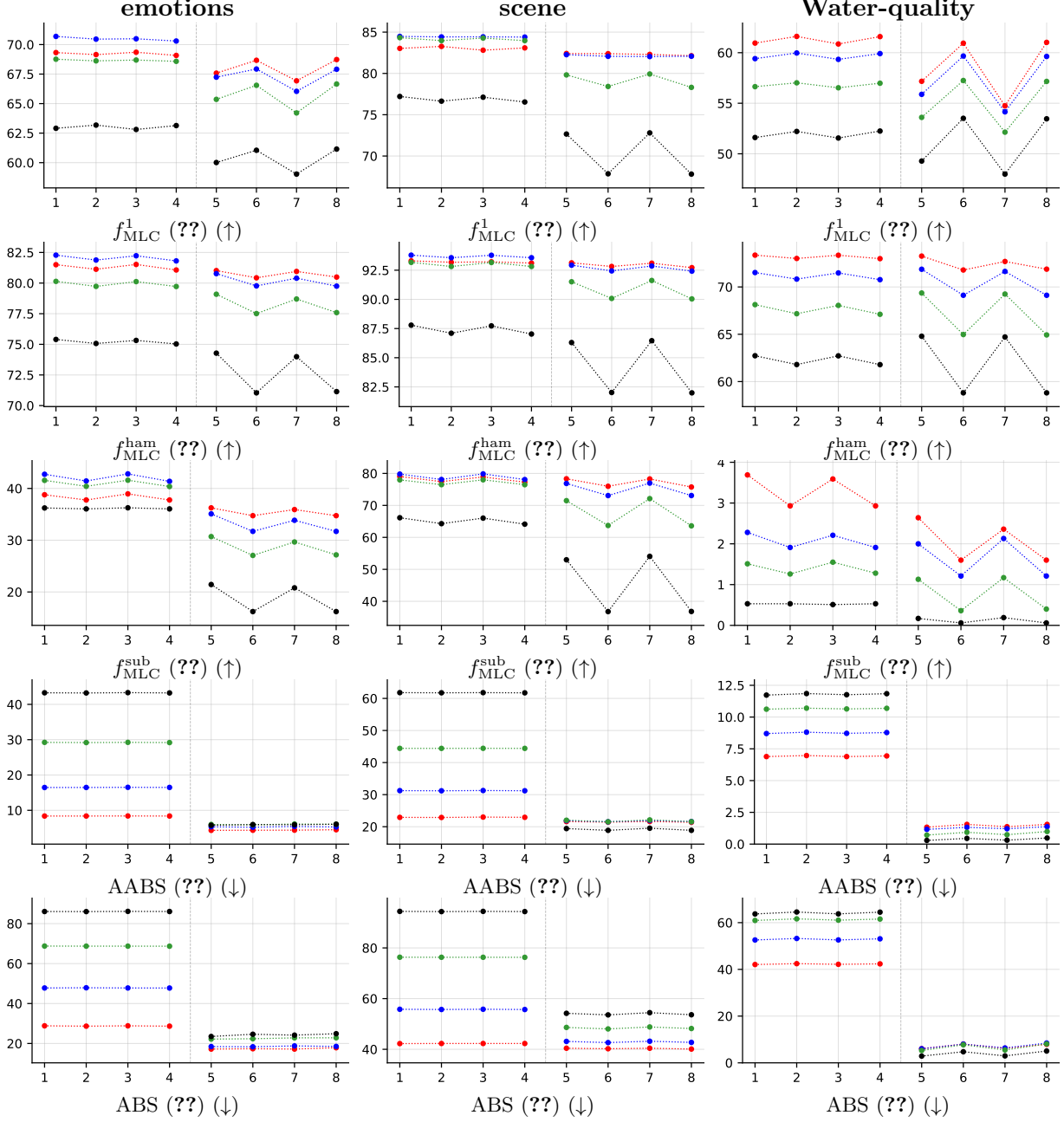


Table 25: (Table G5) Average scores (in %, y -axis) over 5×5 cross-validation folds, plotted against the encoded number of the classifiers (x -axis): (PA-H-2, PA-H, PA-S-2, PA-S) and (PR-H-2, PR-H, PR-S-2, PR-S) are encoded as (1, 2, 3, 4) and (5, 6, 7, 8), respectively. Results are color-coded with respect to the noisy level α as follows: $\alpha = 0.0$, $\alpha = 0.1$, $\alpha = 0.2$ and $\alpha = 0.3$.

A.3 Standard MLC predictions (BinaryVector)

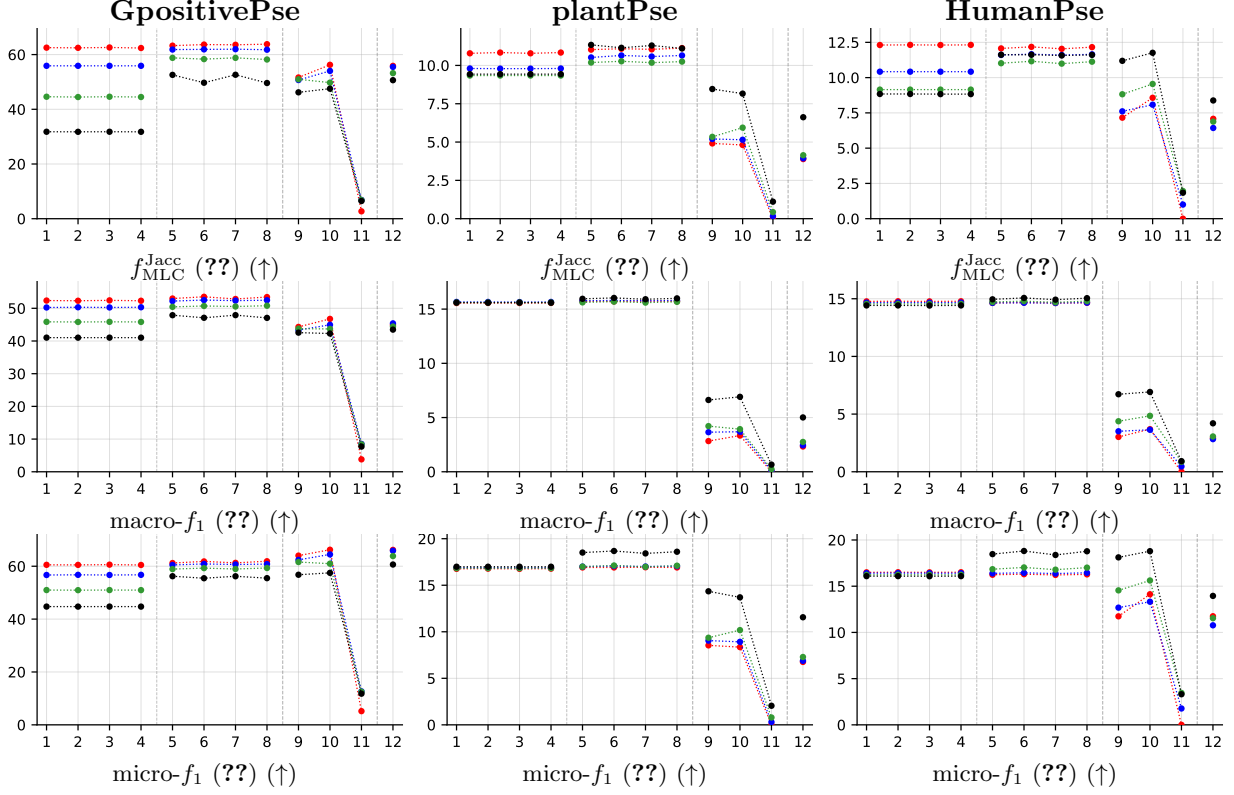


Table 26: (Table G6, new metrics — main datasets) Average scores (in %, y -axis) over 5×5 cross-validation folds, plotted against the encoded number of the classifiers (x -axis): (PA-H-2, PA-H, PA-S-2, PA-S), (PR-H-2, PR-H, PR-S-2, PR-S), and (BR, CC, CLR) are encoded as (1, 2, 3, 4), (5, 6, 7, 8), and (9, 10, 11), respectively. Results are color-coded with respect to the noisy level α as follows: $\alpha = 0.0$, $\alpha = 0.1$, $\alpha = 0.2$ and $\alpha = 0.3$.

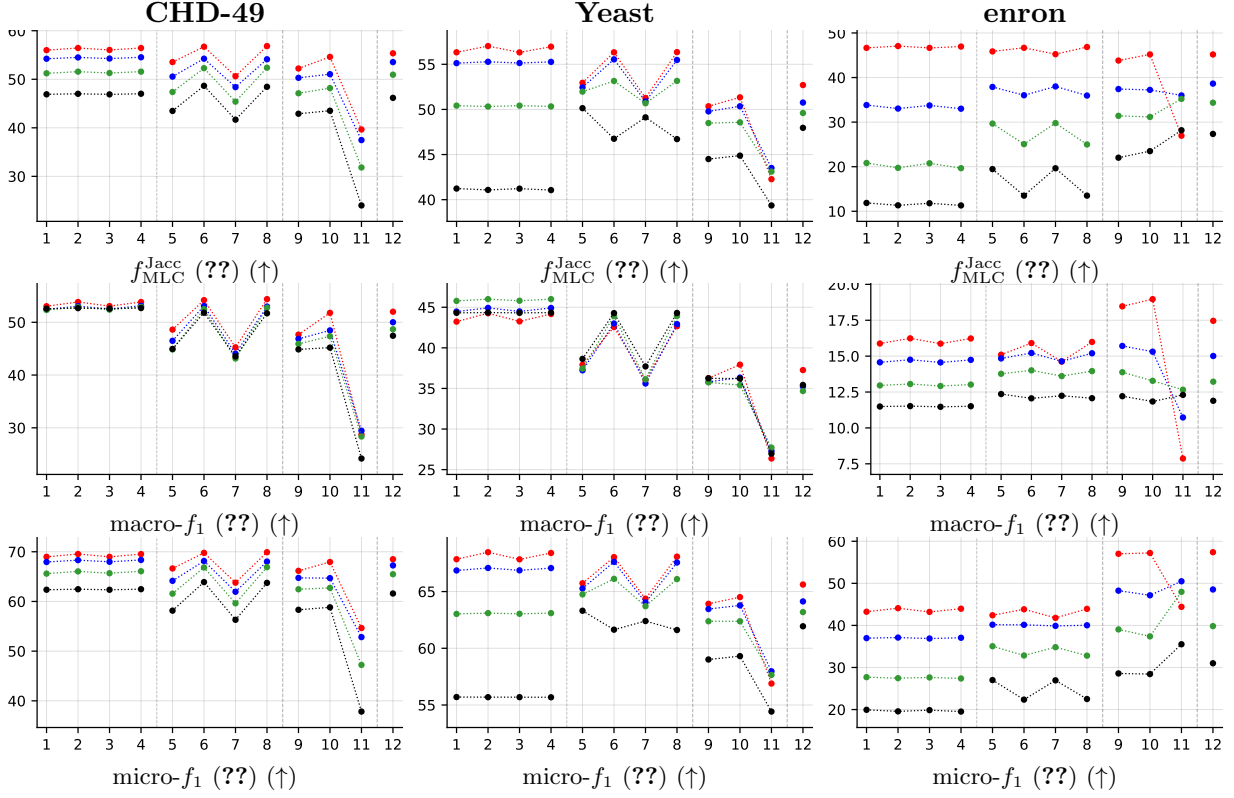


Table 27: (Table G6, new metrics — imbalanced) Average scores (in %, y -axis) over 5×5 cross-validation folds, plotted against the encoded number of the classifiers (x -axis): (PA-H-2, PA-H, PA-S-2, PA-S), (PR-H-2, PR-H, PR-S-2, PR-S), and (BR, CC, CLR) are encoded as (1, 2, 3, 4), (5, 6, 7, 8), and (9, 10, 11), respectively. Results are color-coded with respect to the noisy level α as follows: $\alpha = 0.0$, $\alpha = 0.1$, $\alpha = 0.2$ and $\alpha = 0.3$.

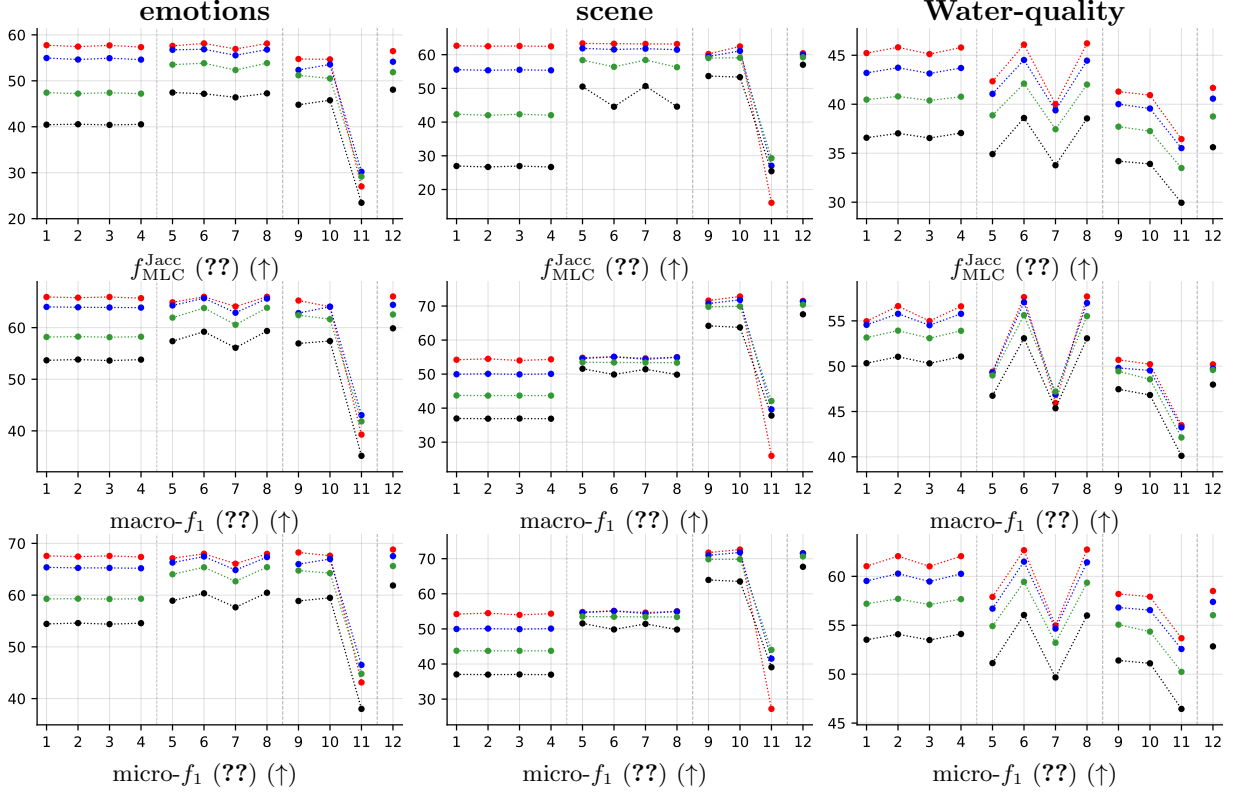


Table 28: (Table G6, new metrics — balanced) Average scores (in %, y -axis) over 5×5 cross-validation folds, plotted against the encoded number of the classifiers (x -axis): (PA-H-2, PA-H, PA-S-2, PA-S), (PR-H-2, PR-H, PR-S-2, PR-S), and (BR, CC, CLR) are encoded as (1, 2, 3, 4), (5, 6, 7, 8), and (9, 10, 11), respectively. Results are color-coded with respect to the noisy level α as follows: $\alpha = 0.0$, $\alpha = 0.1$, $\alpha = 0.2$ and $\alpha = 0.3$.

A.4 Partial abstention (PartialAbstention)

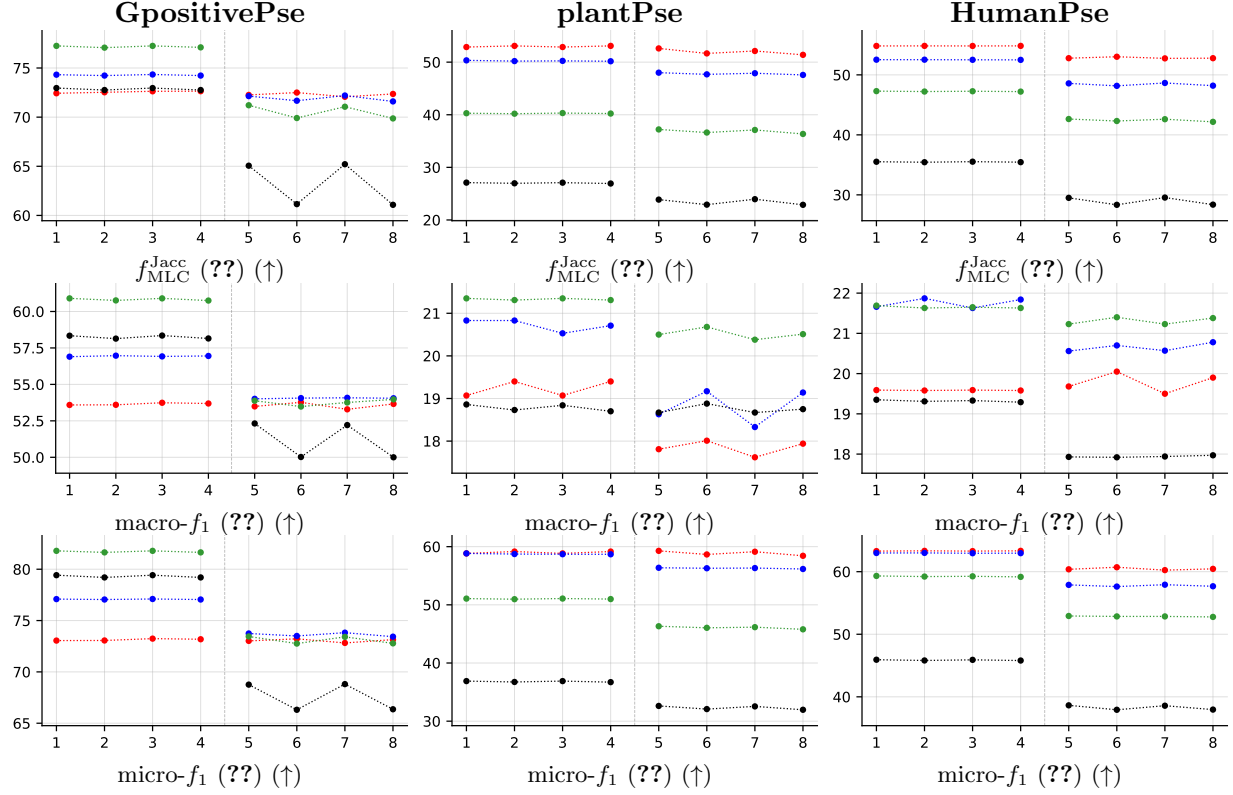


Table 29: (Table G7, new metrics — main datasets) Average scores (in %, y -axis) over 5×5 cross-validation folds, plotted against the encoded number of the classifiers (x -axis): (PA-H-2, PA-H, PA-S-2, PA-S) and (PR-H-2, PR-H, PR-S-2, PR-S) are encoded as (1, 2, 3, 4) and (5, 6, 7, 8), respectively. Results are color-coded with respect to the noisy level α as follows: $\alpha = 0.0$, $\alpha = 0.1$, $\alpha = 0.2$ and $\alpha = 0.3$.

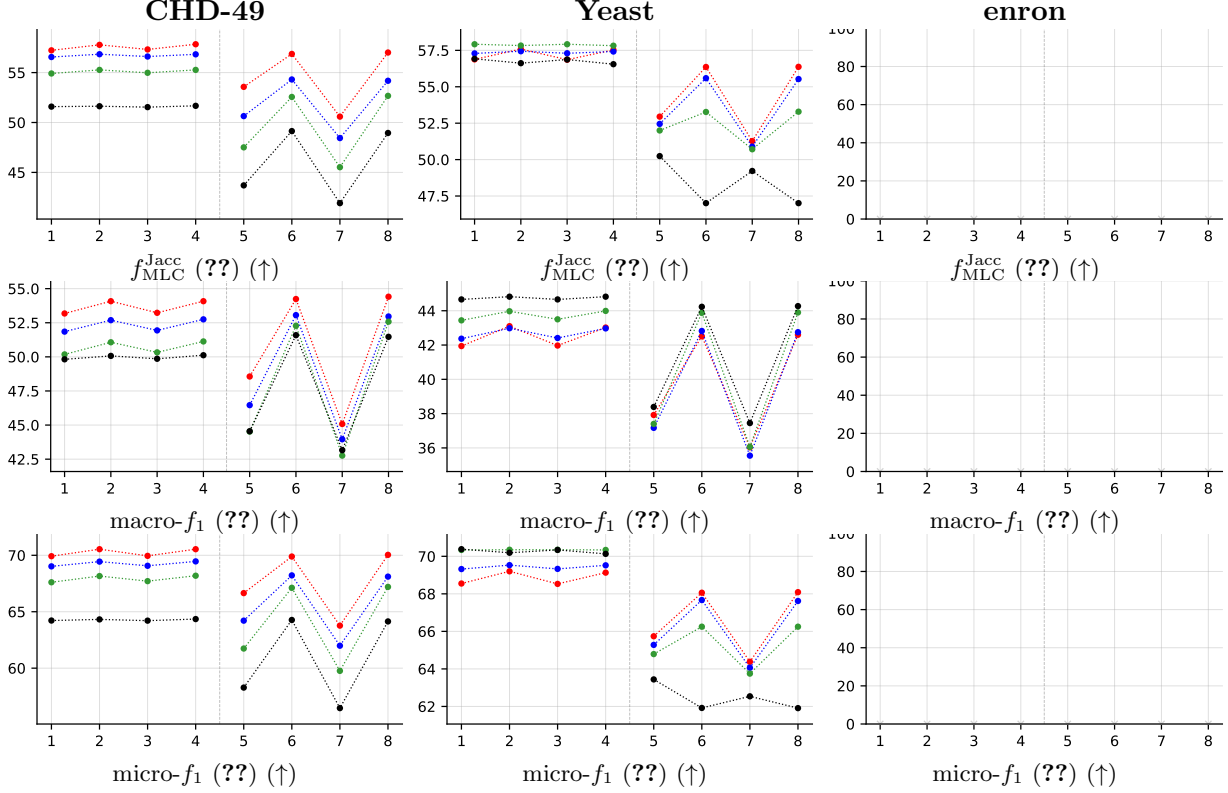


Table 30: (Table G7, new metrics — imbalanced) Average scores (in %, y -axis) over 5×5 cross-validation folds, plotted against the encoded number of the classifiers (x -axis): (PA-H-2, PA-H, PA-S-2, PA-S) and (PR-H-2, PR-H, PR-S-2, PR-S) are encoded as (1, 2, 3, 4) and (5, 6, 7, 8), respectively. Results are color-coded with respect to the noisy level α as follows: $\alpha = 0.0$, $\alpha = 0.1$, $\alpha = 0.2$ and $\alpha = 0.3$.

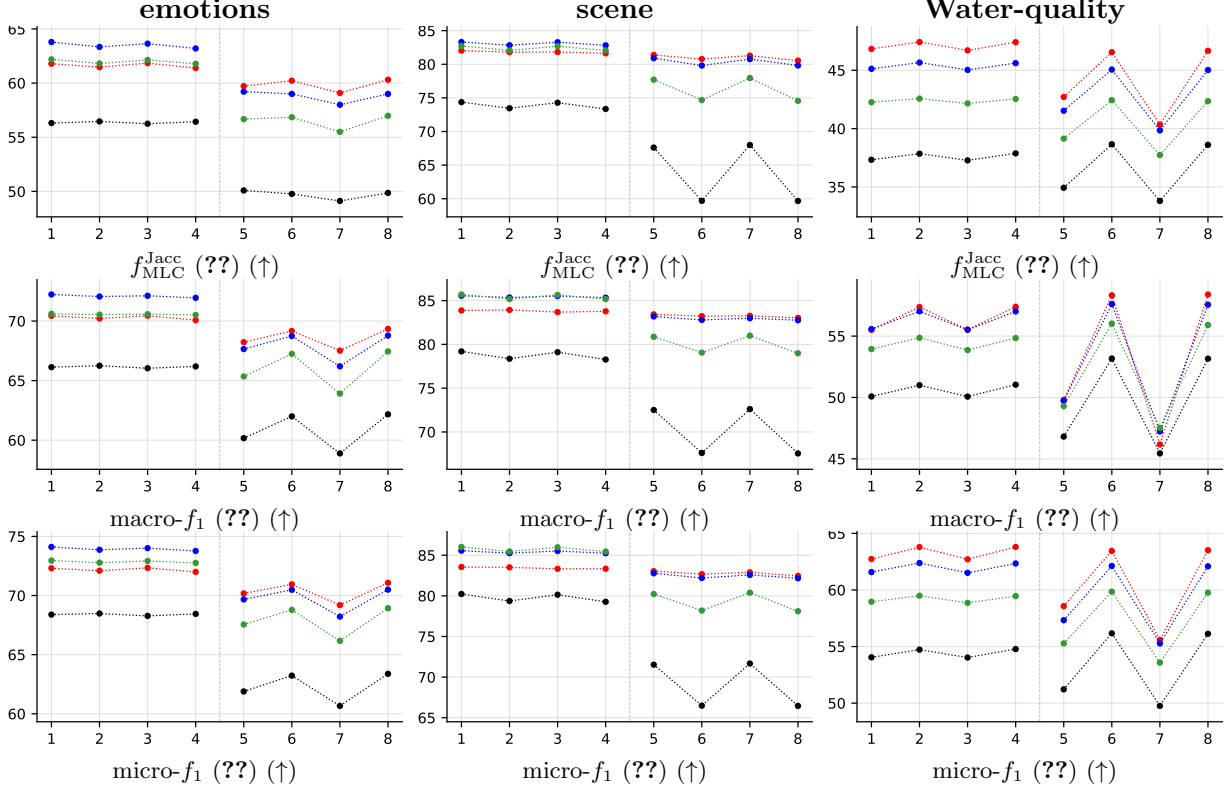


Table 31: (Table G7, new metrics — balanced) Average scores (in %, y -axis) over 5×5 cross-validation folds, plotted against the encoded number of the classifiers (x -axis): (PA-H-2, PA-H, PA-S-2, PA-S) and (PR-H-2, PR-H, PR-S-2, PR-S) are encoded as (1, 2, 3, 4) and (5, 6, 7, 8), respectively. Results are color-coded with respect to the noisy level α as follows: $\alpha = 0.0$, $\alpha = 0.1$, $\alpha = 0.2$ and $\alpha = 0.3$.

A.5 Score-vector ranking metrics (AUROC, AUPRC, etc.) (ScoreVector)

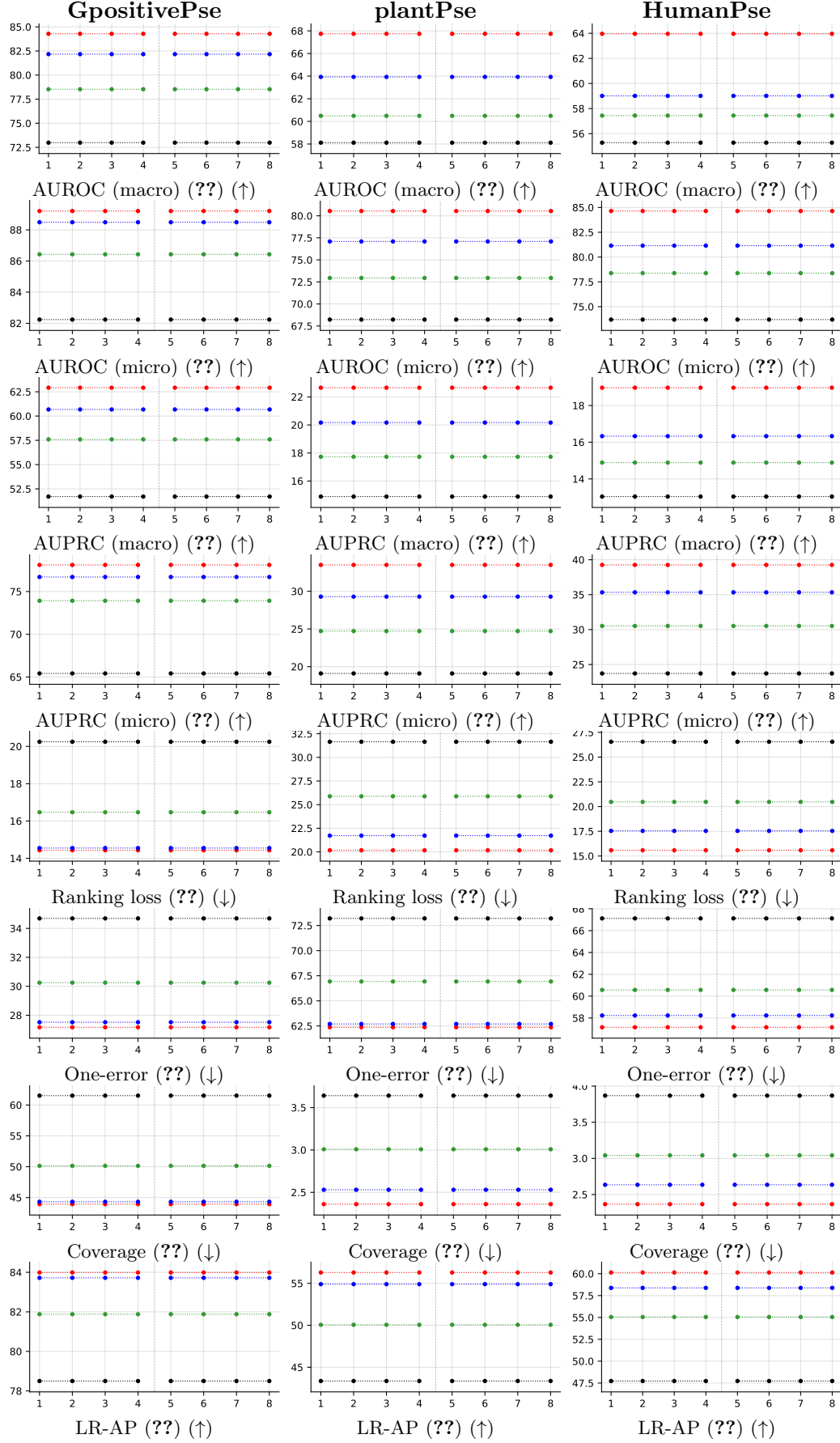


Table 32: (Table G8, new — main datasets) Average scores (in %, y -axis) over 5×5 cross-validation folds, plotted against the encoded number of the classifiers (x -axis): (PA-H-2, PA-H, PA-S-2, PA-S) and (PR-H-2, PR-H, PR-S-2, PR-S) are encoded as (1, 2, 3, 4) and (5, 6, 7, 8), respectively. Results

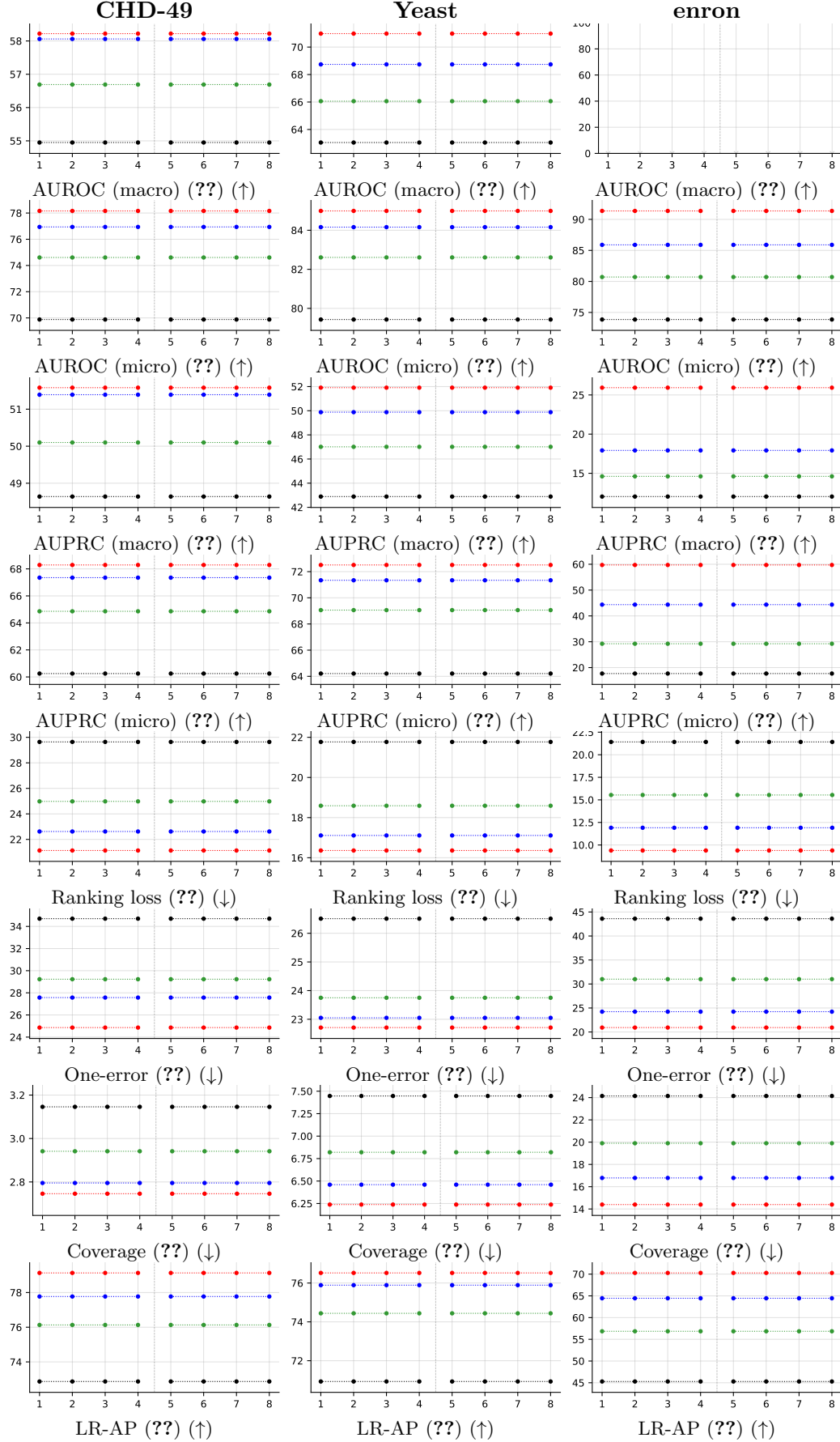


Table 33: (Table G8, new — imbalanced) Average scores (in %, y -axis) over 5×5 cross-validation folds, plotted against the encoded number of the classifiers (x -axis): (PA-H-2, PA-H, PA-S-2, PA-S) and (PR-H-2, PR-H, PR-S-2, PR-S) are encoded as (1, 2, 3, 4) and (5, 6, 7, 8), respectively. Results

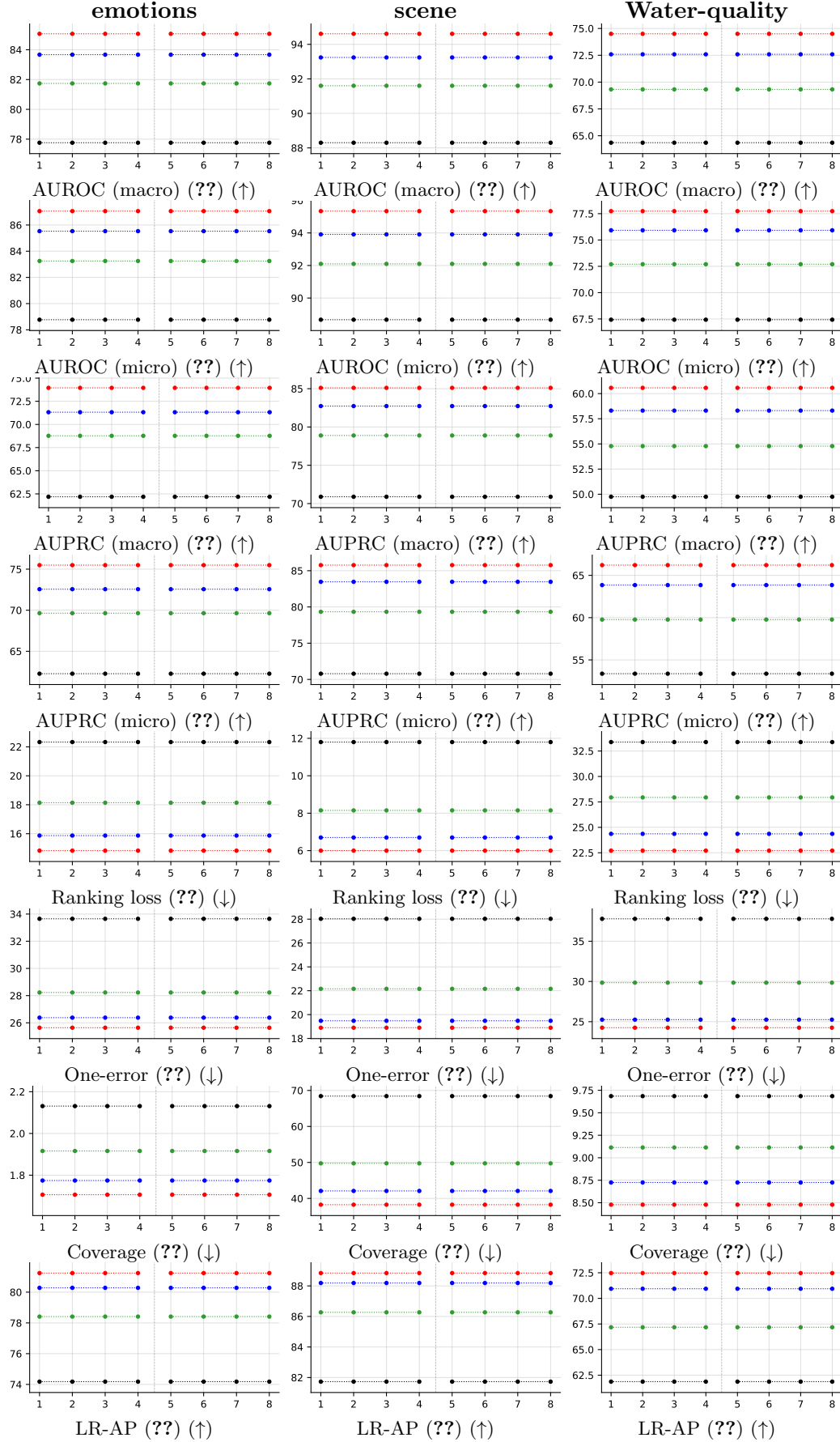


Table 34: (Table G8, new — balanced) Average scores (in %, y -axis) over 5×5 cross-validation folds, plotted against the encoded number of the classifiers (x -axis): (PA-H-2, PA-H, PA-S-2, PA-S) and (PR-H-2, PR-H, PR-S-2, PR-S) are encoded as (1, 2, 3, 4) and (5, 6, 7, 8), respectively. Results

B Per-dataset results (RF base learner)

B.1 GpositivePseAAC ($K = 4$)

RF • GpositivePseAAC • BinaryVector — Standard MLC predictions

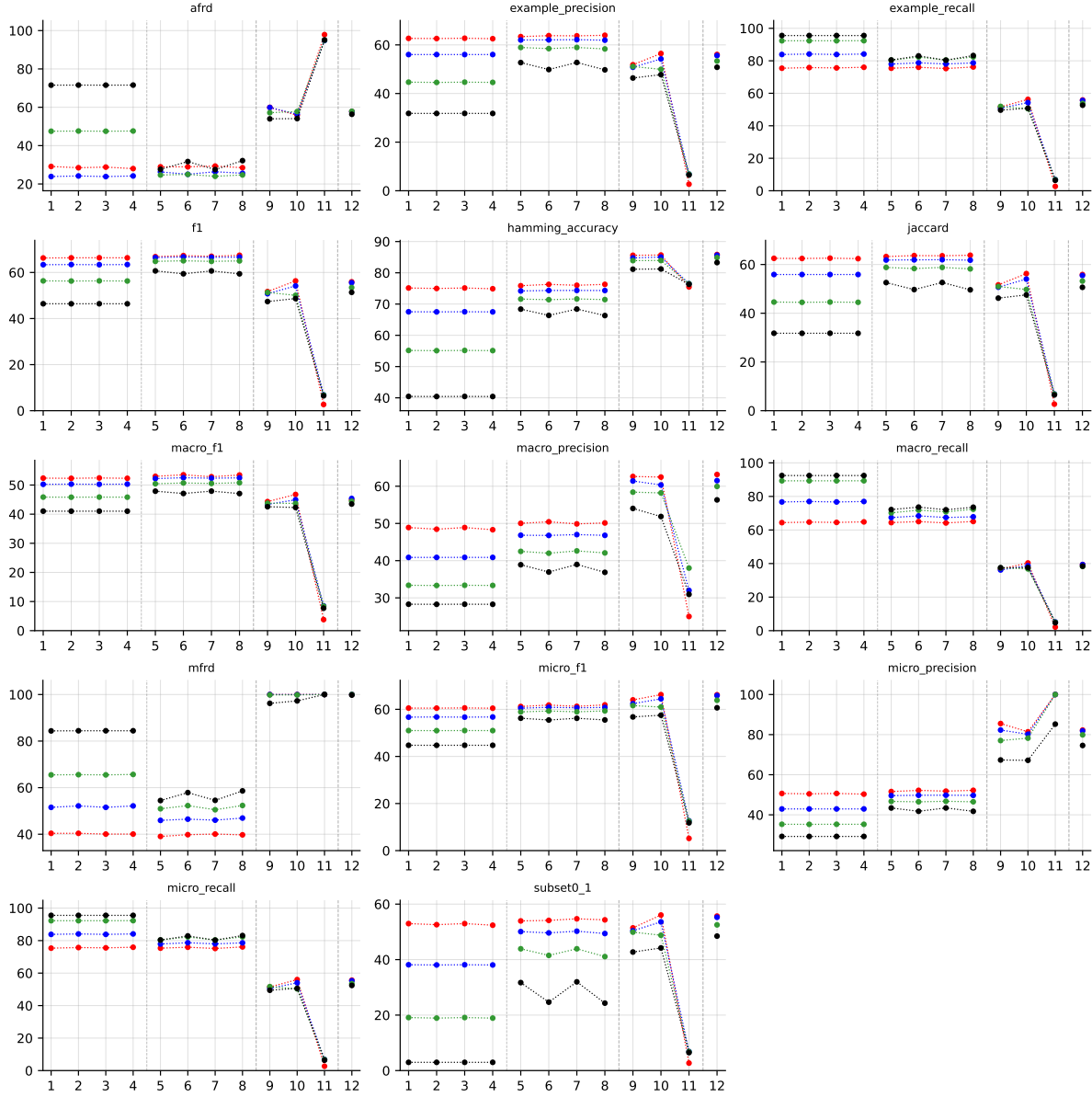


Table 35: Standard MLC predictions (BinaryVector) on GpositivePseAAC (RF base learner).

RF • GpositivePseAAC • PartialAbstention — Partial abstention

RF • GpositivePseAAC • ScoreVector — Score-vector ranking metrics (AUROC, AUPRC, etc.)

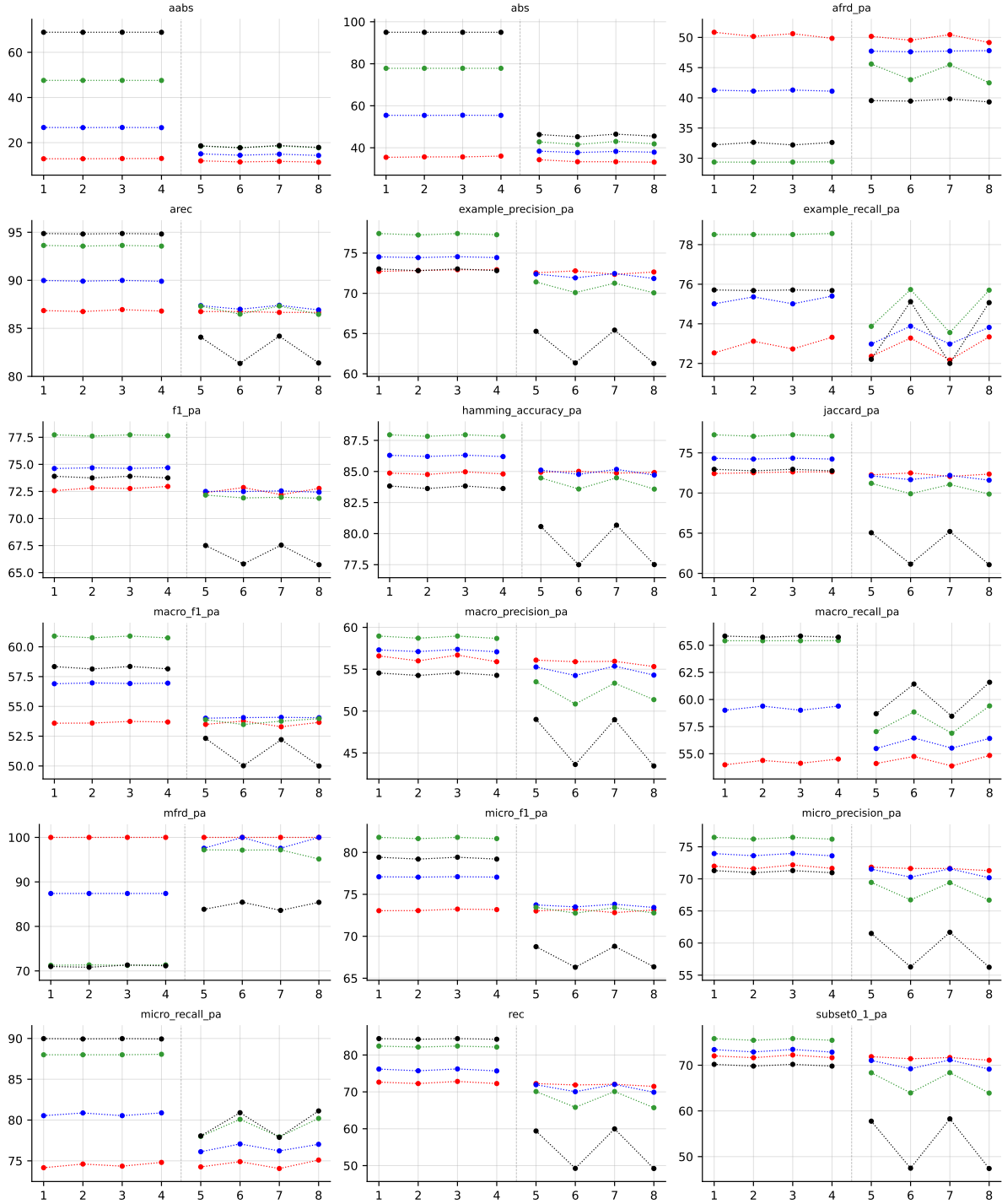


Table 36: Partial abstention (PartialAbstention) on GpositivePseAAC (RF base learner).

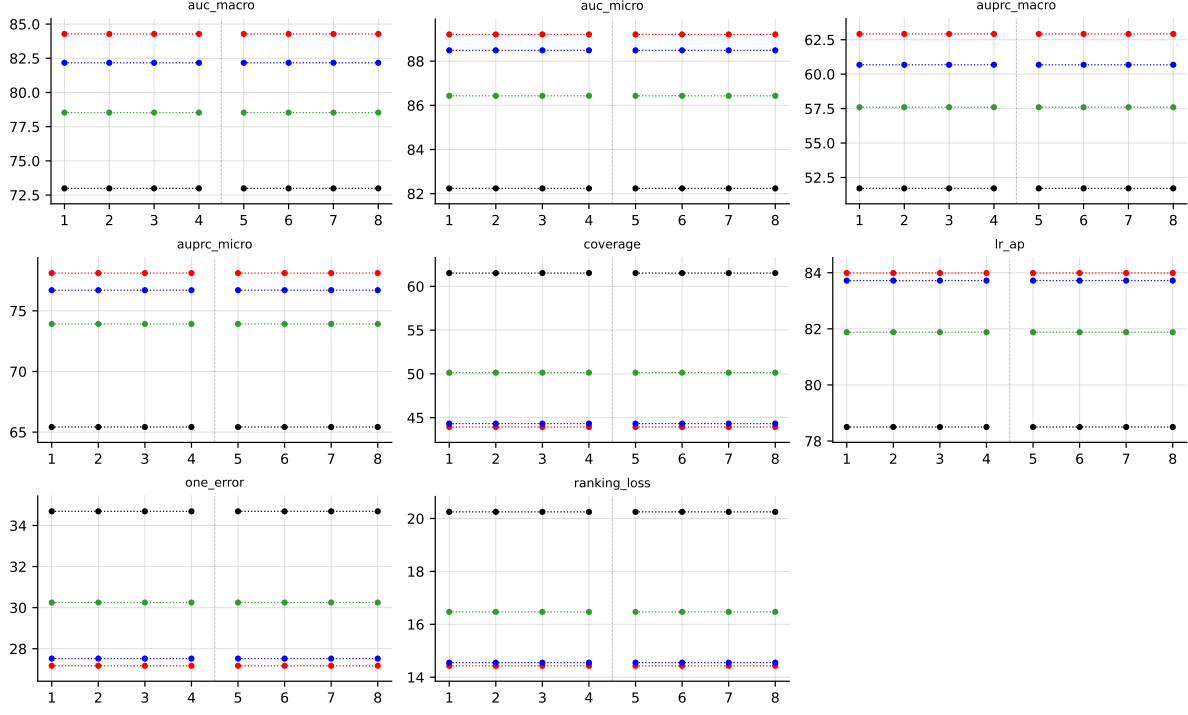


Table 37: Score-vector ranking metrics (AUROC, AUPRC, etc.) (ScoreVector) on GpositivePseAAC (RF base learner).

B.2 Emotions ($K = 6$)

RF • emotions • BinaryVector — Standard MLC predictions

RF • emotions • PartialAbstention — Partial abstention

RF • emotions • ScoreVector — Score-vector ranking metrics (AUROC, AUPRC, etc.)

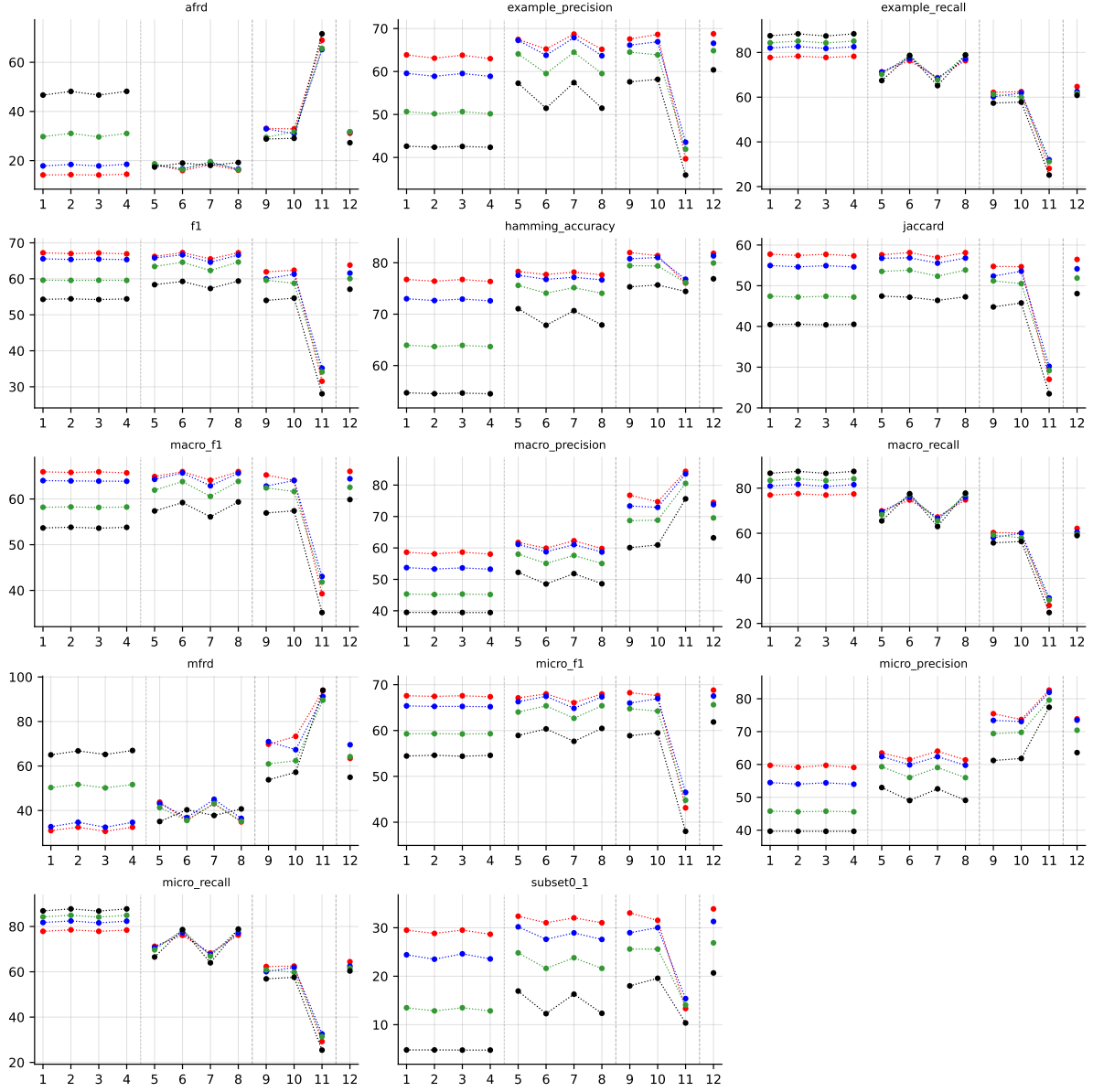


Table 38: Standard MLC predictions (BinaryVector) on emotions (RF base learner).

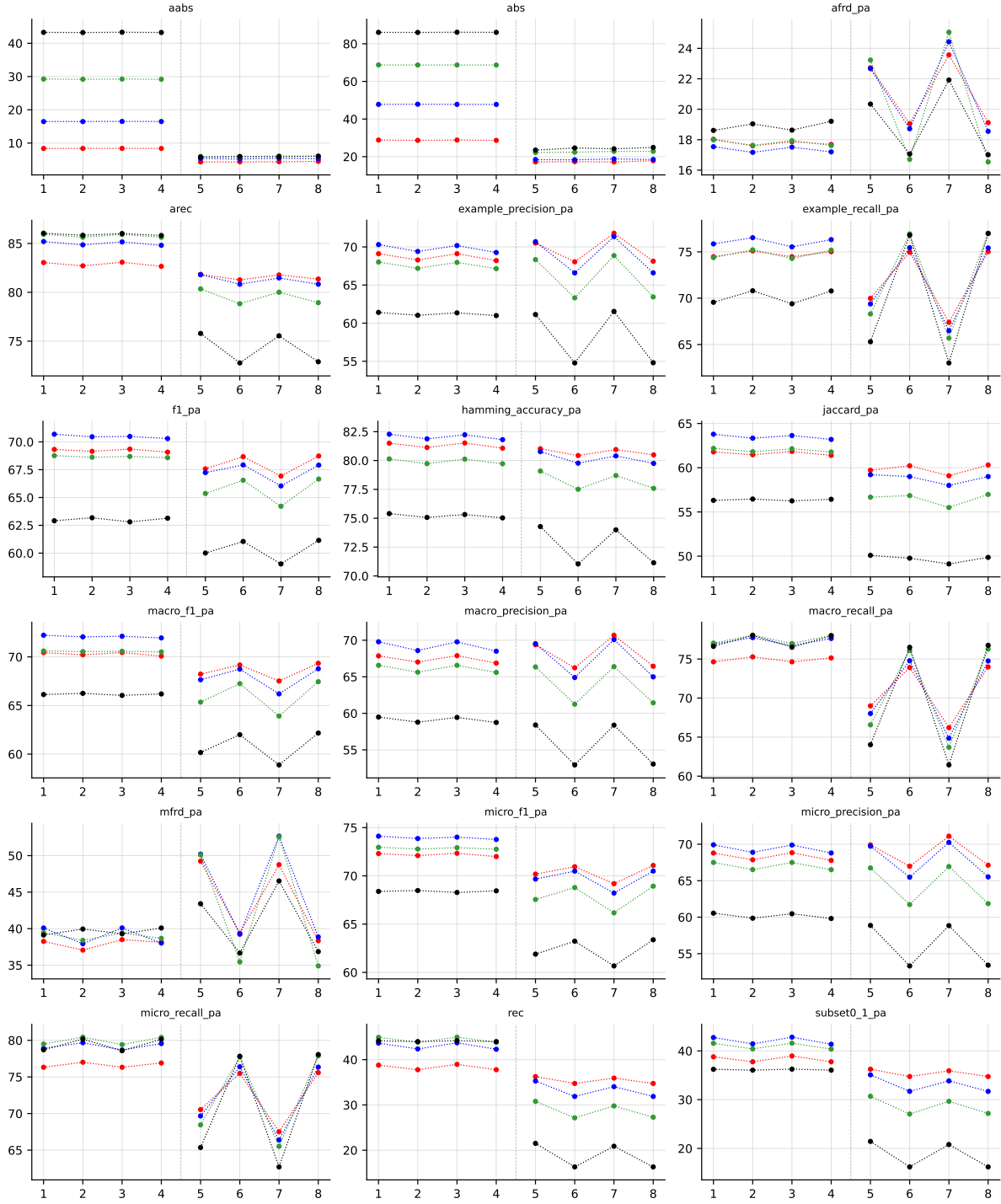


Table 39: Partial abstention (PartialAbstention) on emotions (RF base learner).

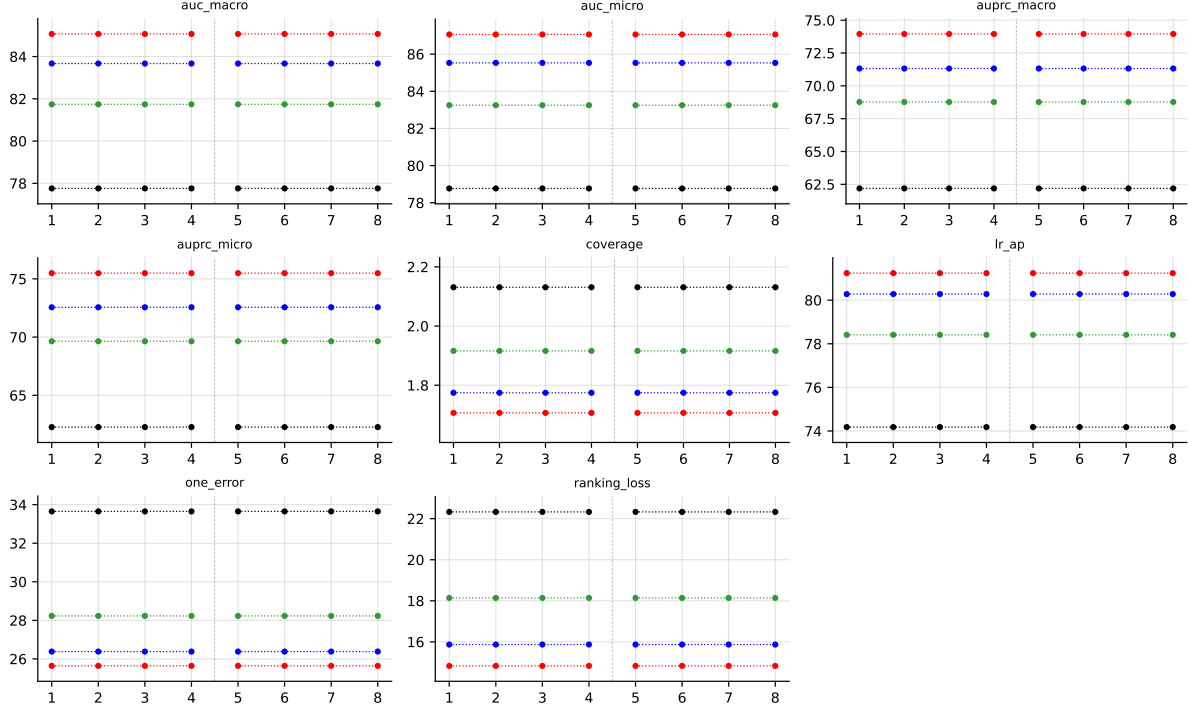


Table 40: Score-vector ranking metrics (AUROC, AUPRC, etc.) (ScoreVector) on **emotions** (RF base learner).

B.3 Scene ($K = 6$)

RF • scene • BinaryVector — Standard MLC predictions

RF • scene • PartialAbstention — Partial abstention

RF • scene • ScoreVector — Score-vector ranking metrics (AUROC, AUPRC, etc.)

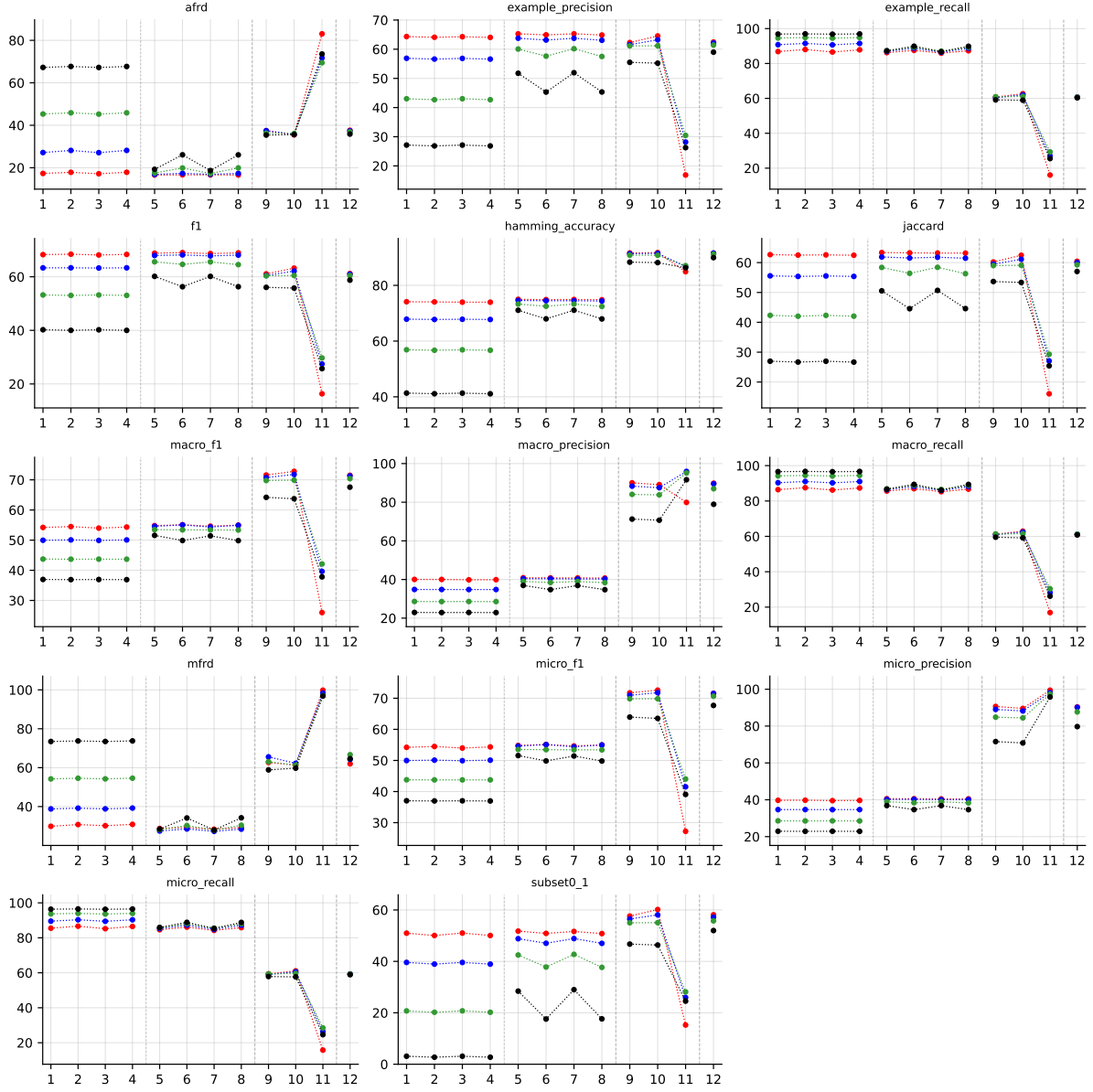


Table 41: Standard MLC predictions (BinaryVector) on scene (RF base learner).

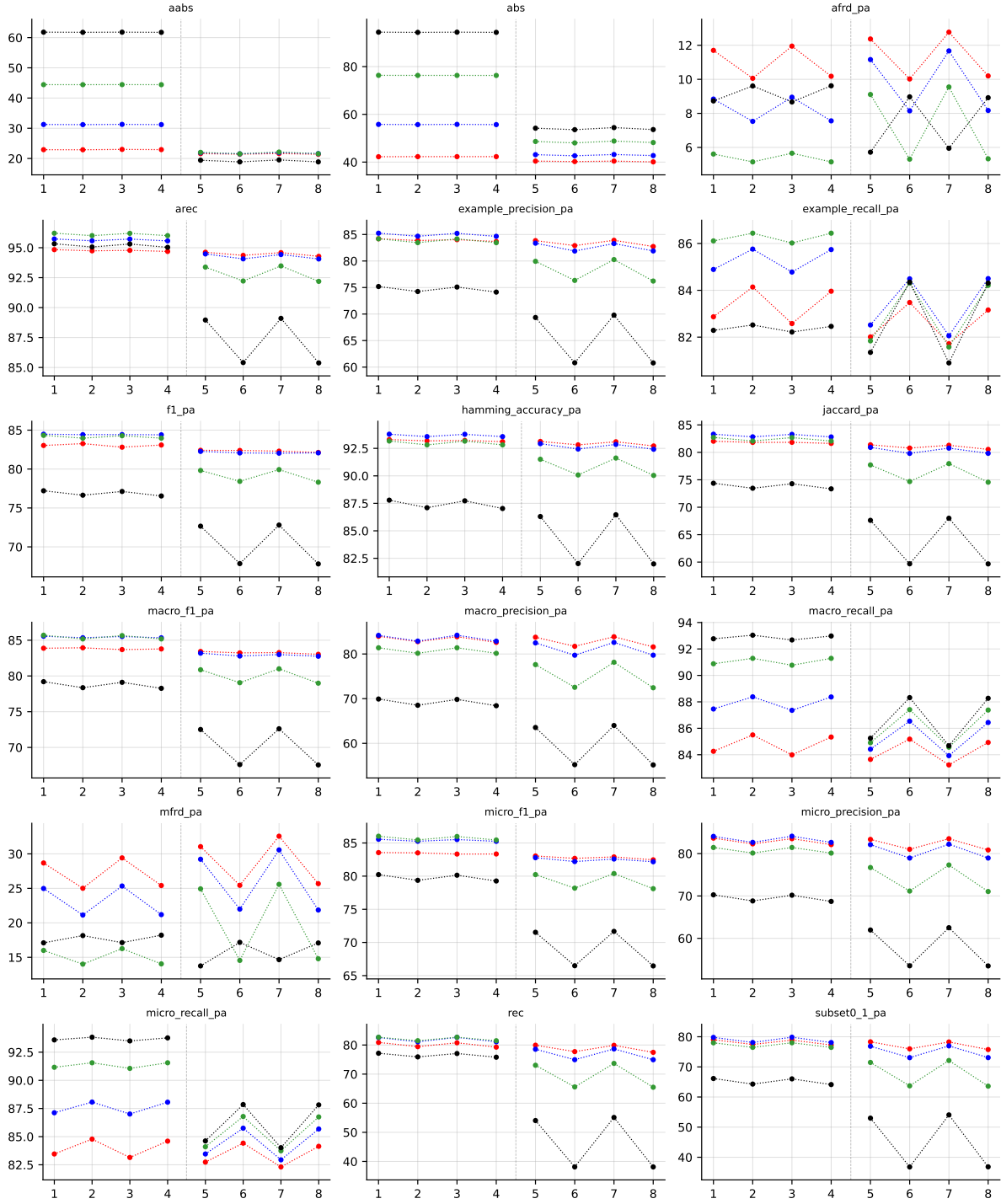


Table 42: Partial abstention (PartialAbstention) on scene (RF base learner).

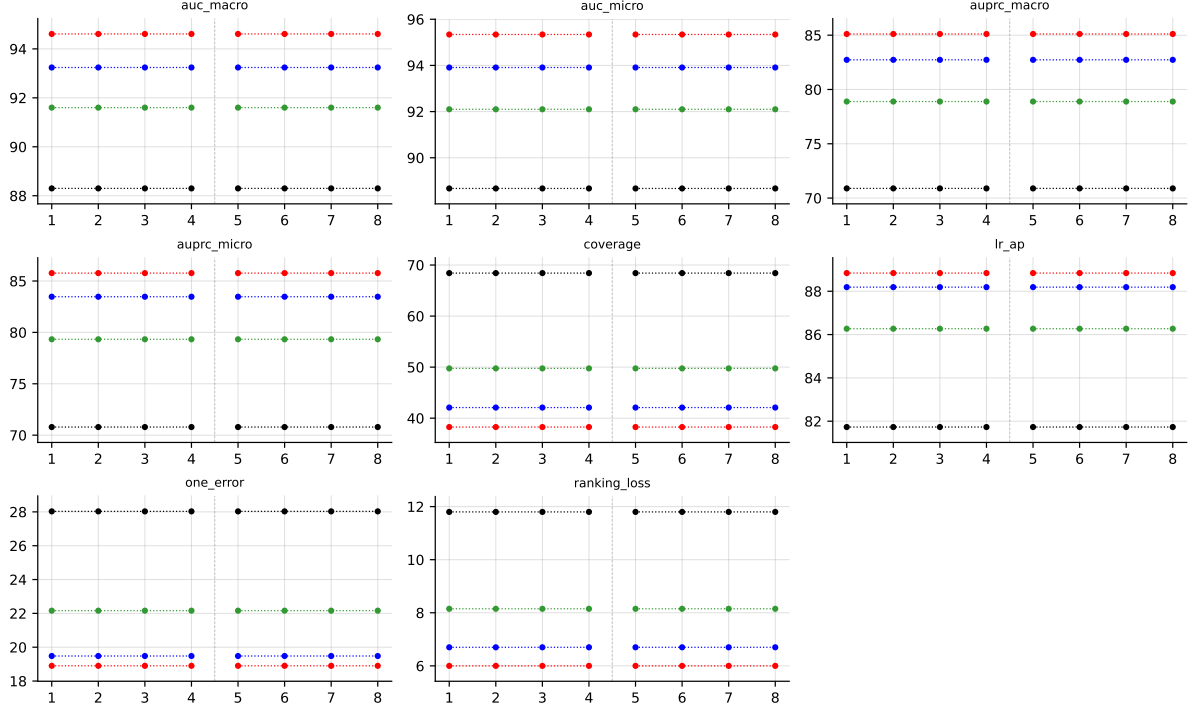


Table 43: Score-vector ranking metrics (AUROC, AUPRC, etc.) (ScoreVector) on `scene` (RF base learner).

B.4 PlantPseAAC ($K = 12$)

RF • PlantPseAAC • BinaryVector — Standard MLC predictions

RF • PlantPseAAC • PartialAbstention — Partial abstention

RF • PlantPseAAC • ScoreVector — Score-vector ranking metrics (AUROC, AUPRC, etc.)

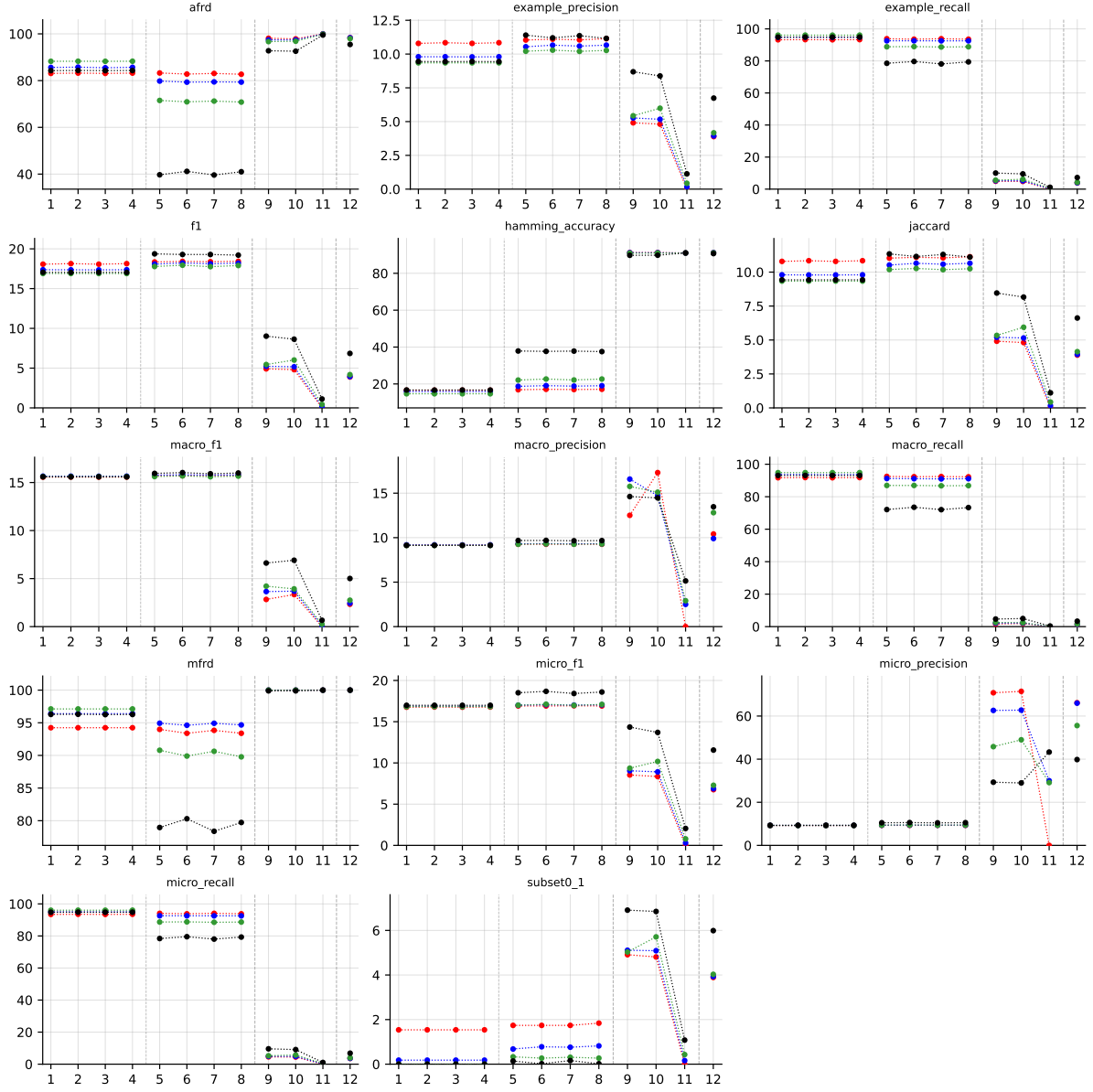


Table 44: Standard MLC predictions (BinaryVector) on PlantPseAAC (RF base learner).

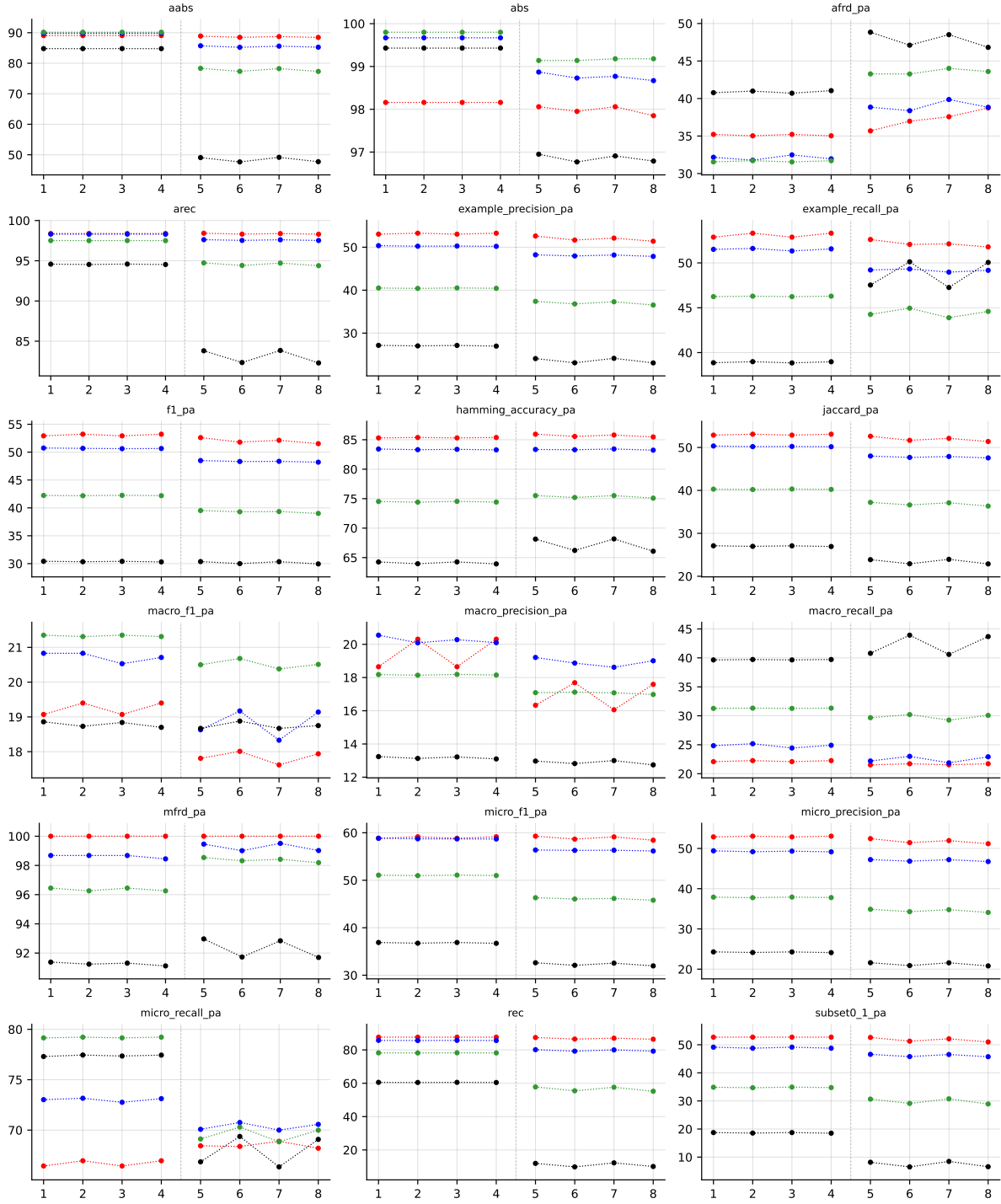


Table 45: Partial abstention (PartialAbstention) on PlantPseAAC (RF base learner).

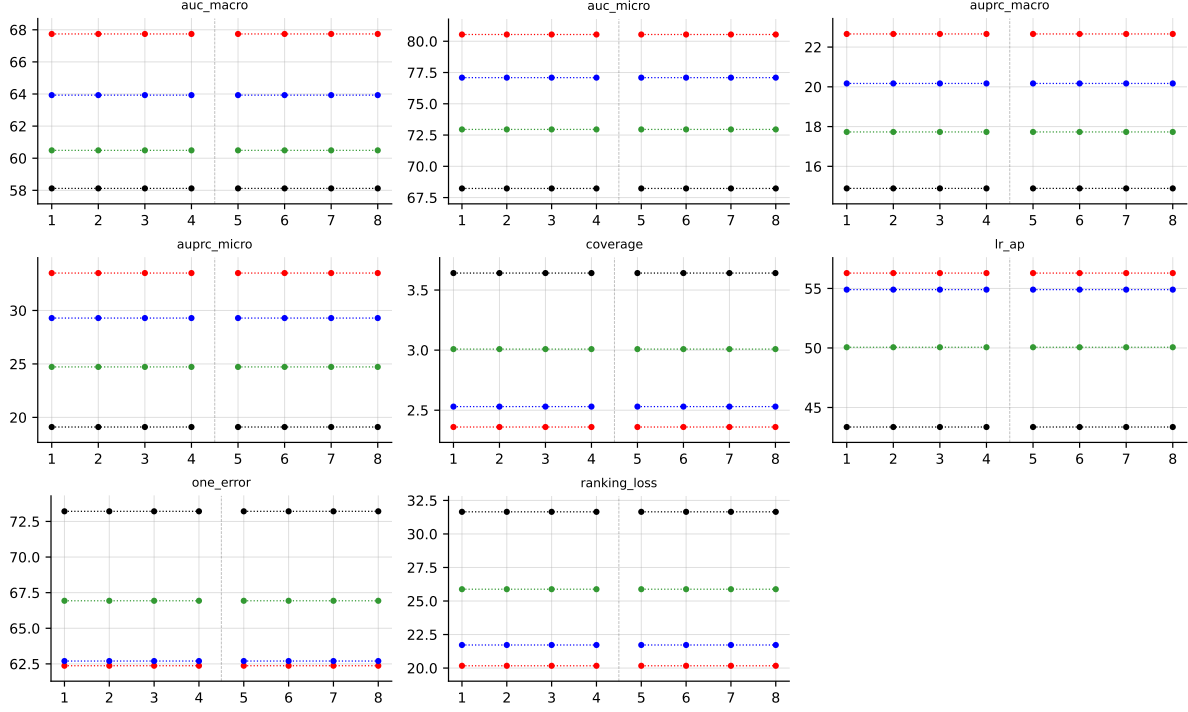


Table 46: Score-vector ranking metrics (AUROC, AUPRC, etc.) (ScoreVector) on PlantPseAAC (RF base learner).

B.5 HumanPseAAC ($K = 14$)

RF • HumanPseAAC • BinaryVector — Standard MLC predictions

RF • HumanPseAAC • PartialAbstention — Partial abstention

RF • HumanPseAAC • ScoreVector — Score-vector ranking metrics (AUROC, AUPRC, etc.)

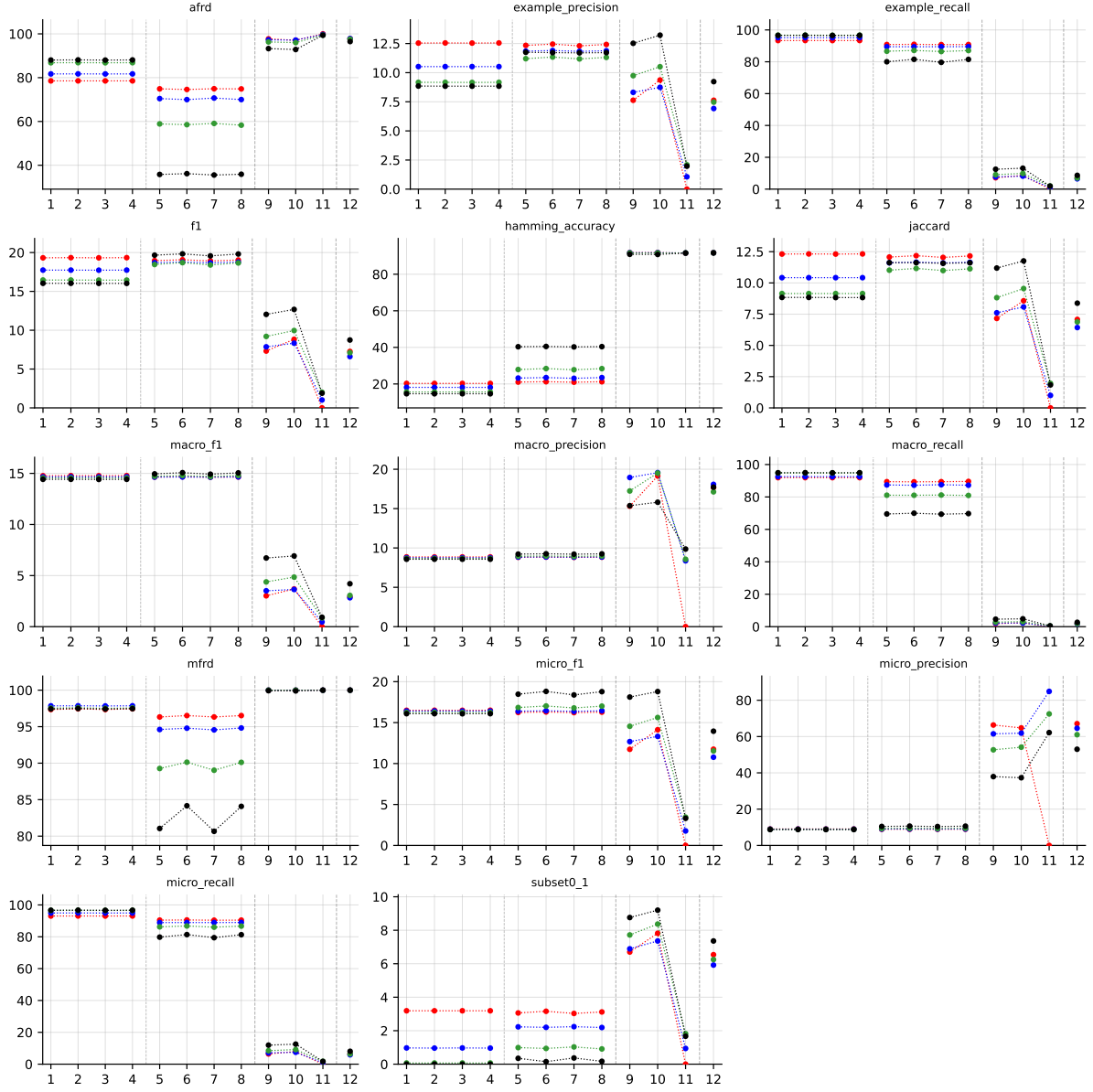


Table 47: Standard MLC predictions (BinaryVector) on HumanPseAAC (RF base learner).

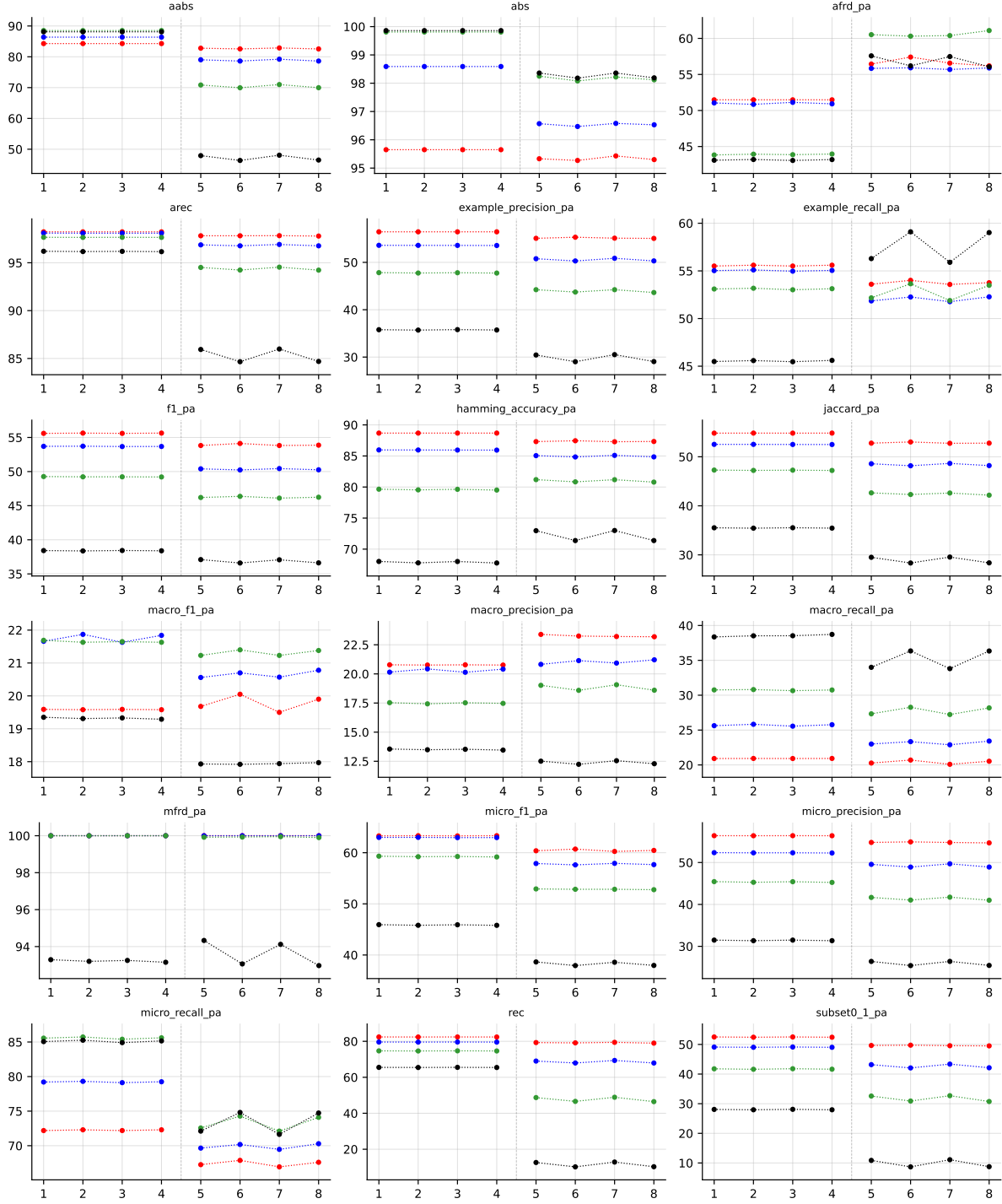


Table 48: Partial abstention (PartialAbstention) on HumanPseAAC (RF base learner).

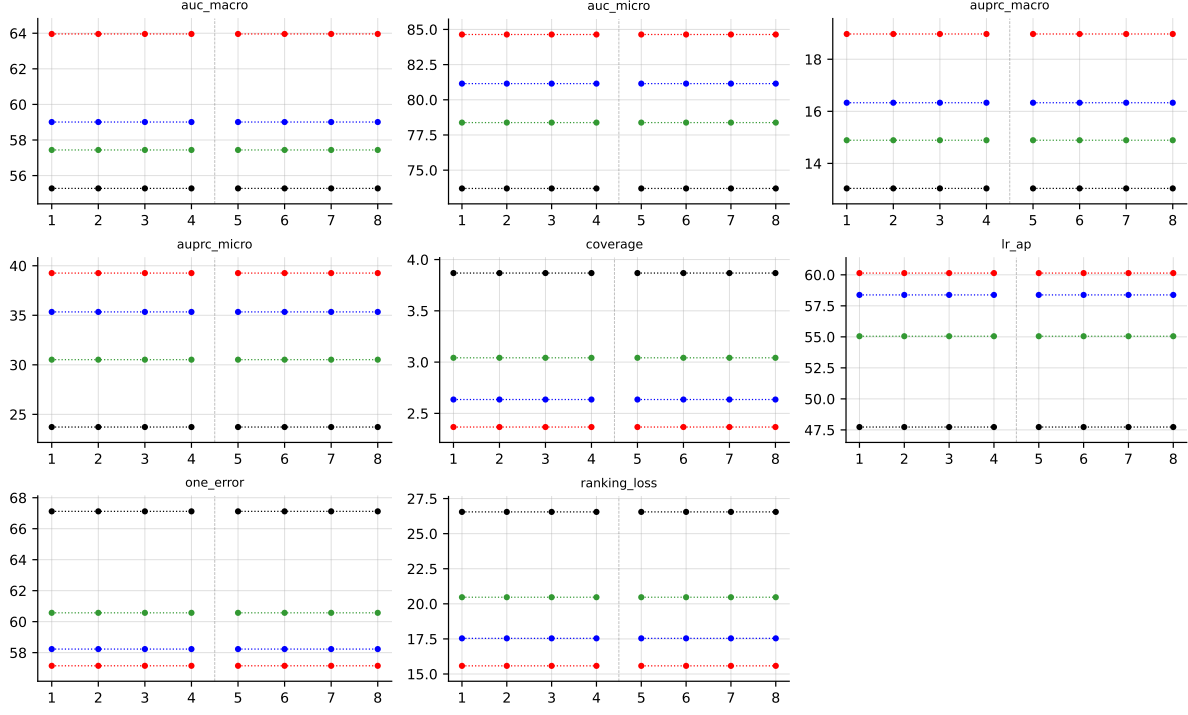


Table 49: Score-vector ranking metrics (AUROC, AUPRC, etc.) (ScoreVector) on HumanPseAAC (RF base learner).

B.6 Yeast ($K = 14$)

RF • Yeast • BinaryVector — Standard MLC predictions

RF • Yeast • PartialAbstention — Partial abstention

RF • Yeast • ScoreVector — Score-vector ranking metrics (AUROC, AUPRC, etc.)

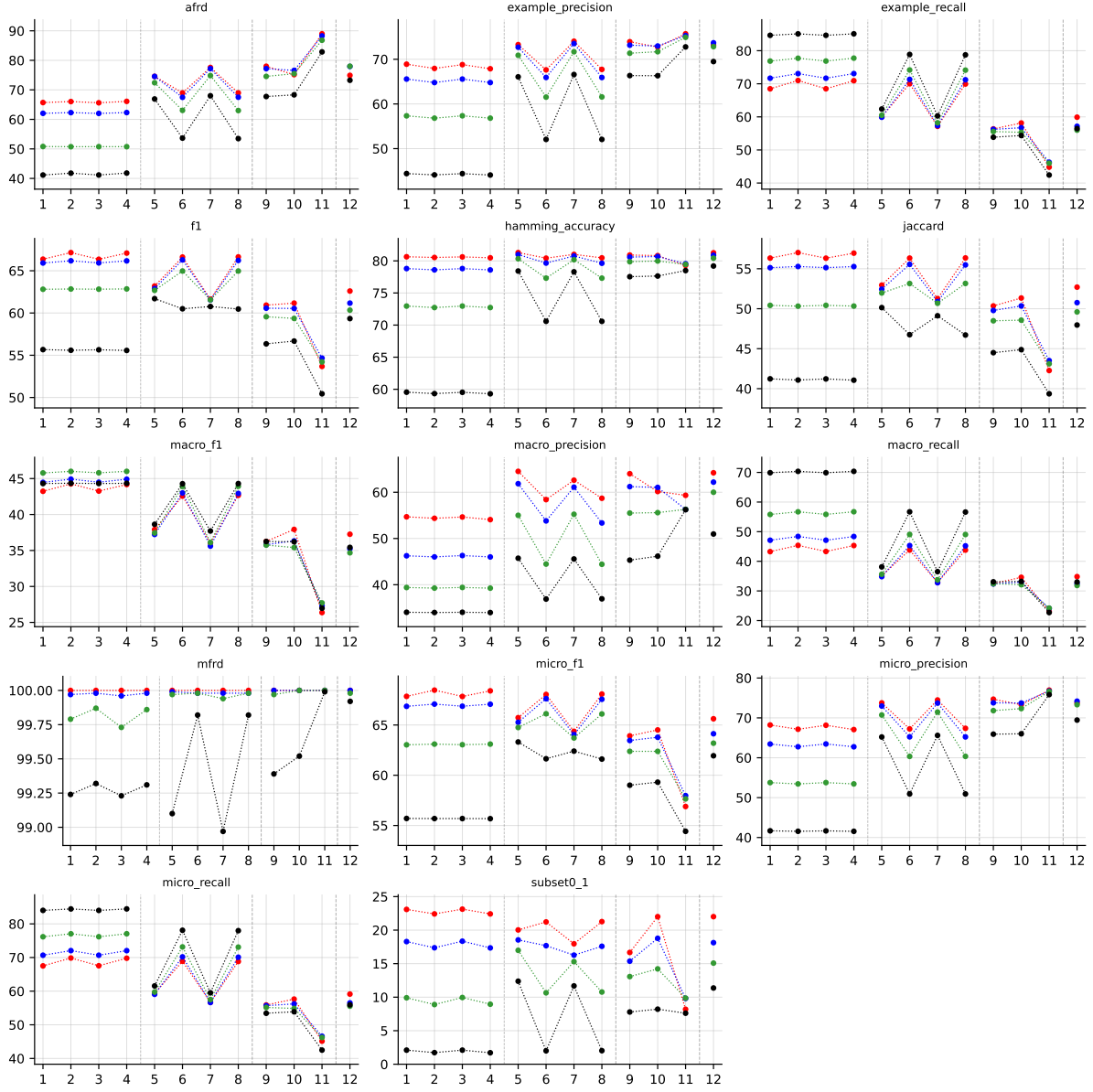


Table 50: Standard MLC predictions (BinaryVector) on Yeast (RF base learner).

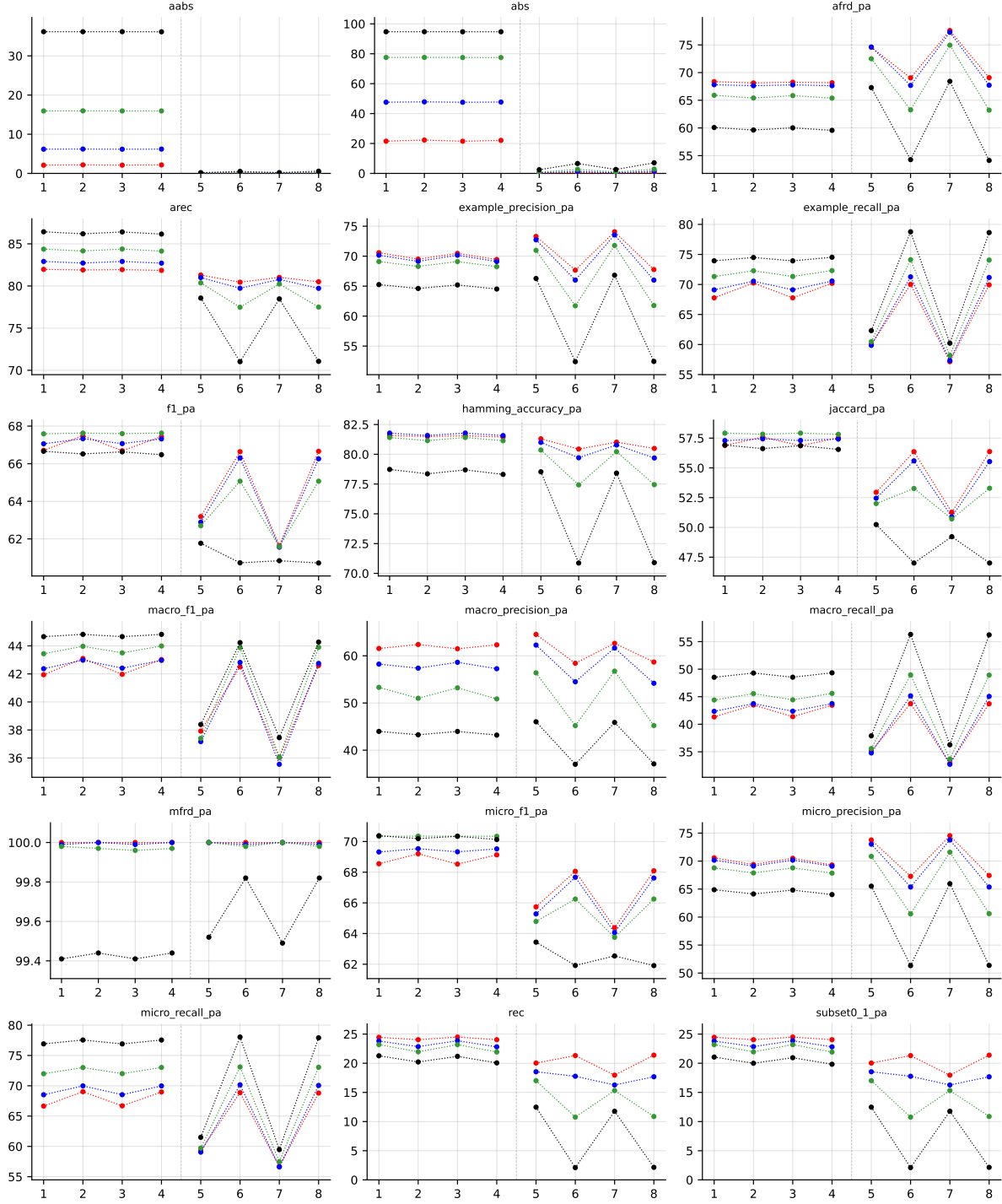


Table 51: Partial abstention (PartialAbstention) on Yeast (RF base learner).

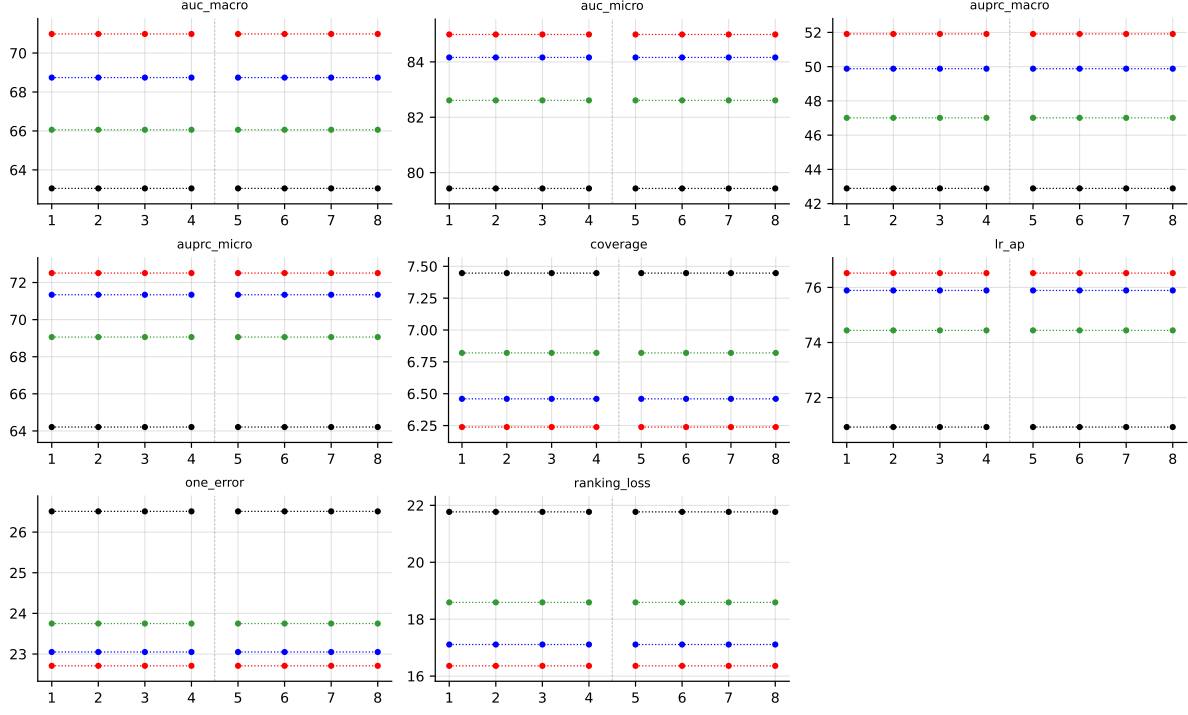


Table 52: Score-vector ranking metrics (AUROC, AUPRC, etc.) (ScoreVector) on Yeast (RF base learner).

B.7 CHD_49 ($K = 6$)

RF • CHD_49 • BinaryVector — Standard MLC predictions

RF • CHD_49 • PartialAbstention — Partial abstention

RF • CHD_49 • ScoreVector — Score-vector ranking metrics (AUROC, AUPRC, etc.)

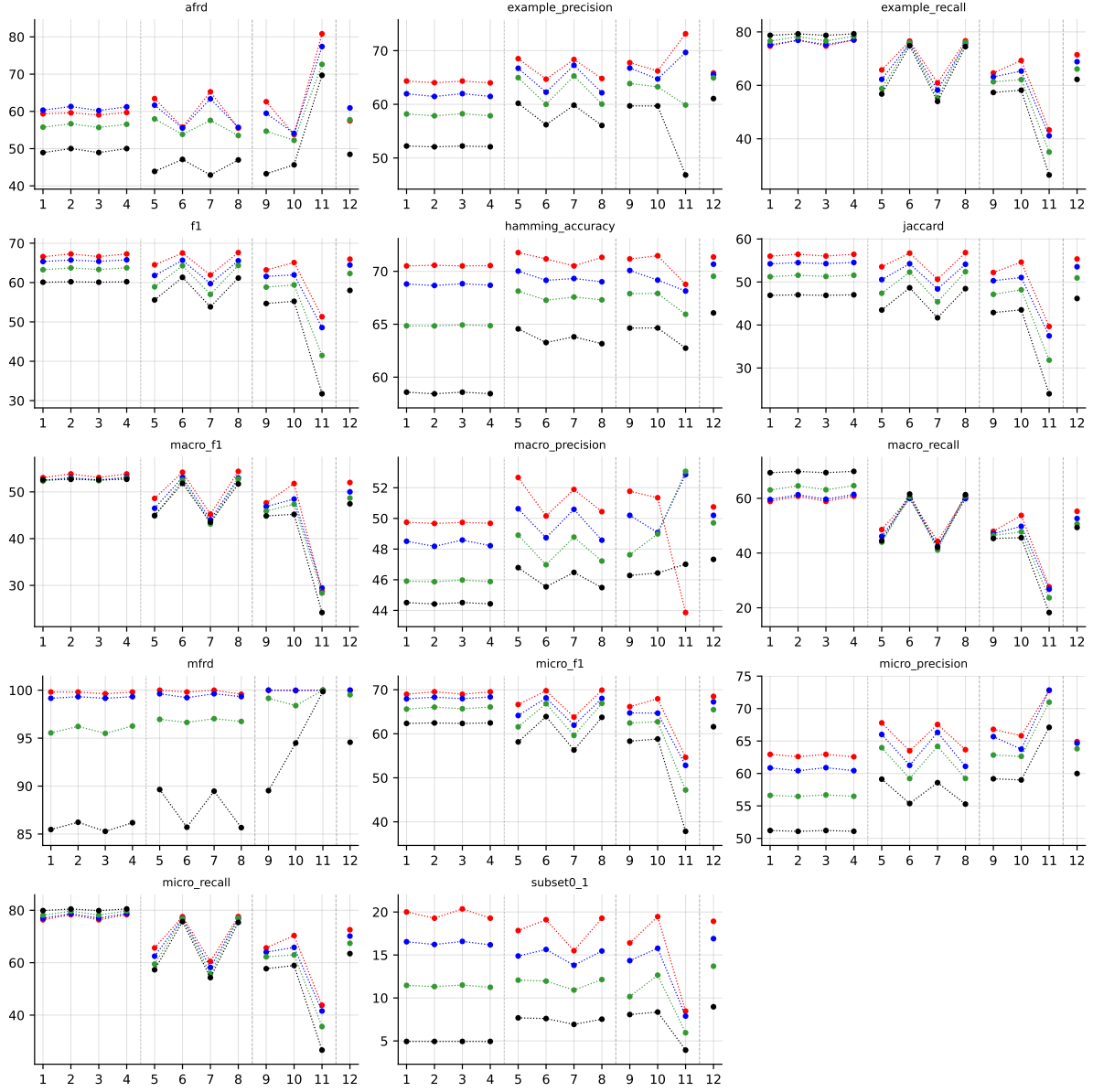


Table 53: Standard MLC predictions (BinaryVector) on CHD_49 (RF base learner).

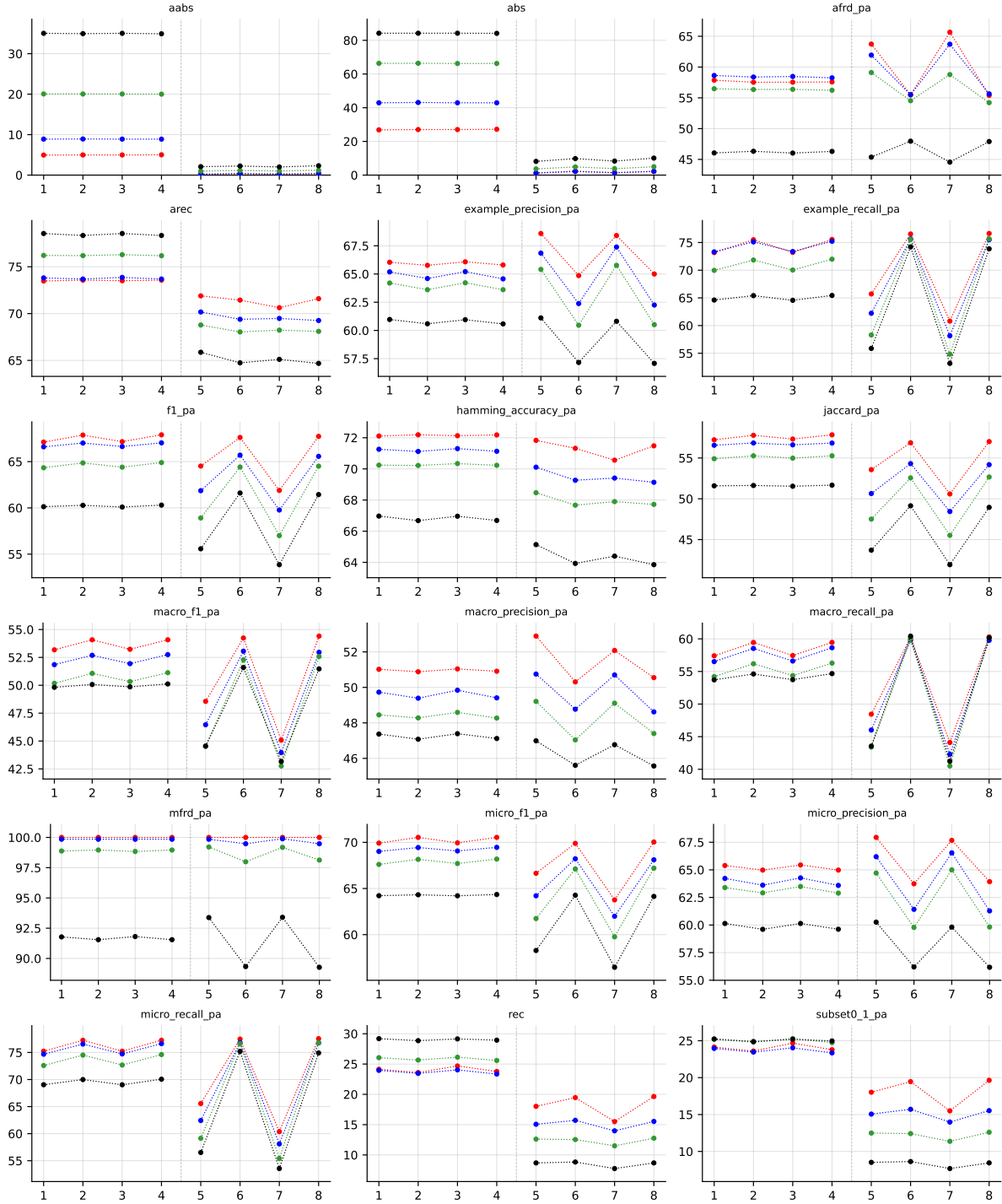


Table 54: Partial abstention (PartialAbstention) on CHD_49 (RF base learner).

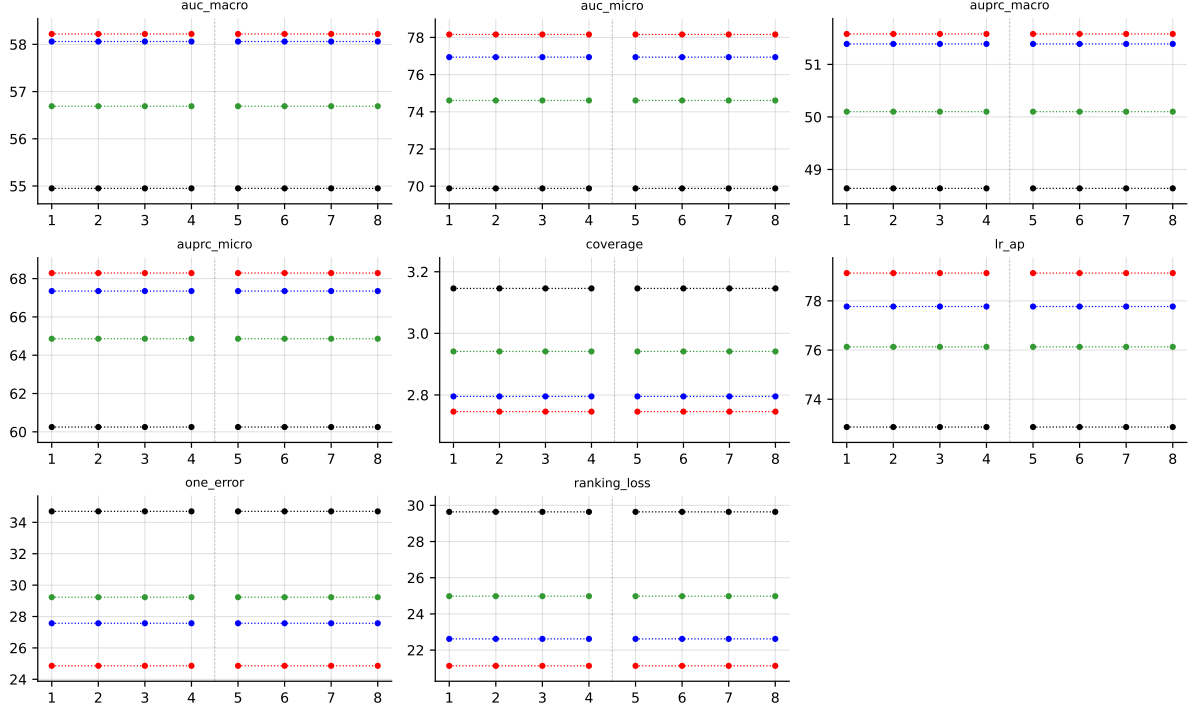


Table 55: Score-vector ranking metrics (AUROC, AUPRC, etc.) (ScoreVector) on CHD_49 (RF base learner).

B.8 Water-quality ($K = 14$)

RF • Water-quality • BinaryVector — Standard MLC predictions

RF • Water-quality • PartialAbstention — Partial abstention

RF • Water-quality • ScoreVector — Score-vector ranking metrics (AUROC, AUPRC, etc.)

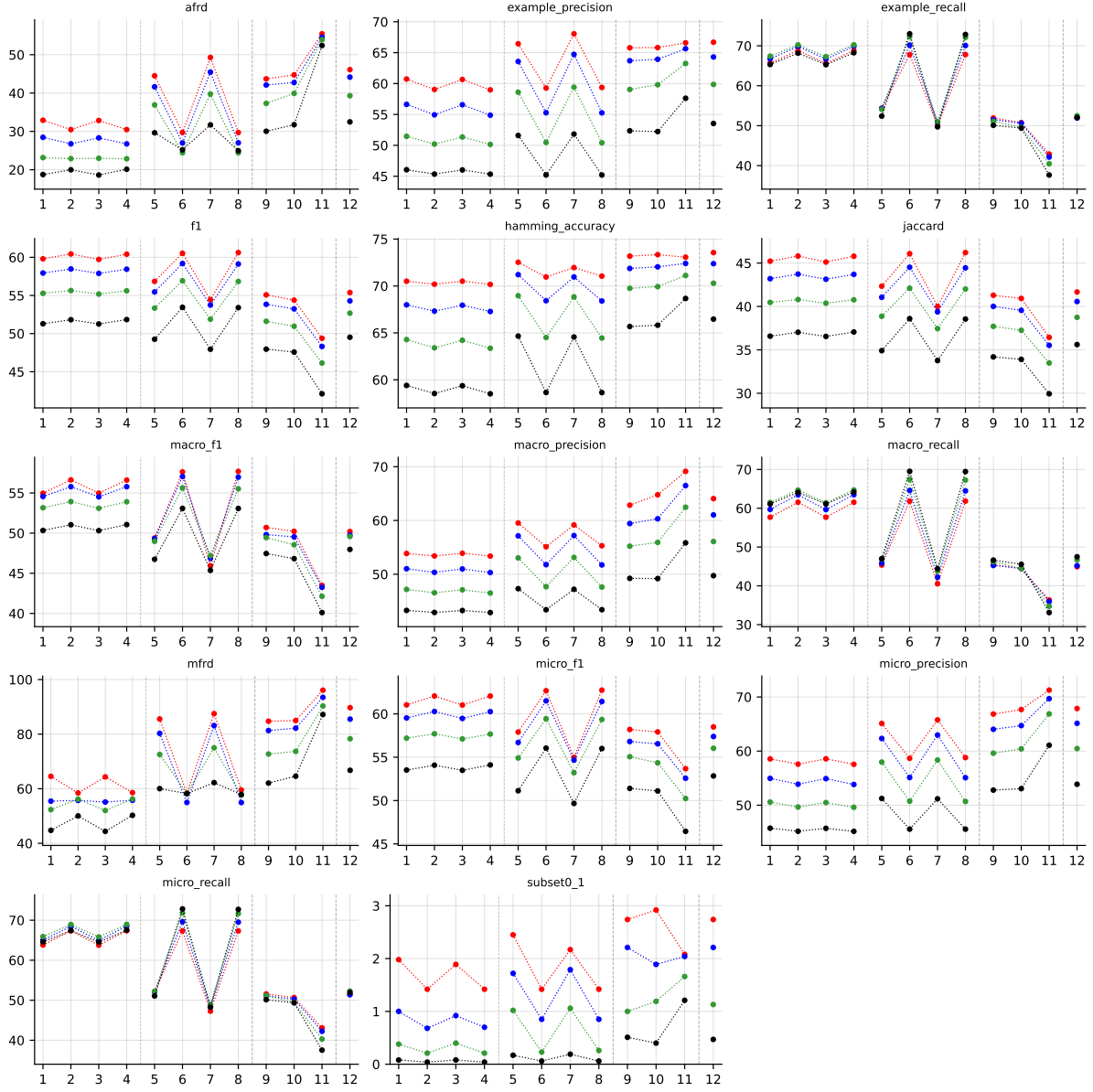


Table 56: Standard MLC predictions (BinaryVector) on Water-quality (RF base learner).

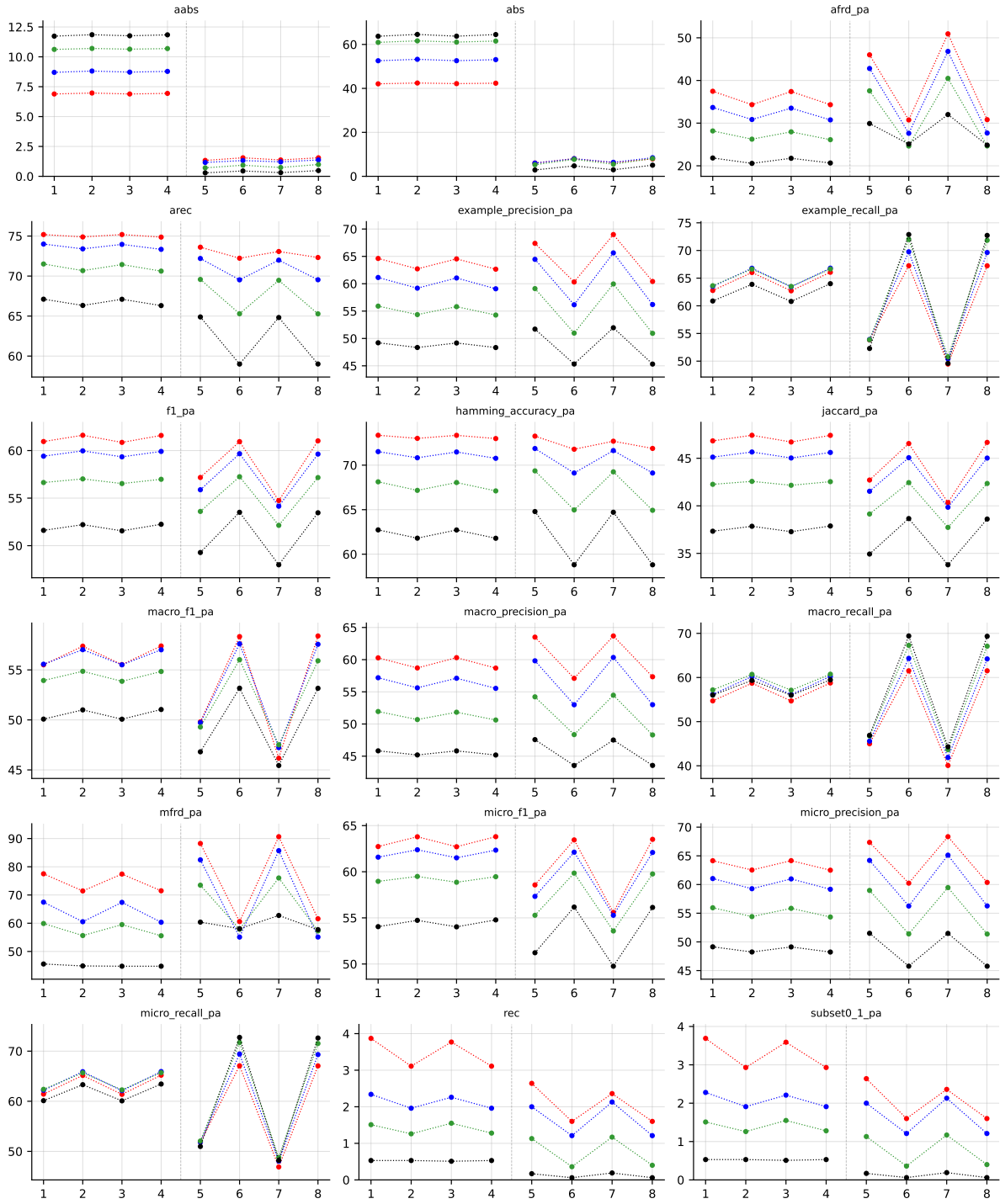


Table 57: Partial abstention (PartialAbstention) on Water-quality (RF base learner).

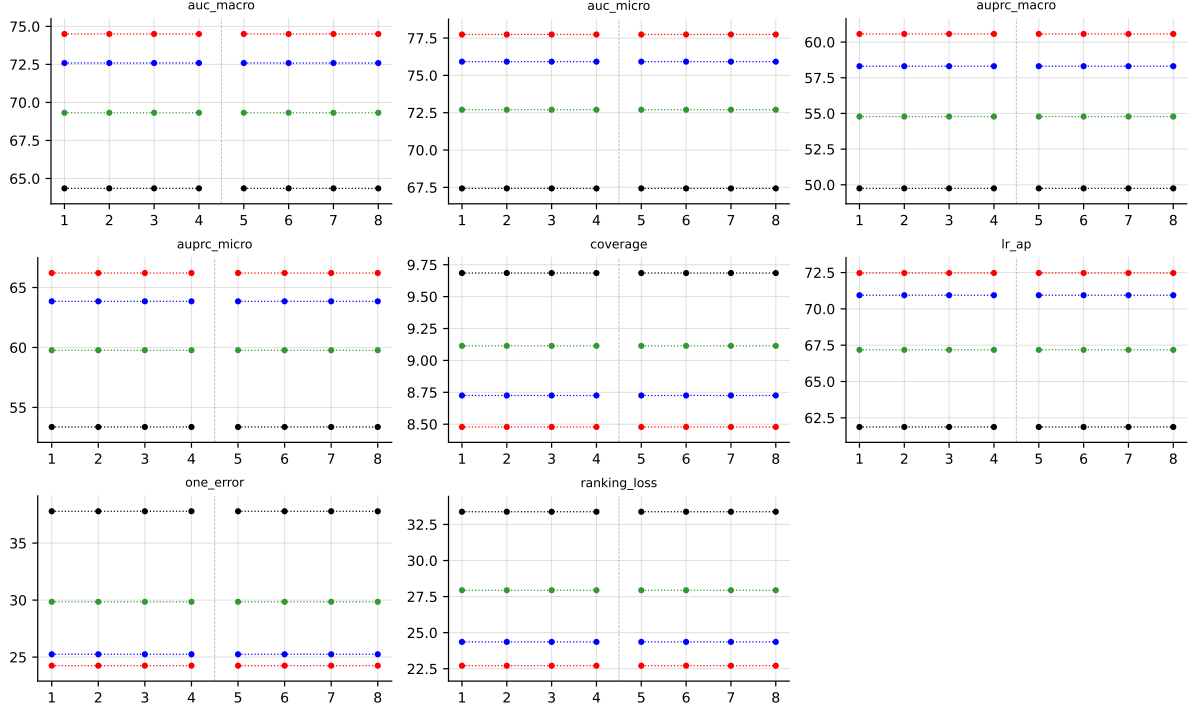


Table 58: Score-vector ranking metrics (AUROC, AUPRC, etc.) (ScoreVector) on Water-quality (RF base learner).

B.9 enron ($K = 53$)

RF • enron • BinaryVector — Standard MLC predictions

RF • enron • PartialAbstention — Partial abstention

RF • enron • ScoreVector — Score-vector ranking metrics (AUROC, AUPRC, etc.)

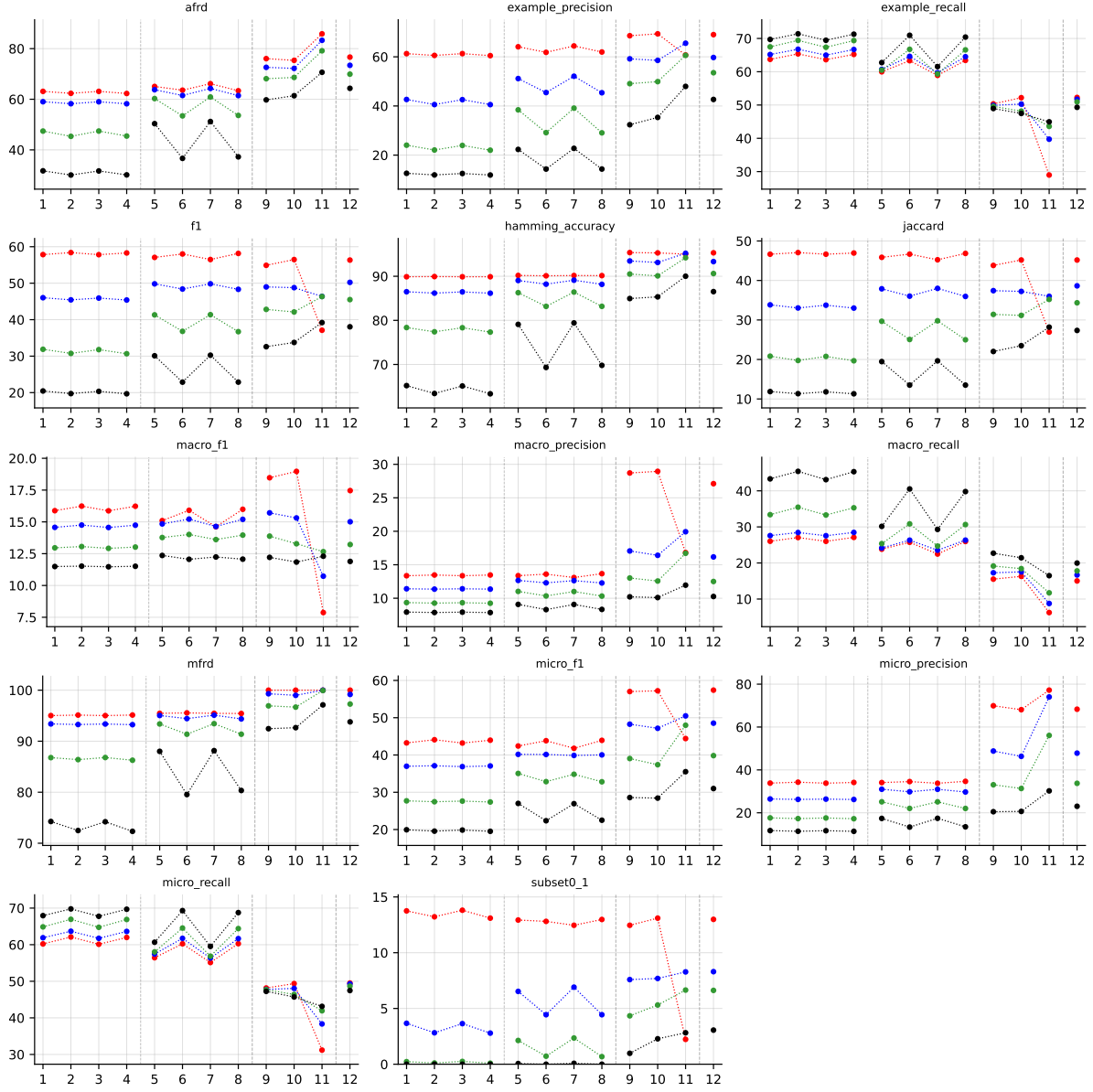


Table 59: Standard MLC predictions (BinaryVector) on **enron** (RF base learner).

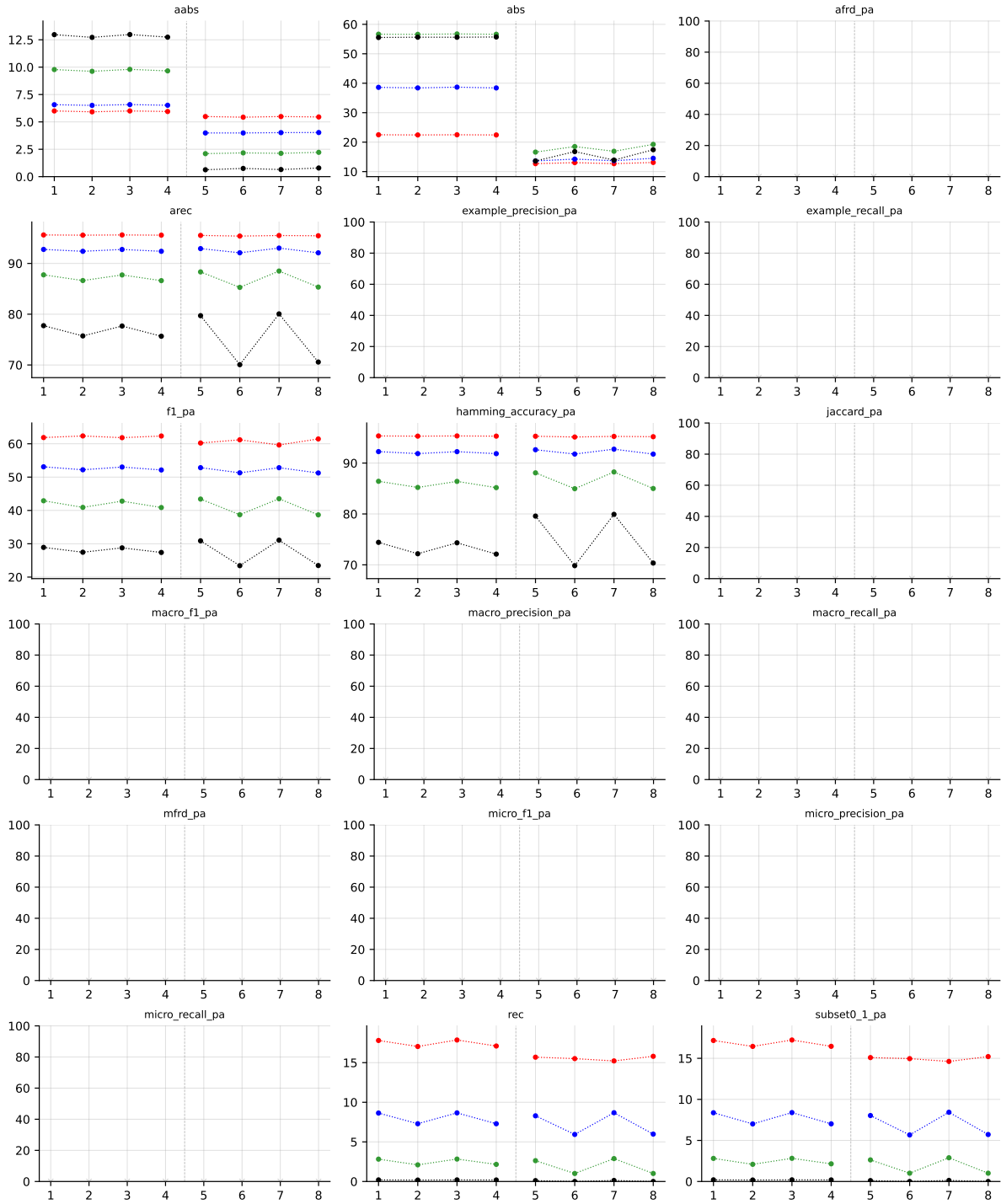


Table 60: Partial abstention (PartialAbstention) on enron (RF base learner).

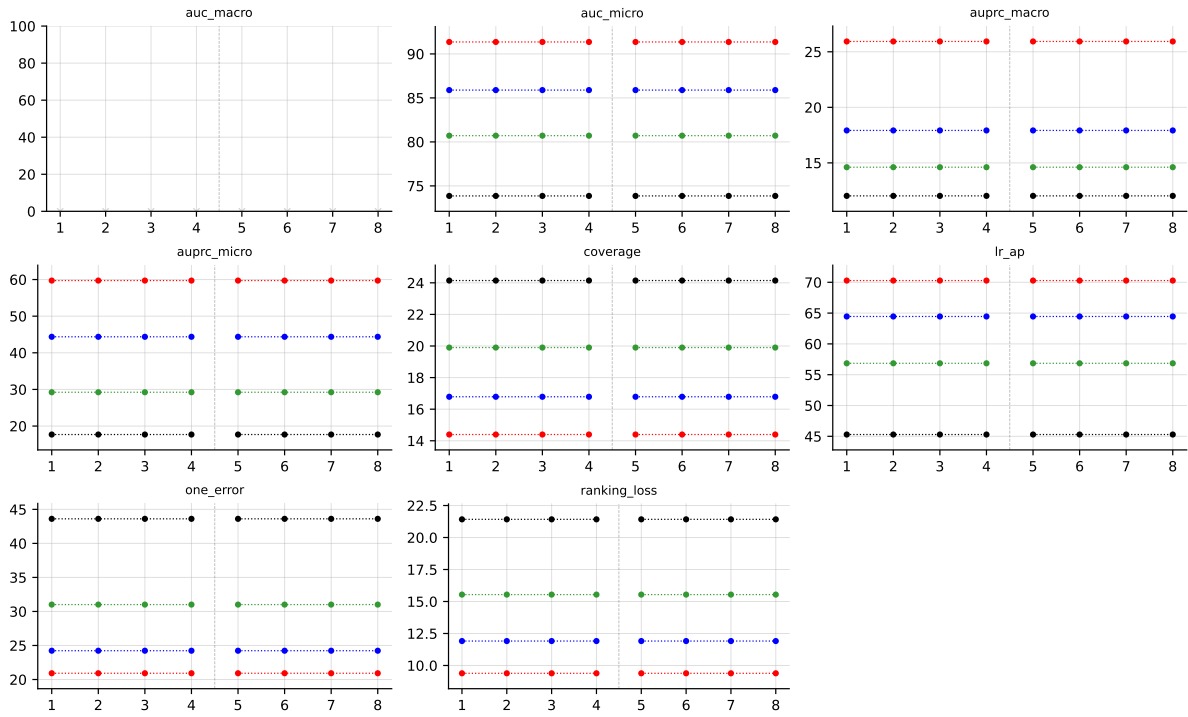


Table 61: Score-vector ranking metrics (AUROC, AUPRC, etc.) (ScoreVector) on **enron** (RF base learner).

C LGBM aggregated results

LGBM • Aggregate • BinaryVector — Standard MLC predictions

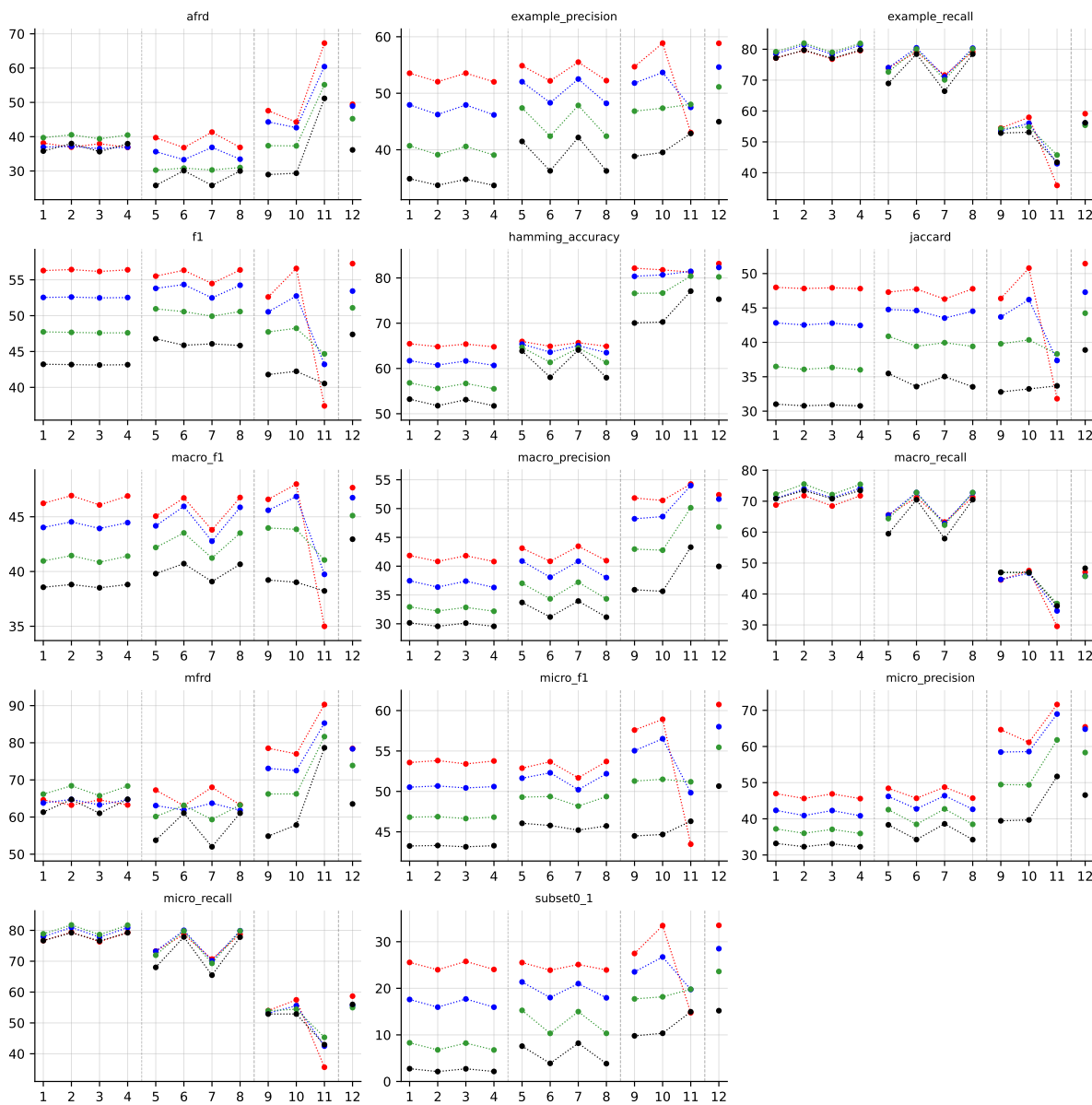


Table 62: Standard MLC predictions (BinaryVector), LGBM base learner, averaged across datasets (8 of 9: enron LGBM still training). Method indices and line colors follow the encoding paragraph above.

LGBM • Aggregate • PartialAbstention — Partial abstention

LGBM • Aggregate • ScoreVector — Score-vector ranking metrics (AUROC, AUPRC, etc.)

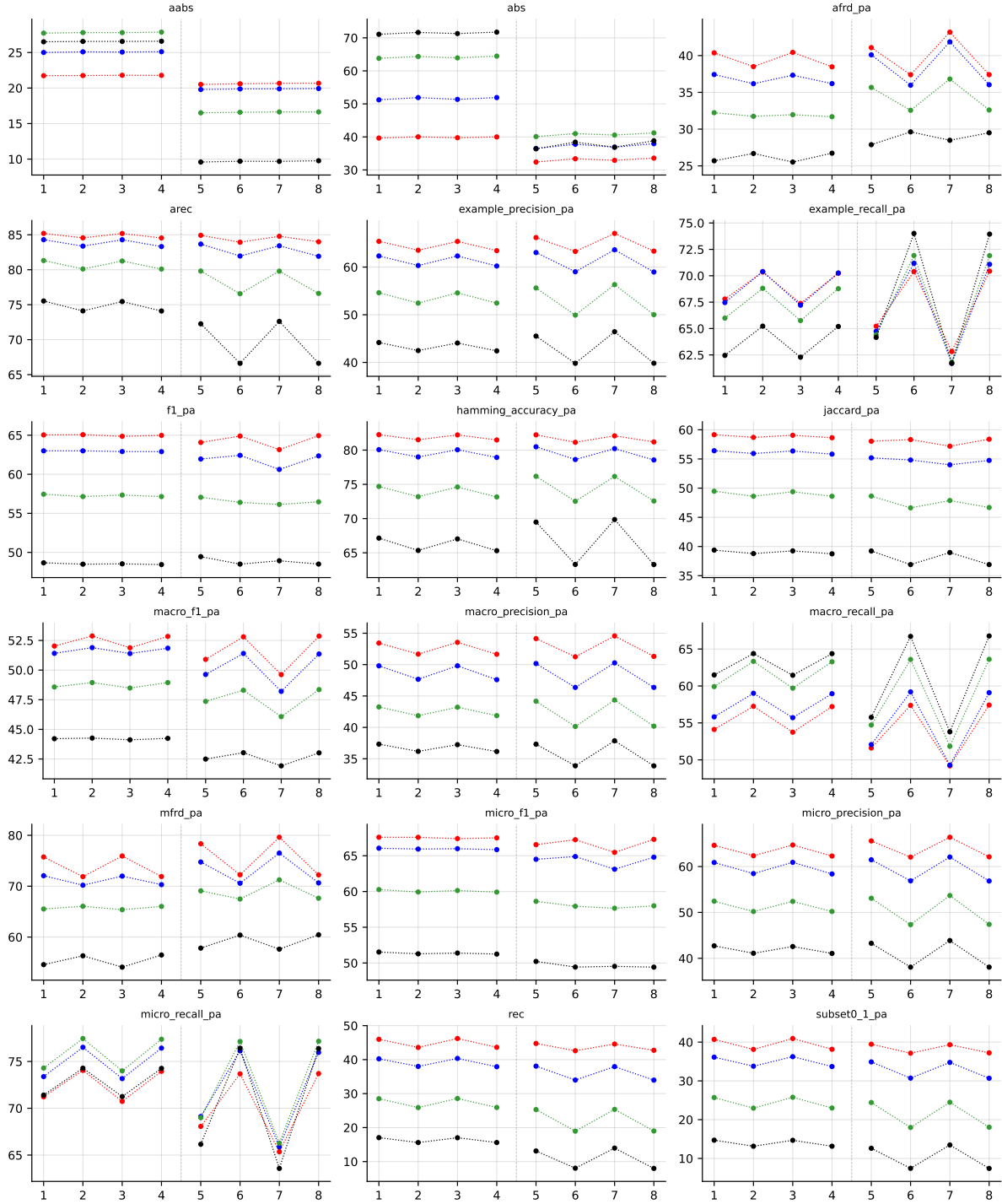


Table 63: Partial abstention (PartialAbstention), LGBM base learner, averaged across datasets (8 of 9: enron LGBM still training). Method indices and line colors follow the encoding paragraph above.

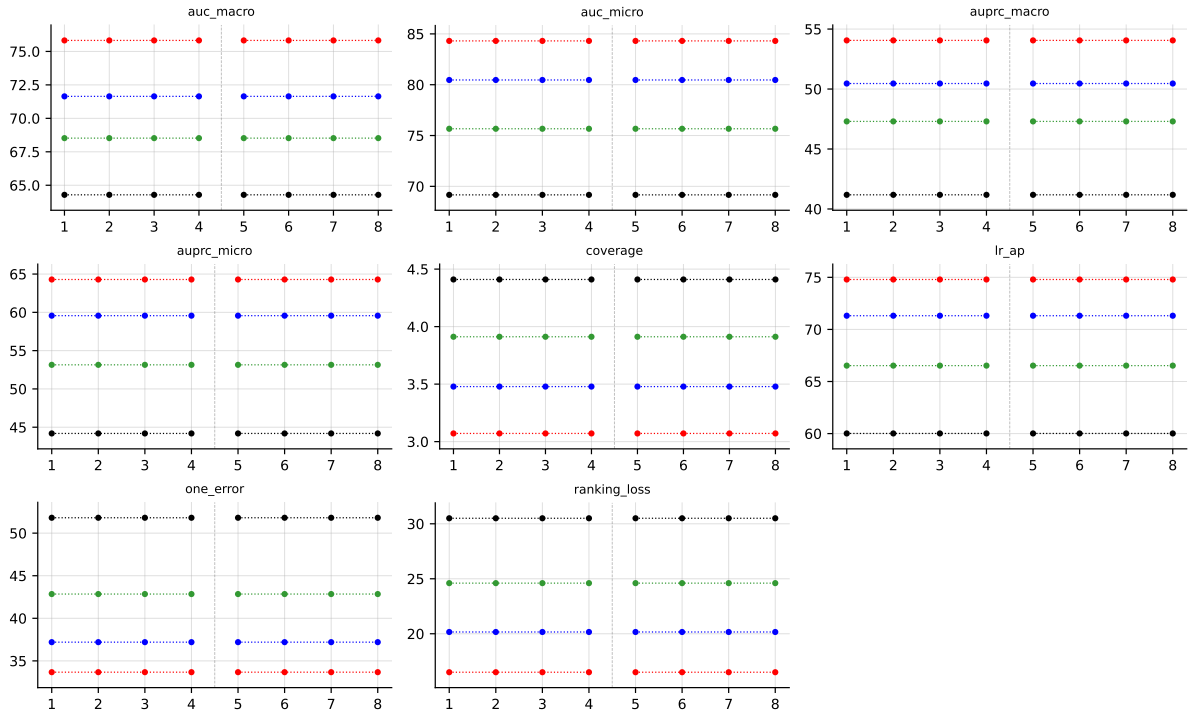


Table 64: Score-vector ranking metrics (AUROC, AUPRC, etc.) (ScoreVector), LGBM base learner, averaged across datasets (8 of 9: enron LGBM still training). Method indices and line colors follow the encoding paragraph above.